

22 Whiting in North Sea and eastern English Channel

whg.27.47d – *Merlangius merlangus* in Subarea 4 and Division 7.d

This section contains the assessment and forecast relating to whiting in the North Sea (ICES Subarea 4) and eastern English Channel (ICES Division 7.d). The current assessment is formally classified as an update assessment. The next benchmark is planned for 2025/2026.

22.1 General

22.1.1 Stock definition

A summary of available information on stock definition can be found in the Stock Annex and in the WKNSEA 2018 benchmark report working documents (WKNSEA; ICES, 2018). A complex population structure for whiting in the North Sea has been proposed, based on studies about whiting movements, life-history traits, genetic data, identification of spawning aggregation, as well as on population temporal asynchrony observed in SSB, recruitment and egg abundance between areas. The benchmark concluded that literature and provided data did not suffice to revise assessment units for this stock. As before, the new assessment was run for the combined North Sea and eastern English Channel (27.4 and 27.7d). Exploratory SURBAR assessments were run for individual components (northern and southern component) and compared to the combined stock. Stock dynamics of the stock components show greater synchrony in recent periods.

22.1.2 Ecosystem aspects

No new information was presented at the WG. A summary of available information on ecosystem aspects is presented in the Stock Annex prepared by ICES WKROUND (ICES, 2013).

22.2 Fisheries

Information on the fishery (and its historical development) is contained in the Stock Annex prepared by ICES WKNSEA (ICES, 2018).

22.3 ICES advice

ICES advice for 2023

In April 2022, ICES concluded as follows:

ICES advises that when the MSY approach is applied, catches in 2023 should be no more than 110 172 tonnes. ICES notes the existence of a precautionary management plan, developed and adopted by one of the relevant management authorities for this stock. Management should be implemented at the stock level.

ICES advice for 2024

In October 2024, ICES concluded as follows:

ICES advises that when the MSY approach is applied, catches in 2024 should be no more than 111 953 tonnes. ICES notes the existence of a precautionary management plan, developed and adopted by one of the relevant management authorities for this stock. Management should be implemented at the stock level.

ICES advice for 2025

In October 2024, ICES concluded as follows:

ICES advises that when the MSY approach is applied, catches in 2024 should be no more than 188 148 tonnes. ICES notes the existence of a precautionary management plan, developed and adopted by one of the relevant management authorities for this stock.

22.4 Management

Management of whiting is implemented by TAC and technical measures. The TACs for this stock are split between two areas: (i) Subarea 4 and Division 2.a (EU waters), and (ii) Divisions 7b–k. Since 1996 the North Sea and eastern English Channel whiting assessments have been combined into one.

Since 2018, with introduction of the landing obligation the EU legislation specifies that the TAC accounts for total human consumption catch in Subarea 4, including discards and landings below minimum landings size (BMS) but excluding industrial bycatch (IBC). IBC of whiting, mainly occurring in the Danish fishery, is regulated separately relating to TAC of the respective target species in the industrial fishery (EU, 2025). In contrast, the TACs for whiting in Subarea 4 and Division 7.d were set by fisheries managers in recent years based on total catches of whiting which include IBC (EU–Norway–UK, 2023). This discrepancy between management and legislation may lead to excess catches of whiting in the North Sea.

The TAC in Subarea 4 has increased in recent years, for 2023 it was 34 294 t, for 2024 it was 76 697 t and for 2025 it is 111 861 t. Up to 2023, there was no separate TAC for Division 7.d; landings from this Division were counted against the TAC for Divisions 7.b–k combined (8450 t in 2023). Since 2024, the TAC is given for Division 7.d separately (EU–Norway–UK, 2023). The TAC for Division 7.d for 2024 was 18 899 t and for 2025 is 27 564 t.

Since 2009, the landings data have been collated separately for each area. In previous years, the human consumption landings in Subarea 4 were estimated between 65% and 95% of the combined area totals. In 2023 about 91% and in 2024 about 90% of the total human consumption landings originated from Subarea 4.

The minimum landing size for whiting in Subarea 4 and Division 7.d is 27 cm. The minimum mesh size for targeting whiting in Subarea 4 is 120 mm and in Division 7.d is 80 mm.

Whiting are a by-catch in some *Nephrops* fisheries that use a mesh size of 80 mm, although landings are restricted through bycatch regulations. They are also caught in flatfish fisheries that use a smaller mesh size. Industrial fishing with small-meshed gear is permitted, subject to by-catch limits of protected species. Regulations also apply to the area of the Norway pout box, preventing industrial fishing with small meshes in an area where the by-catch limits are likely to be exceeded. Industrial bycatch occurred mainly in Subarea 4 by Danish industrial fisheries. In 2016–2018, some very minor catches in the Norwegian fishery have been reported as BMS, these may be considered industrial bycatch but were not reported as such.

Conservation credit scheme

Since 2008, real time closures (RTCs) have been implemented under the Scottish Conservation Credits Scheme (CCS). The CCS has two central themes aimed at reducing the capture of cod through (i) avoiding areas with elevated abundances of cod through the use of Real Time Closures (RTCs) and (ii) the use of more species selective gears. Within the scheme, efforts are also being made to reduce discards generally. In 2009, 144 RTCs were implemented, and the CCS was adopted by 439 Scottish and around 30 English and Welsh vessels. In 2010, there were 165 closures, and from July 2010, the area of each closure increased (from 50 square nautical miles to 225 square nautical miles). In more recent years, the following numbers of closures were implemented: 185 (2011), 173 (2012), 166 (2013), 94 (2014), 97 (2015) and 114 (2016). Although the scheme is intended to reduce mortality on cod, it undoubtedly has an effect on the mortality of associated species such as whiting. However, the scheme was suspended 20 November 2016 and there are no plans for its reintroduction.

In 2016, 14 Scottish demersal whitefish vessels participated in a trial Fully Documented Fishery (FDF) scheme, following similar schemes during 2010–2015. The uptake of the scheme declined due to concerns about monitoring of discards under the EU Landing Obligation. The cod-specific FDF scheme terminated at the end of 2016, due to the suspension of most aspects of the EU Cod Recovery plan which removed the opportunity for countries to provide additional quota for participants. However, a new Scottish FDF scheme has commenced, which is being run along similar lines, and which is intended to monitor discarding of saithe and monkfish. Since 2017 there were no data submissions to InterCatch on discard rates from the FDF fleets for whiting.

22.5 Data available

22.5.1 Catch

Since 2009, international data on landings and discards have been collated through the InterCatch system. As additional categories logbook registered discards and BMS landings can be submitted. In 2024 data, no logbook registered discards were submitted. Minor whiting landings have been reported as BMS landings into InterCatch since 2016. In 2024 data, these mostly originated from the UK (England, Scotland) by otter bottom trawls (OTB_DEF_>=120 and OTB_CRU_70-99; 47 t). Generally, BMS was treated as discards as in previous years. For this assessment, InterCatch data for the years 2021–2024 were raised following new data submissions. The catch data for 2021–2023 showed minor change (increase 3.1–6%) due to revisions in Dutch and English discard submissions.

In 2024 data, 60% of the landings (here total landings include industrial bycatch) had associated discard data imported to InterCatch. The landings of métiers for which discard data was provided in 2024 are illustrated in Figure 22.1. Discards were raised from discard ratios from Subarea 4 and Division 7.d combined. Data were stratified by gear type (TR1 and TR2) and half year to raise discards for fleets without imported discards (i.e., quarter 1 and 2 together, and quarter 3 and 4 together). For other gear types, discards were raised using discard rates from all available fleets. The raised discards amounted to 50% of total discards (Table 22.3b). Industrial bycatch landings were excluded from the discard raising, as no discards are assumed to occur in that fleet. Throughout this report minor BMS landings were grouped together with discards for age allocations as well as estimation of mean weights-at-age.

Figure 22.2a shows métier-specific landings in percent of the total landings in 2024 for whiting in Subarea 4 and Division 7.d, for fleets sampled for age compositions in landings and unsampled fleets. The figure also shows the cumulative landings when sampled and unsampled fleets

are ordered by landings yield. Sampled fleets comprise around 65% of the overall landings, from 23 different métiers (Table 22.3.c).

However, although the unsampled fleets provide considerable landings overall (35%), most métiers provide less than 5% of the overall landings each. A métier summarized as miscellaneous landings of industrial bycatch (MIS_MIS_0_0_0_IBC) provides 11% of the total landings (4.7% of total catch), all of which occurred in the Danish fishery and were generally not sampled. Generally, no age samples are provided for the Dutch métiers, but which contribute an increasing amount of reported landings (SSC_DEF_70-99_0_0_all) and discards (TBB_DEF_70-99_0_0_all) in recent years.

For raising discard rates from sampled to unsampled fleets all samples were used with splitting of fleets on the basis of gear type. Discard rates for unsampled whiting fleet components were obtained from discards reported by France, UK (England, Scotland), Netherlands, Denmark, Belgium and Germany.

Of the total discards, 50% were imported into InterCatch. 33% of the imported discards were sampled for age distributions (Table 22.3c). The 11 métiers providing discard samples and unsampled métiers are listed in Figure 22.2b.

Official reported landings by country, WG estimates of total catch and catch component yields, as well as TACs covering the respective areas are given in Table 22.1 for the North Sea (Subarea 4) and in Table 22.2 for the eastern English Channel (Division 7.d).

Figure 22.3 plots the trends in the commercial catch for each component in Subarea 4 and Division 7.d combined. In 2024, the amount of estimated discards exceeded the landings.

22.5.2 Age compositions

In 2024, age compositions for the landings and discards were based on samples provided by France, UK (England, Scotland) and Denmark. Age compositions are applied to landings with splitting of fleets on the basis of half year (quarter 1 & 2 vs. 3 & 4) and gear type (TR1 and TR2), while discards age compositions are allocated using all discard samples with splitting of fleets on the basis of gear type (TR1) and half year (quarter 1 & 2 vs. 3 & 4). For the remaining gear types, age compositions were allocated using all available samples, for landings and discards separately.

Limited sampling of the industrial bycatch component resulted in the 2006 data appearing as an outlier and the 2007 to 2010 data were deemed unreliable. This applies to both the age compositions and the estimates of mean weights at age. Thus, the data for 2006 to 2010 were replaced with estimates derived from the years 1990 to 2005 (as described in the Stock Annex). For the industrial bycatch in 2011 and 2012, age compositions were inferred in InterCatch from corresponding age samples taken from small-mesh fisheries of France and the UK. In recent years, age compositions for industrial bycatch are estimated from all samples (landings and discards) without splitting of fleets. Minor BMS landings (below minimum landing size) were not sampled. BMS was treated the same as discards, and age compositions are inferred from discard samples only. BMS and discards were combined as discards.

Total international catch numbers-at-age (Subarea 4 and Division 7.d combined) as estimated by ICES are presented in Table 22.4. Numbers for human consumption landings, discards, and industrial bycatch are given in tables 22.5 to 22.7. Total catches, and catch components, as estimated by ICES are listed in Table 22.12. In 2024, discards represented 57% of the total catches. In recent years the catch level has stabilized to a certain extent. There continued to be a high proportion of discarding of whiting up to age 2 (Figure 22.4). In recent years, also whiting aged 3-5 have been increasingly discarded.

22.5.3 Weight-at-age

Mean weights-at-age (Subarea 4 and Division 7.d combined) in the catch are presented in Table 22.8. Mean weights-at-age (both areas combined) in human consumption landings are presented in Table 22.9, and for the discards and industrial by-catch in the North Sea in Tables 22.10 and 22.11, respectively. Weights-at-age are depicted graphically in Figure 22.5, which indicates an increasing trend (with annual fluctuations) in mean weight-at-age in the landings, discards and total catch for ages > 2 since the early 2000s. In recent years, mean weights at age have stabilized on the higher level. Mean weights at age in landings have decreased for age 0 in the late 2000s.

Unrepresentative sampling of industrial bycatch in 2006 to 2010 resulted in poor estimates of the mean weights-at-age and these have been replaced by the mean weight at age for the period 1995 to 2005 (zero weights are taken as missing values). From 2009 onwards, the sampled weights-at-ages of total catches were used to estimate weights-at-ages of industrial bycatch.

Stock mean weights at age are estimated from commercial catch weights-at-age scaled to the level of weights-at-age estimated in IBTS Q1 (WKNSEA; ICES 2018, Figure 22.6). Unsmoothed values of weights-at-age are used in the assessment (Table 22.13). For the calculation of SSB in 2025, the recent 3-year average of stock-weights-at age was used in the assessment model.

22.5.4 Maturity

Values for proportion mature at age are estimated using IBTS Q1, shown in Table 22.14a and Figure 22.7. The estimation procedure is detailed in the Stock Annex. Values prior 1991 are assumed constant using estimated values of 1991, due to data quality issues and high variability in results for the earlier time period. The same maturation proportion was assumed for individuals 6 years and older.

The number of individuals sampled at age has slightly decreased for age 1 and 3 in 2024 but is still considered representative. This reduction is due to a combination of factors, low 2023 cohort, bad weather conditions during the survey (Table 22.14b).

In 2022 and 2024, a high proportion mature at age 1 was estimated (A_{50} below 1). A new maturity scales was introduced in 2021, since then some countries submit more than 1 stage of immatures (A, Ba) allowing for higher resolution maturity data. Submission of mature stage B (combining immatures Ba and matures Bb) may impact maturity ogives by overestimation of the number of mature individuals. However, estimated proportions of matures at age were smoothed before inclusion in the assessment model, which limits the impact of these particular years. At the next benchmark, increased uncertainty in recent maturity data will be considered.

For the calculation of SSB in 2025, the recent 3-year (2022-2024) average of maturity-at-age was used in the assessment model.

22.5.5 Natural mortality

At WGNSSK 2024, the most recent natural mortality estimates were included in the assessment model and the reference points were revised as a result (ICES, 2024a, b). Estimates of natural mortality (M) are taken from the 2023 update key run from of the SMS multispecies model (WGSAM; ICES, 2024a). Natural mortality estimates were smoothed before inclusion in the assessment model (Table 22.15, Figure 22.8). The same natural mortality was assumed for individuals 6 years and older. Data is available up to 2022. For the assessment, natural mortality estimates for 2023-2025 are assumed identical to 2022.

22.5.6 Research vessel data

Up until 2019, the historical time-series of survey indices has been calculated using a manual substitution procedure. The data obtained with this manual procedure is only available until Q3 2019. Since 2020, survey indices are recalculated using a new automated substitution procedure to fill ALK key in areas with low sample size. This new automated method is seen as an improvement to data quality and transparency of the procedure. A comparison of the historical survey indices obtained with the old manual method and the historical survey indices recalculated with the new automated method show that the new method revealed that assessment outputs obtained with the new methods result in lower Mohn's rho values for SSB, F and recruitment. The new data series therefore appear to lead to more consistent assessment results (see Annex 9, ICES 2021a). As a result, for the 2021 assessment on whiting in 27.4 and 7d it was decided to use the historical time-series of survey indices obtained with the new automated substitution procedure. The time series is extended each year since 2021 with new indices calculated using the new automated substitution procedure, as provided by DATRAS.

Survey tuning indices are presented in Table 22.16a and b. The indices used in the assessment are ages 1–5 from the IBTS–Q1 and ages 0–5 from IBTS–Q3 surveys, from 1983–2025 and 1991–2024, respectively. The report of the 2001 meeting of WGNSSK (ICES, 2002) and the ICES advice for 2002 provide arguments for the exclusion of commercial CPUE tuning series from calibration of the catch-at-age analysis. Such arguments remain valid and only survey data have been considered for tuning purposes. All available tuning series are presented in the Stock Annex.

In Figure 22.9, survey distribution maps based on the IBTS–Q1 survey in the North Sea, for ages 1–3+ of the first quarter (Q1) 2021–2025, are presented. In recent years, high density of age 1 whiting appears mostly in the southern North Sea. Whiting aged 3+ can be found in high abundance throughout the Greater North Sea. Figure 22.10, the third quarter is represented (Q3) for ages 0–3+ for the years 2021–2024. Similarly, as in Q1, in recent years low recruitment at age 0 is seen in the northern North Sea, while significant survey catches at age 0 occurred in the southern North Sea. High densities of fish aged 3+ can be observed throughout the North Sea in recent years.

22.6 Benchmarks

The ICES Benchmark Workshop on North Sea Stocks 2018 (WKNSEA) was held at ICES in Copenhagen in early 2018. Analyses focused on a number of key issues (maturity, natural mortality, stock-weights at age, stock identity, assessment model) details can be found in WKNSEA report (ICES, 2018) and stock annex.

No changes were made to the use of survey indices. Catch data was updated in InterCatch following a data call for 2009–2016. A new stratification design to allocate discard ratios and age distributions was introduced, details of the allocation scheme can be found in the Stock Annex and in Section 22.5. The assessment model was updated from XSA to SAM and new reference points were estimated.

As before, Area 27.4 represents the management unit with TAC advice to be given. WGNSSK and WKNSEA recommended, that the stock identity issue should be reviewed in the future when firm evidence become available. Until then it is recommended to monitor area-specific stock development based on survey data when it is available (see Section 22.15). The feasibility of combining Division 3.a with Subarea 4 components was explored, but data showed there were biological reasons to leave the components as separate stocks.

In April 2021, an interbenchmark was carried out to assess the impact of new natural mortality estimates from WGSAM (ICES, 2021b) on the assessment, and the reference points previously defined were updated as a result (IBPNSSWhiting; ICES, 2021c). In April 2022, the SAM assessment model settings were updated to accommodate a plusgroup at age 6 (previously age 8) and F averaged across age 2 to 5 (previously 2 to 6). This was done in order to address retrospective patterns caused by a drop in the abundance of age 7 and 8 individuals which resulted in a much higher F , and lower SSB, compared to previous assessments. The reference points were updated during the WGNSSK 2022 meeting (Annex 8, ICES, 2022).

At WGNSSK 2024, the most recent natural mortality estimates provided by WGSAM (ICES, 2024a) were included in the assessment model and the reference points were revised as a result (ICES, 2024b).

The next benchmark is planned for 2025/2026, aimed to update survey indices to modelled estimates, reconsider other input data (maturity) and SAM model settings (plusgroup), as well as consider SAM stochastic forecast.

22.7 Data analyses

22.7.1 Exploratory survey-based analyses

In Figure 22.11, time-series of survey log CPUE at age (ages 1–5+) are presented, which suggest that while broad trends are captured in a consistent way by the two surveys, finer-scale details of year-class strength may not be.

Catch-curve analyses for the surveys are shown in Figure 22.12. These show consistent tracking of year classes (since catch curves are mostly smooth) and consistent selection with some exceptions in recent years. The catchability of the IBTS–Q1 seems to have changed since 2007, underestimating the size of the 2006-year class at age 1. The 2007 to 2012, 2017 and 2023 year classes also seem to have been underestimated at age 1. The IBTS–Q3 survey shows low mortality for the 2006-year class, and a potential underestimate of the 2011, 2017- and 2018-year class at age 0. However, numbers at age 2 in the 2007- and 2013 year class may well be an overestimate.

In both quarters, indices at younger age have increased in recent years after low values in the late 2000s and early 2010s.

The consistency within surveys is assessed using correlation plots in Figures 22.13 and 22.14. These indicate that the IBTS–Q1 and Q3 surveys both show good internal consistency across ages. The log CPUE plots by survey (Figure 22.15) support the conclusion of good internal consistency. Only in recent years, age 1 differs somewhat from overall pattern.

Figures 22.16–22.18 summarize the results of a SURBAR analysis using the available IBTS surveys. These show a well-specified analysis in which the data agree broadly with the separability assumptions in the model and uncertainty bounds are fairly tight. Mortality has been on a relatively lower level since the early 2000s. Recruitment (age 1) in 2020, 2021 and 2025 is estimated to have been much higher than in previous years, while SSB and TSB have increased to historically highest values. The log survey residuals (Figure 22.17) suggest in most recent years some negative residuals in Q1 and positive residuals in Q3 that suggest a trade-off between surveys in the SURBAR model.

22.7.2 Exploratory catch-at-age-based analyses

Catch curves for the catch data are plotted in Figure 22.19 and show numbers-at-age on the log scale linked by cohort. This shows partial recruitment to the fishery up to age 2 for some cohorts, particularly in the more recent time period.

The negative gradients of log catches per cohort, averaged over ages 2–4 are given in Figure 22.20. The gradients appear to have been decreasing since 1990 and are fluctuating around a mean level for more recent cohorts that is lower than the mean level prior to 1990, suggesting a fishing mortality likely to be lower than in the past for the cohorts since 2000. For the 2020 cohort the negative gradient of commercial catch data was lowest in the series. Slopes for the catch curves were less steep for this cohort, indicating relatively higher CPUE at higher ages.

Within cohort correlations between ages are presented in Figure 22.21. In general, catch numbers correlate well between cohorts with the relationship breaking down as cohorts are compared across increasing age gaps. Correlation were non-significant comparing age groups 0, 1 and 3 to age group 6+. This is due to the increased catches of older fish over the years and decreasing trends for younger age groups, as well as variability in the composition of the plusgroup over time (Figure 22.19).

22.7.3 Conclusions drawn from exploratory analyses

Catch curve analysis and correlation plots show that in general both surveys and catch data track cohorts well and are internally consistent (Figures 22.12–14, 22.19–21). For the 2006–2012 year classes and in recent years, the IBTS Q1 appears to be underestimating the abundance of age 1 whiting in some years (Figure 22.12). In previous assessments, this had implications for the estimation of recruitment and can result in a considerable retrospective bias in recruitment.

22.7.4 Final assessment

The final assessment used SAM (stockassessment.org) fitted to the combined landings, discards and industrial bycatch data for the period and two survey tuning indices. The used time range for input data for SAM was agreed at WKNSEA and is detailed in the stock annex (ICES, 2018). During WGNSSK 2022, it was agreed to update the model with a plus group at age 6 and F averaged over ages 2 to 5 in order to address retrospective patterns (Annex 8, ICES, 2022). The 2025 assessment model, including input data, results and diagnostics can be found on www.stockassessment.org as “NSwhiting_2025”. For comparison, also a SAM run of the 2024 assessment using new catch data for 2021–2023 is also available (“NSwhiting_2024_new-catch21”).

The settings as given by the configuration file, updated during WGNSSK 2022, are provided below (further details can be found in the Stock Annex).

Catch-at-age data	1978–2024	ages 0–6+
Survey: IBTS Q1	1983–2025	ages 1–5
Survey: IBTS Q3	1991–2024	ages 0–5
 \$minAge		
# The minimim age class in the assessment		
0		
 \$maxAge		
# The maximum age class in the assessment		
6		
 \$maxAgePlusGroup		
# Is last age group considered a plus group (1 yes, or 0 no).		
1		


```

$keyLogFsta
# Coupling of the fishing mortality states (normally only first row is used).
  0  1  2  3  4  5  5
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$corFlag
# Correlation of fishing mortality across ages (0 independent, 1 compound symmetry, or 2 AR(1))
2

$keyLogFpar
# Coupling of the survey catchability parameters (normally first row is not used, as that is covered
by fishing mortality).
-1 -1 -1 -1 -1 -1 -1
-1  0  1  2  3  3 -1
  4  5  6  7  8  8 -1

$keyQpow
# Density dependent catchability power parameters (if any).
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyVarF
# Coupling of process variance parameters for log(F)-process (normally only first row is used)
  0  0  0  0  0  0  0
-1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1

$keyVarLogN
# Coupling of process variance parameters for log(N)-process
0 1 1 1 1 1 1

$keyVarObs
# Coupling of the variance parameters for the observations.
  0  1  1  1  1  1  4
-1  2  2  2  2  2 -1
  3  3  3  3  3  3 -1

$obsCorStruct
# Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US" for unstructured). |
Possible values are: "ID" "AR" "US"
"ID" "AR" "AR"

$keyCorObs
# Coupling of correlation parameters can only be specified if the AR(1) structure is chosen above.
# NA's indicate where correlation parameters can be specified (-1 where they cannot).
#0-1 1-2 2-3 3-4 4-5 5-6 6-7 7-8
NA NA NA NA NA NA
-1  0  0  0  0 -1
  1  1  2  2  2 -1

$stockRecruitmentModelCode
# Stock recruitment code (0 for plain random walk, 1 for Ricker, and 2 for Beverton-Holt).
0

$noScaledYears
# Number of years where catch scaling is applied.
0

$keyScaledYears
# A vector of the years where catch scaling is applied.

$keyParScaledYA
# A matrix specifying the couplings of scale parameters (nrow = no scaled years, ncols = no ages).

$fbarRange
# lowest and highest age included in Fbar
2 5

$keyBiomassTreat
# To be defined only if a biomass survey is used (0 SSB index, 1 catch index, and 2 FSB index).
-1 -1 -1

$obsLikelihoodFlag
# Option for observational likelihood | Possible values are: "LN" "ALN"
"LN" "LN" "LN"

$fixVarToWeight
# If weight attribute is supplied for observations this option sets the treatment (0 relative weight,
1 fix variance to weight).
0

```

The results of the final assessment run are illustrated in Figure 22.22.

Fishing mortality estimates at age from final SAM run are presented in Table 22.17. Estimated stock numbers at age are given in Table 22.18. The assessment summaries are presented in Table 22.19 for recruitment, SSB, mean F, and TSB including upper and lower ranges. Catch biomass with lower and upper range as estimated in SAM are given in Table 22.20.

Final SAM run model parameters are given in Table 22.21.

Estimated correlations between ages are illustrated in Figure 22.23. The correlations reflect SAM settings of autocorrelations and parameter coupling, assuming independence in the catch fleet and correlation between ages in each survey fleet coupled for ages 2+.

The joint-sample residuals for the unobserved processes stock size N show no apparent cohort effects across ages. For the fishing mortality process, in recent years the negative residuals for $\log(F)$ are apparent with some tendency for a year effect (Figure 22.24). This might relate to relatively low fishing mortality estimated in the recent period.

Standardized one-observation-ahead residuals are presented for each fleet in Figure 22.25. These look reasonable with some negative residuals in the plusgroup in catch and IBTS Q1 in recent years. The IBTS-Q3 survey appear to fit slightly better to the model than the IBTS-Q1 survey, which demonstrate some year effects in the 2010s and towards the end of the time-series. This indicates that the model is effectively paying less attention to the Q1 survey than to the Q3 survey, and this is visible in Figures 22.27 and 22.28 which show the comparison of predicted and observed points for each survey fleet. The single fleet SAM runs (leave-one-out runs) were conducted to compare trends in estimated stock dynamics by using either survey data for quarter 1 or 3 separately. Summary plots of these runs together with the final run are presented in Figure 22.29. These leave-one-out runs show that leaving out IBTS Q3 has a stronger effect on the assessment results than leaving out IBTS Q1. With IBTS Q1 data alone, stock size is underestimated. This impact is comparable to the negative residuals observed for IBTS Q1 in the SURBAR results (Figure 22.17). The population trends from each survey are consistent. The mean F and SSB estimates are consistent across the time-series with only some differences in most recent year's estimates. Recruitment peaks are less pronounced when using only IBTS Q1 data in the model, since indices at age 0 are only available from IBTS Q3. The run using only quarter 3 matches more closely the final SAM run with both surveys included, in particular for recruitment.

A retrospective analysis is shown in Figure 22.30. The retrospective patterns show that results were robust to removing up to 5 years of recent data. Mohn's rho measures the retrospective bias, values are given in Table 22.21 and confirm the relatively low retrospective bias in recruitment, SSB and fishing mortality. The Mohn's Rho value for SSB is within the acceptable bounds set by WKFORBIAS (ICES, 2020a). Retrospective peels are generally covered by the confidence interval, apart from one peel for recruitment.

The spawning stock recruitment relationship shows no apparent pattern, confirming that the assumed random walk in recruitment in the model is appropriate (Figure 22.31). Standard deviation for $\log(\text{Recruitment})$ is relatively low for most of the time series (just above 0.1). Only in recent years the values increase somewhat, with 2024 value just below 0.3, as expected since recruitment estimates become more uncertain at the end of the time series (Figure 22.32). The 2024 recruitment value is still considered reasonable for inclusion in the forecast calculations.

The age composition of SSB is illustrated in Figure 22.33. As the stock size increased in recent years, a large proportion of the spawning stock biomass is found in age group 4+. In 2025, the plusgroup (6+) currently contains just below 20% of the SSB, an increase from 2024.

Finally, Figure 22.34 compares the SURBAR results with the final SAM assessment. Dynamics in SAM and SURBAR are similar with higher variability in the SSB estimates from SURBAR,

confirming the strong increase in SSB in recent years. The comparison of recruitment (at age 1) also shows similar dynamics. The mean Z (total mortality, ages 2–4) estimates from SURBAR show higher mortalities since 1990 than SAM and some increase in mortality in recent years, but the level and trends are similar. In recent years, estimated total mortality are similar across the two assessment methods.

22.8 Historical stock trends

Historical trends for catch, mean F , SSB and recruitment are presented in Figure 22.22. These show that mean F has been declining since 1990 and reached the minimum of time-series in 2023 of 0.089. The SSB was at extremely high levels before 1983 (no survey information included prior 1983). SSB estimates since 2020 show a dramatic increase with recent values on par with the early 1980s estimates, although the values suggest that this increase may be starting to level off. Recruitment has been fluctuating around a recent (post 2001) lower average but is showing an increase in recent years. Recruitment is estimated to be relatively high in 2019 and 2020, and recruitment in 2024 is estimated to be above average. In the recent years, total catches remained stable. The comparison of historical assessments, historical retros indicates that fishing mortality is well estimated with some underestimation of SSB as the stock size increased (Figure 22.35).

The update of catch data for 2021–2023 had only a minor impact on assessment results (Figure 22.46).

22.9 Biological reference points

The 2013 benchmark meeting (WKROUND; ICES, 2013) attempted to calculate F_{MSY} for North Sea whiting, but concluded that this value was inestimable using standard equilibrium considerations and would need to be determined as part of a management strategy evaluation. After the considerable revisions in the 2012 assessment, caused by new estimates of natural mortality, the target F of 0.3 was no longer considered applicable. The management plan was re-evaluated in October 2013 (ICES, 2013) and ICES advised that updating the target F from 0.3 to 0.15 within the management plan. New revisions of natural mortalities were presented at WGSAM 2014. An interbenchmark was performed for whiting in the North Sea and Division 7.d in early 2016 (ICES, 2016a). This included EqSim runs and an MSE. It was found that a target F of 0.15 together with a TAC constraint of 15% according to the EU–Norway Management Plan may not be sufficient to keep SSB above B_{lim} . It was concluded to use instead the MSY approach with target F of 0.15.

In the WKNSEA 2018 benchmark new data and assessment model were introduced, EqSim was run to determine new reference points (ICES, 2018). $F_{p.05}$ was calculated by running EqSim to ensure that the long-term risk of $SSB < B_{lim}$ of any F used does not exceed 5% when applying the advice rule. Accordingly, F_{MSY} had to be set to $F_{p.05} = 0.172$. Following a request by EU and Norway the management strategy was tested using a full MSE (ICES, 2019a, b). At the time results indicated that the MSY approach with $F_{MSY} = 0.172$ was not precautionary in the long-term.

At WGNSSK 2020, it was recommended to use new survey indices provided by DATRAS for the whiting assessment in 2020 and onwards (see Section 22.5.5). The reference points $B_{lim} = 119\,970$ and $F_{MSY} = 0.172$, set for North Sea whiting at the benchmark 2018 (ICES, 2018), were suggested to remain unchanged. Retrospectives and Mohn's ρ indicated that using the completely new survey indices leads to more consistent assessments with lower retro than using a survey series combining old (up to 2019) and new method (Q1 2020) (see Annex 9, ICES, 2020b).

In April 2021, an interbenchmark was carried out to include new natural mortality estimates from WGSAM (ICES, 2021b). EqSim was run to determine new reference points, and the reference points previously defined during the 2018 benchmark were updated as a result (ICES,

2021c). The new F_{MSY} value is 0.371 and the new B_{lim} value is 103 560. In April 2022, following the update of the SAM assessment model new reference points were estimated using EqSim. In WGNSSK 2024, natural mortality estimates were updated following a new NS SMS key run by ICES WGSAM (see section 27.27). Current reference points are listed in Table 22.23.

22.10 Short-term forecasts

A short-term forecast was carried out based on the final SAM assessment. SAM estimates of population numbers from the beginning of 2025 were used as input population numbers for ages 1 and older in 2025. Recruitment (age 0) assumptions are detailed in Table 22.24. For the intermediate year and the following two forecast years, the geometric mean of recruitment from 2003–2024 is used.

The exploitation pattern is chosen as the mean exploitation pattern over the most recent three years 2022–2024. The mean exploitation pattern was scaled to the mean F_{2-5} in 2024 for forecasts. Partial F at age for each catch component was estimated by splitting the forecast F at age using the mean proportion in the catch of each catch component over the years 2022–2024. The F at age used in the forecast is compared with the F at age estimates for 2022–2024 in Figure 22.36.

Mean weights at age are generally consistent over the recent period but there is variability at several ages (Figure 22.5 and 22.6). To avoid introducing bias, therefore, the average of estimates of 2022–2024 are used for the purposes of forecasting. The strong trend as observed between 2000 and 2015 is not apparent in the recent three years.

The inputs to the short-term forecast are given in Table 22.25, and results are presented in Table 22.26. As in previous years, the MFDP program was used to carry out the forecasts, accounting for separate fleet for industrial bycatch.

No TAC constraint was applied in the intermediate year since the TAC was not fully taken in recent years and the TAC is not considered restrictive for this stock.

Assuming mean F_{2025} equal to mean F_{2024} (using the average selectivity over the last 3 historical years) results in human consumption catch in the intermediate year 2025 of 32 418 tonnes from a total catch of 34 523 tonnes (Table 22.26).

Carrying the same fishing mortality forward into 2026 (the status quo F option, F_{sq}) would result in human consumption catch of 31 374 tonnes out of total catches of 33 503 tonnes and would result in an SSB of 413 121 tonnes in 2026 and 403 077 tonnes in 2027 (an SSB change of -2.4%).

Since SSB in 2026 is predicted to be higher than $MSY B_{trigger}$, following the MSY approach allows for applying F_{MSY} leading to an F_{target} of 0.68.

Applying the F_{MSY} of 0.68 in 2026 would generate human consumption catches of 197 033 tonnes out of total catches of 198 609 tonnes and result in an SSB of 286 198 tonnes in 2027 (a 31% decrease in SSB relative to 2026 of 413 121 tonnes). In 2027, SSB would be above B_{lim} and $MSY B_{trigger}$.

Change in advice

The change in advice for 2026 (+5.6% when compared to 2025) is caused by an increase in recruitment and SSB.

Recruitment in 2024 is estimated to be higher than previously assumed (Figure 22.40, 20.41). Both the stock weights-at-age (Fig. 22.42) and the selectivity (Fig. 22.44) used in last year's and this year's forecasts are similar. The assumptions used in both last year's and this year's forecasts are summed up in Table 22.33 and Figure 22.45. The difference in assumed target fishing mortality in 2025, is due to current fishing mortality being well below the MSY reference point.

22.11 MSY estimation and medium-term forecasts

No medium-term forecasts or MSY estimation were conducted during the WG meeting.

22.12 Quality of the assessment

Previous meetings of WGNSSK and the benchmark workshop (ICES, 2009; ICES, 2013) have concluded that the historical survey data and commercial catch data contain different signals concerning the stock. Analyses by Working Group members and by the ICES Study Group on Stock Identity and Management Units of Whiting (SGSIMUW; ICES, 2005) indicate that data since the early to mid-1990s are sufficiently consistent to undertake a catch-at-age analysis calibrated against survey data from 1990. WKNSEA (ICES, 2018) considered the question of time-series length again and concluded that the divergence between survey-based and catch-based analysis are not sufficient to exclude pre-1990 data. Survey data was included since 1983 with standardization of survey design.

Given the spatial structure of the whiting stock and of the fleets exploiting it, it is important to have data that covers all fleets. Considering that age 1 and age 2 whiting make up a large proportion of the total stock biomass, good information of the discarding practices of the major fleets is important.

The survey information for Division 7.d were not available in a form that could be used by WGNSSK. Due to the recent changes in distribution of the stock, tuning information from this area would be extremely useful, and could improve the estimate of recruitment in the most recent year. However, previous analyses of the survey in Division 7.d showed it did not track cohorts well (ICES, 2009).

Age distributions and mean weights at age have been estimated for the industrial bycatch from 2006 to 2010. This was due to low sampling levels of the Danish industrial bycatch fisheries. In recent years, no samples of industrial bycatch were available. Age distributions and weights at age were inferred from sampling of both landings and discards from other fleets.

In 2017, French samples for quarter 1 and 2 particularly in Subdivision 7.d are sparse due a disruption in the onshore sampling scheme. Therefore, a percentage of data was simulated randomly from previous year's data. This affected about 8% of total catch weight (landings more than discards, in particular TR2 fleet in 7.d). Decreased sampling intensity in quarter 1 and 2 of recent years (particularly in 2020, 2024) have been addressed by updating the catch raising scheme (using half year rather than quarterly discard raising).

In the past, there have been issues with regard to the age readings of North Sea whiting as compared to other gadoids (Norway as compared to Netherlands and UK (Scotland)). This applies in particular to the age readings used for the IBTS indices. An otolith workshop, WKARWHG2, took place in late 2016, to improve consistency in preparation techniques and readings (ICES, 2016b). This exercise showed an improvement in age reading compared to the same read in the 2015 exchange. A recommendation was made to investigate the quality of age readings further. The historical performance of the assessment is summarized in Figure 22.35. The difference in stock numbers and fishing mortality is due to updates of the SAM assessment model and natural mortality estimates in 2021, 2022 and 2024.

There is uncertainty in recent maturity estimates likely overestimating maturity for young ages which affects the reference points. The IBTS working group and WGBIOP are aware and are

currently investigating the use of the new maturity scales across countries. Maturity data as input to the assessment model will be reconsidered at the next benchmark.

22.13 Status of the stock

For North Sea whiting, SSB has an upwards trend in recent years trend. SSB is estimated to be above $MSY B_{trigger}$, B_{pa} and B_{lim} (Figures 22.22, 22.35). The stock, at the level of the entire North Sea and eastern English Channel, was at an historical low level in the late 2000s, and the recent increase in SSB is in large part due to relatively improved perception of recruitment in 2014–2016, 2019–2020 and 2024. All indications are that fishing mortality has been declining over most of the time-series, currently fluctuating around a low level. Since 1988, fishing mortality has been below $F_{MSY} = 0.68$. While landings and industrial bycatch have been relatively stable over time with a slight decrease in recent years, discards have been a bit more variable and increased more recently. The development of whiting biomass depends on the size of recruitment. Recruitment in 2019, 2020 and 2024 are estimated to be high and on par with levels observed prior to 2002. Stock biomass estimated for 2025 increased and is well above $MSY B_{trigger}$, close to the historic maximum.

22.14 Management considerations

In 1996, 2006, 2011, 2012 and 2017, the whiting stock produced the lowest recruitments in the series (below 8 billions). In recent years an increased proportion of whiting mature already at age 1 and grow quickly at young ages; therefore, an increase in SSB is seen soon after a good recruitment year. Managers should consider the age structure of the population as well as the SSB, since at low stock sizes short term forecasts are highly sensitive to recruitment assumptions. Due to the high stock size and relatively high number of older fish (around 50% of SSB at age 4+, Figure 22.33), there is currently little concern for this stock.

Catches of whiting, including discards and industrial bycatch, have been declining since the 1980s from more than 200 000 tonnes to around 30 000 tonnes in recent years (Figure 22.47).

Catch rates from localized fleets may not represent trends in the overall North Sea and eastern English Channel. The localized distribution of the stock is known to be resulting in substantial differences in the quota uptake rate. This is likely to result in localized discarding problems that should be monitored carefully.

Whiting are caught in mixed demersal roundfish fisheries, fisheries targeting flatfish, the *Nephrops* fisheries, and the industrial fishery. The current minimum mesh-size in the targeted demersal roundfish fishery in the northern North Sea has resulted in reduced discards from that sector compared with the historical discard rates. Mortality may have increased on younger ages due to increased discarding in recent years as a result of recent changes in fleet dynamics of *Nephrops* fleets and small mesh fisheries in the southern North Sea. The industrial bycatch of whiting in the sprat, Norway pout and sandeel fisheries is dependent on activity in that fishery, which has recently declined after strong reductions in the fisheries. Industrial bycatches are considered relatively low in the forecast.

Catches of whiting in the North Sea are also likely to be affected by the effort reduction seen in the targeted demersal roundfish and flatfish fisheries, although this will in part be offset by increases in the number of vessels switching to small mesh fisheries. It is important to consider both the species-specific assessments of these species for effective management, but also the broader mixed-fisheries context. This is not straight forward when stocks are managed via a series of single-species management plans that do not incorporate such mixed stocks

considerations. WGMIXFISH monitors the consistency of the various single species management plans and TAC advice under current effort schemes, in order to estimate the potential risks of quota over and under shooting for the different stocks, and it was demonstrated that the current basis for whiting advice was not consistent with other single-stock management objectives. It is recommended that the ongoing discussions about the whiting management plan takes into account such mixed-fisheries considerations before implementation.

The stock dynamics of North Sea whiting are largely driven by recruitment and natural mortality. The most significant measure would be to improve selectivity and reduce under-sized catches in those fisheries with high rates of discarding. Currently, other choke species in the mixed fishery have helped to keep fishing mortality low on this whiting stock. Low fishing mortality together with high recruitment has led to stock biomass increase in recent years.

BMS landings reported to ICES in 2015–2024 were low. In 2024, whiting was fully under Landings Obligation with a *de minimis* exemption for whiting caught with bottom trawls in ICES Division 4.c. Nevertheless, reported BMS was very low, and discarding was still observed in the sampled fleets and are assumed to take place also in unsampled fleets. The amount of reported BMS is expected to increase in the next years as the landing obligation continues to be implemented.

ICES has developed a generic approach to evaluate whether new survey information that becomes available in autumn forms a basis to update the advice. A reopening in autumn will not be necessary for this stock since stock biomass is well above MSY $B_{trigger}$.

22.15 SURBAR Northern and Southern stock component

Exploratory SURBAR assessments were run for individual components (northern and southern component, Figure 22.37) using component area-specific DATRAS survey indices provided by ICES (Tables 22.27–28) and estimated area-specific maturity ogives (Tables 22.29–30, Figure 22.39 a, b). Stock weights-at-age were assumed to be the same in northern, southern components and the combined area. The stock dynamics for the combined stock were more similar to the northern component and more variable in the southern one. Nevertheless, stock dynamics in northern and southern were comparable in recruitment, TSB and SSB; while mean total mortality showed stronger fluctuations for the southern component (Figure 22.38). The SURBAR analyses indicate that the southern stock component (as well as the combined stock) is at a historically high level of SSB and unlikely to be negatively affected by management decisions based on the combined analyses dominated by the northern component. The northern component has seen high SSB in the 2020s, comparable to historic high levels in the mid 1990ies. In recent years, recruitment has been relatively low for the northern component, and while still on high level also SSB and TSB showed a declining trend in recent years (Figure 22.38).

22.16 Issues for future benchmarks

The stock was benchmarked in 2018, implementing a new assessment model, natural mortality estimates, maturity ogive and stock weights at age estimation. The stock identity issue was revisited and decided to continue with the assessment area previously used (North Sea and eastern English Channel). The discard raising and age allocations method in InterCatch was revised to account for fleet differences (TR1/TR2, seasonal) in discard rate and age distributions. An inter-benchmark was performed in 2021 to include new mortality estimates from WGSAM (ICES, 2021b), and reference points were updated accordingly (ICES, 2021c). Following updates to the SAM assessment model during WGNSSK 2022, reference points were updated (see Annex 8,

ICES, 2022). At WGNSSK 2024 new mortality estimates from WGSAM were included in the model (ICES, 2024a), and reference points were updated accordingly (ICES, 2024b).

The next benchmark is planned for 2025/2026.

22.16.1 Data and assessment

Stock weights-at-age are estimated each year by scaling the catch-at-weight time-series by using the NS-IBTS quarter 1 weights-at-age (shorter time-series). Even though the entire time-series of stock weights-at-age is re-estimated each year, so far historical values did not change. If estimated stock weights-at-age in the historical time period differ significantly from one year to the next, the estimation should be reconsidered, i.e. only add newly estimated most recent data point (not an issue this year). At the next benchmark, scaling factors by age could be set constant and be applied in the following years, as values have remained stable over the last years.

Natural mortality at age: When new natural mortality estimates (WGSAM) become available these data need to be included and potentially reference points may need to be revised (updated this year).

Maturity at age: In recent years the number of mature fish at age 1 has been rather high. There appears to be some uncertainty that is currently being investigated by IBTSWG and WGBIOP. Maturity estimation will be reconsidered at the next benchmark.

Stock identity: In the last benchmark, stock identity was considered for North Sea whiting distinguishing a northern and a southern stock component. Analysis (see Section 22.1.1) suggest similar dynamics in the northern and southern component with dynamics being dominated by the northern component. At this point in time, a separate assessment is not considered necessary from reviewed literature and SURBAR analyses. Consideration of whiting in 3.a has been given at the last benchmark, with the decision not to include this area in the assessment. Additional evidence and data should be revisited at the next benchmark to confirm the decision.

Survey indices: There has been a new French data upload for the historical time-series. The use of a delta GAM method to calculate indices with up-to-date data should be explored. This would allow incorporation of survey uncertainty into the assessment. The availability and impact of alternative indices (HERAS, DRS) can be explored.

SAM assessment: the use of unsmoothed maturity and natural mortality estimates as input for the assessment model, in order to use the new SAM method to estimate missing historical values, should be explored. Alternative plusgroup (previously 8+) and alternative settings in SAM should be explored.

22.16.2 Reference points and Forecast

Currently a deterministic forecast continues to be done in MFDP. A SAM forecast can be considered which allows fleet separation (human consumption and industrial bycatch fleet) and stochasticity.

22.17 Update of catch data 2021-2023

New estimates of catch at age, catch weights at age (also impacting stock weights at age) for North Sea whiting were obtained for the years 2021-2023 from InterCatch following data resubmission from UK (England) (both landings and discards) and the Netherlands (discards). This led to a minor increase in catches of 3.5% in 2021, 3.1% in 2022 and 6% in 2023.

Results from different SAM assessment models were compared in order to assess the impact of the new catch data (Figure 22.46).

- WGNSSK 2024 SAM assessment
- WGNSSK 2024 SAM assessment with the updated catch data and stock weights-at-age
- WGNSSK 2025 SAM assessment (which includes updated and new data)

The assessments with new and old catch data are similar for the 2024 assessment runs; similar catches, recruitment, fishing mortality and with only a minor impact on SSB. The 2024 runs are compared to the current 2025 assessment, which all show good agreement.

22.18 References

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22.19 Tables and Figures

Table 22.1. Whiting in Subarea 4 and Division 7.d. Whiting in Subarea 4. Nominal landings (in tonnes) as officially reported to ICES, ICES estimates of catch components, and TACs. *Before 2015, the official landings from Denmark are likely to exclude Industrial bycatch. n/a=not available

Year	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Belgium.4	1040	913	1030	944	1042	880	843	391	268	529	536
Denmark.4	1206	1528	1377	1418	549	368	189	103	46	58	105
Faroe.4	26	0	16	7	2	21	0	6	1	1	0
France.4	4951	5188	5115	5502	4735	5963	4704	3526	1908	n/a	2527
Germany.4	692	865	511	441	239	124	187	196	103	176	424
Lithuania.4	0	0	0	0	0	0	0	0	0	0	0
Nether-lands.4	3273	4028	5390	4799	3864	3640	3388	2539	1941	1795	1884
Norway.4	55	103	232	130	79	115	66	75	65	68	33
Sweden.4	16	48	22	18	10	1	1	1	0	9	4
UK.4											
Eng-land.Wales.4	2338	2676	2528	2774	2722	2477	2329	2638	2909	2268	1782
Scotland.4	27486	31257	30821	31268	28974	27811	23409	22098	16696	17206	17158
Total.land-ings.4	41083	46606	47042	47301	42216	41400	35116	31573	23937	n/a	24453

Year	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Unallo- cated.land- ings.4	-1097	396	1832	691	346	850	-434	633	247	n/a	173
ICES.land- ings.4	42180	46210	45210	46610	41870	40550	35550	30940	23690	25700	24280
ICES.dis- cards.4	52270	30840	28470	41400	31840	28940	27130	16660	12480	22110	21931
ICES.ibc.4	51337	39755	25045	20723	17473	27379	5116	6213	3494	5038	9160
ICES.catch.4	145787	116805	98725	108733	91183	96869	67796	53813	39664	52848	55371
TAC.4.2a											30000

Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Belgium.4	454	270	248	144	105	93	45	116	162	147	74	45
Denmark.4	105	96	89	62	57	251	78	42	79	158	135	131
Faroe.4	0	17	5	0	0	0	0	0	2	0	0	0
France.4	3455	3314	2675	1721	1261	2711	3336	3076	2305	2644	2794	1925
Germany.4	402	354	334	296	149	252	76	76	124	156	111	25
Lithuania.4	0	0	0	0	0	0	0	0	0	0	0	0
Nether- lands.4	2478	2425	1442	977	805	702	618	656	718	614	514	471
Norway.4	44	47	38	23	16	17	11	92	73	118	28	94
Sweden.4	6	7	10	2	0	2	1	2	4	8	6	4
UK.4						11632	12110	10391	8853	7845	8892	9893
England. wales.4	1301	1322	680	1209	2560							
Scotland.4	10589	7756	5734	5057	3441							
Total. land- ings.4	18834	15608	11255	9491	8394	15660	16275	14451	12320	11690	12554	12588
Unallo- cated. land- ings.4	-426	738	805	541	-2286	563	609	972	-124	-1111	-706	-356
ICES. landings.4	19260	14870	10450	8950	10680	15097	15666	13479	12444	12801	13260	12944
ICES. discards.4	16130	17144	26135	18142	10300	14018	5206	8356	6597	8451	7989	9307

Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
ICES.ibc.4	940	7270	2730	1210	890	2190	1240	0	1344	1907	1035	1117
ICES.catch.4	36330	39284	39315	28302	21870	31305	22112	21835	20385	23159	22283	23368
TAC.4.2.a	29700	41000	16000	16000	28500	23800	23800	17850	15173	12897	14832	17056

Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Belgium.4	33	46	70	65	71	71	142	211	121	117	163	153
Denmark.4	124	160	2375	4727	2804	2026	2113	3520	2493	1311	2957	2073
Faroe.4	0	0	0	8	1	0	80	25	0	0	0	0
France.4	942	1884	1131	1232	952	918	890	677	478	447	253	394
Germany.4	44	31	73	111	82	99	81	277	201	103	228	259
Lithuania.4	0	0	0	0	0	0	0	20	0	0	0	0
Nether-lands.4	495	464	581	644	791	684	911	799	726	734	951	1404
Norway.4	560	918	1088	1150	993	1025	1102	1674	1098	909	1014	315
Sweden.4	1	2	0	6	11	8	18	28	67	40	214	132
UK.4	11162	10290	10015	9412	9263	10689	11846	12191	11591	10373	11474	9635
Eng-land.wales.4												
Scotland.4												
Total.land-ings.4	13361	13795	15333	17355	14968	15520	17183	19422	16775	14037	17254	14365
Unallo-cated.land-ings.4	-456	-52	2101	5113	3140	2942	1649	3444	2691	926	2810	1866
ICES.land-ings.4	13817	13847	13232	12242	11828	12578	15534	15978	14084	13111	14444	12499
ICES.dis-cards.4	4608	7016	12265	10413	9799	8026	7581	9749	10723	9687	9206	16181
ICES.ibc.4	1654	1623	2097	4551	2635	1658	1864	3115	2080	831	2601	1733
ICES.catch.4	20079	22486	27593	27206	24262	22263	24979	28842	26887	23629	26251	30413
TAC.4.2.a	18932	16092	13678	13678	16003	22057	17191	17158	21306	26636	34294	76697

Table 22.2. Whiting in Subarea 4 and Division 7.d. Whiting in Division 7.d. Nominal landings (in tonnes) as officially reported to ICES, ICES estimates of catch components, and TACs. n/a= not available

Year	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Belgium.7.d	83	83	66	74	61	68	84	98	53	48	65	75
France.7.d	n/a	n/a	5414	5032	6734	5202	4771	4532	4495	n/a	5875	6338
Nether-lands.7.d	0	0	0	0	0	0	1	1	32	6	14	67
Germany.7.d	0	0	0	0	0	0	0	0	0	0	0	0
Ireland.7.d	0	0	0	0	0	0	0	0	0	0	0	0
UK.7.d												
Eng-land.wales.7.d	239	292	419	321	293	280	199	147	185	135	118	134
Scotland.7.d	0	0	24	2	0	1	1	1	0	0	0	0
Total.land-ings.7.d	n/a	n/a	5923	5429	7088	5551	5056	4779	4765	n/a	6072	6614
Unallo-cat.land-ings.7.d	n/a	n/a	183	219	468	161	106	159	165	n/a	1772	814
ICES.land-ings.7.d	3480	5720	5740	5210	6620	5390	4950	4620	4600	4430	4300	5800
ICES.dis-cards.7.d	3330	4220	4090	2970	3850	3240	3370	3000	3210	3570	4129	3109
ICES.catch.7.d	6810	9940	9830	8180	10470	8630	8320	7620	7810	8000	8429	8909
TAC.7b.k											22000	21000

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Belgium.7.d	58	67	46	45	73	75	68	71	88	78	66	95
France.7.d	5172	6654	5006	4638	3487	3135	2875	6248	5512	4833	3093	3076
Nether-lands.7.d	19	175	132	128	117	118	162	112	275	282	437	650
Germany.7.d	0	0	0	0	0	0	0	0	0	0	0	0
Ireland.7.d	0	0	0	0	0	0	0	0	0	0	0	0
UK.7.d					72	63	87	138	258	271	261	472
Eng-land.wales.7.d	112	109	99	90								
Scotland.7.d	0	0	0	0								

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Total.land-ings.7.d	5361	7005	5283	4901	3749	3391	3192	6569	6133	5464	3857	4293
Unalloc.land-ings.7.d	-439	1295	933	111	306	137	-1279	649	-967	315	-556	-15
ICES.land-ings.7.d	5800	5710	4350	4790	3443	3254	4471	5920	7100	5149	4413	4308
ICES.dis-cards.7.d	1356	604	907	2219	2291	1763	1943	2086	4532	3183	2389	2186
ICES.catch.7.d	7156	6314	5257	7009	5734	5017	6414	8006	11632	8332	6802	6494
TAC.7b.k	31700	31700	27000	21600	19940	19940	19940	16949	14407	16568	19053	24500

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Belgium.7.d	90	121	146	128	138	144	45	56	47	29	22
France.7.d	2126	3102	2771	2378	2720	2106	1309	1724	1329	870	818
Nether-lands.7.d	663	565	556	593	484	611	330	229	260	243	415
Germany.7.d	0	0	0	0	0	0	1	3	7	3	4
Ireland.7.d	0	0	0	0	0	0	0	0	0	2	0
UK.7.d	345	379	259	358	283	259	302	417	323	134	140
Eng-land.wales.7.d											
Scotland.7.d											
Total.land-ings.7.d	3224	4167	3732	3457	3625	3120	1987	2429	1966	1281	1399
Unalloc.land-ings.7.d	99	190	32	103	143	145	76	215	-30	-40	39
ICES.land-ings.7.d	3125	3977	3700	3354	3482	2975	1911	2214	1996	1321	1360
ICES.dis-cards.7.d	2709	4627	2313	1550	2562	2499	4046	5207	3137	1741	4799
ICES.catch.7.d	5834	8604	6013	4904	6044	5474	5957	7421	5133	3061	6159
TAC.7b.k	20668	17742	22778	27500	22213	19184	10863	10259	8352	8450	23709

Table 22.3.a. Whiting in Subarea 4 and Division 7.d. Description of InterCatch raising procedure. SOP.

Catch Category	SOP
BMS landing	1.105
Discards	1.185
Landings (incl. IBC)	1.02
Logbook Registered Discard	NA

Table 22.3.b. Whiting in Subarea 4 and Division 7.d. Description of InterCatch raising procedure using Table 2 of CatchAndSampleData.Tables.txt. Summary of imported and raised data (uploads in weight).

Catch Category	Raised or Imported	CATON tonnes	Percent
BMS landing	Imported_Data	47.29	100
Discards	Raised_Discards	8776	50
Discards	Imported_Data	8886	50
Landings	Imported_Data	15648	100
Logbook Registered Discard	Imported_Data	0	NA

Table 22.3.c. Whiting in Subarea 4 and Division 7.d. Description of InterCatch raising procedure using Table 2 of CatchAndSampleData.Tables.txt. Summary of the imported or raised, and sampled or estimated data (uploads in weight).

Catch Category	Raised or Imported	Sampled or estimated distribution	CATON tonnes	Percent
Logbook Registered Discard	Imported_Data	Estimated_Distribution	0	NA
Landings	Imported_Data	Sampled_Distribution	10136	65
Landings	Imported_Data	Estimated_Distribution	5512	35
Discards	Raised_Discards	Estimated_Distribution	8776	50
Discards	Imported_Data	Estimated_Distribution	6025	34
Discards	Imported_Data	Sampled_Distribution	2861	16
BMS landing	Imported_Data	Estimated_Distribution	47.29	100

Table 22.3d. Whiting in Subarea 4 and Division 7.d. Description of InterCatch raising procedure using Table 2 of CatchAndSampleData.Tables.txt. Summary of the imported or raised, and sampled or estimated data by area (uploads in weight).

Catch Category	Raised or Imported	Sampled or Estimated	Area	CATON	percent
Logbook Registered Discard	Imported_Data	Estimated_Distribution	27.7.d	0	NA

Catch Category	Raised or Imported	Sampled or Estimated	Area	CATON	percent
Landings	Imported_Data	Estimated_Distribution	27.7.d	774.1	55
Landings	Imported_Data	Sampled_Distribution	27.7.d	634.9	45
Discards	Raised_Discards	Estimated_Distribution	27.7.d	3676	93
Discards	Imported_Data	Estimated_Distribution	27.7.d	159.2	4
Discards	Imported_Data	Sampled_Distribution	27.7.d	118.8	3
BMS landing	Imported_Data	Estimated_Distribution	27.7.d	0	NA
Logbook Registered Discard	Imported_Data	Estimated_Distribution	27.4.c	0	NA
Landings	Imported_Data	Estimated_Distribution	27.4.c	1139	100
Discards	Raised_Discards	Estimated_Distribution	27.4.c	1897	100
BMS landing	Imported_Data	Estimated_Distribution	27.4.c	0	NA
Logbook Registered Discard	Imported_Data	Estimated_Distribution	27.4.b	0	NA
Landings	Imported_Data	Estimated_Distribution	27.4.b	423.2	100
Discards	Raised_Discards	Estimated_Distribution	27.4.b	691.1	100
BMS landing	Imported_Data	Estimated_Distribution	27.4.b	1.003	100
Logbook Registered Discard	Imported_Data	Estimated_Distribution	27.4.a	0	NA
Landings	Imported_Data	Estimated_Distribution	27.4.a	362.7	100
Discards	Raised_Discards	Estimated_Distribution	27.4.a	232.1	100
BMS landing	Imported_Data	Estimated_Distribution	27.4.a	0.0056	100
Logbook Registered Discard	Imported_Data	Estimated_Distribution	27.4	0	NA
Landings	Imported_Data	Sampled_Distribution	27.4	9501	77
Landings	Imported_Data	Estimated_Distribution	27.4	2813	23
Discards	Imported_Data	Estimated_Distribution	27.4	5866	54
Discards	Imported_Data	Sampled_Distribution	27.4	2742	25
Discards	Raised_Discards	Estimated_Distribution	27.4	2279	21
BMS landing	Imported_Data	Estimated_Distribution	27.4	46.28	100

Table 22.4. Whiting in Subarea 4 and Division 7.d. Total catch numbers-at-age (thousands). Age 6 is a plus-group. Estimated by ICES, input data for SAM. Ages 0–6+ are included in the final assessment. Model input.

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1978	687238	418909	313391	242369	90047	7564	7564	1851	253	11	9	4	0	0	0	0	9692
1979	476383	615525	467538	218283	100976	29267	3111	1657	264	35	1	4	0	0	0	0	5072
1980	332209	265359	416009	286077	90719	52969	10752	1153	689	58	14	5	1	0	0	0	12672
1981	516869	162899	346343	266518	102295	27776	12297	3540	244	45	37	1	0	0	0	0	16164
1982	101057	192641	114443	245247	88137	26796	6909	2082	400	53	26	4	1	0	0	0	9475
1983	668604	205647	184747	118411	131507	37231	8688	1780	793	101	35	0	0	0	0	0	11397
1984	157819	323408	175965	124886	49504	59817	13860	2964	410	182	21	0	0	0	0	0	17437
1985	186723	203321	141716	82037	37847	14420	17446	3329	805	89	9	1	0	0	0	0	21679
1986	225202	576732	167078	169578	46516	13368	3487	3975	497	71	0	1	0	0	0	0	8031
1987	84863	267051	368230	122748	85240	11391	4555	928	930	98	7	0	0	0	0	0	6518
1988	416924	430344	307429	179503	39635	17902	2174	544	59	72	37	0	0	0	0	0	2886
1989	87325	331672	173676	191942	78464	14367	5051	517	291	37	6	1	0	0	0	0	5903
1990	289174	258102	501373	127967	84147	31102	1933	719	93	16	0	0	0	0	0	0	2761
1991	1057999	135797	194921	184960	36290	25554	5339	526	249	17	1	0	0	0	0	0	6132
1992	259390	230302	167479	87820	91081	11654	6634	2546	104	7	1	0	0	0	0	0	9292
1993	628301	223424	172049	125599	46181	45300	3898	1501	682	56	15	0	0	0	0	0	6152
1994	218287	191544	158369	97559	51041	18683	17905	1258	441	73	0	0	0	0	0	0	19677

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1995	1597900	148169	144023	112416	35649	15061	5117	4472	314	101	54	0	0	0	0	0	10058
1996	96515	86318	118910	99644	48304	14087	4638	1282	897	166	24	6	2	0	0	0	7015
1997	19001	60946	80471	84336	41975	18303	3333	1012	305	135	16	0	0	0	0	0	4801
1998	72289	92556	50362	43424	36295	17628	6343	1417	306	66	34	0	0	0	0	0	8166
1999	76975	189162	95415	45920	33921	18271	7443	2021	565	95	12	0	0	0	0	0	10136
2000	1970	82546	129582	63706	23913	16199	8758	4309	969	244	47	3	0	0	0	0	14330
2001	18012	52567	83085	52076	20800	9256	4826	2233	896	246	124	2	0	0	0	0	8327
2002	135848	51338	62462	84600	34659	8099	2048	1461	621	102	13	9	9	0	0	0	4263
2003	60744	83680	111144	55866	41841	14217	2359	473	329	50	16	1	0	0	0	0	3228
2004	34210	47966	23009	32557	30400	21755	8342	1352	198	93	12	1	4	0	0	0	10002
2005	17622	47805	34626	12204	18146	14931	8979	3041	540	83	29	1	0	0	0	0	12673
2006	15673	73908	42199	21651	8642	15077	11822	4618	1300	142	14	0	0	0	0	0	17896
2007	2490	39041	34001	24900	9906	4008	7657	5268	2560	476	82	0	0	0	0	0	16043
2008	5631	62163	28301	22741	13571	4305	1847	3954	2134	631	143	43	0	0	0	0	8752
2009	12139	57412	31004	15181	12782	7432	3380	2153	2601	1801	1967	20	1	0	0	0	11923
2010	3930	33756	33320	25516	9932	7776	6263	2136	4347	1491	1053	30	1	0	3	0	15324
2011	3563	31377	42201	28903	12537	3813	3178	2090	877	472	1293	31	1	0	0	0	7942
2012	3548	53445	32509	18882	14862	6952	2773	1558	1213	624	482	15	37	0	0	0	6702

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2013	4341	20378	15548	25362	15593	10812	3343	1048	643	660	292	0	0	0	0	0	5986
2014	6225	29785	14623	17450	19683	11351	4710	2038	1018	641	431	0	0	0	0	0	8838
2015	7705	48349	53345	15714	10220	14163	5068	2086	1210	607	401	4	0	0	0	0	9376
2016	17208	27639	36165	36788	9129	7813	6046	2548	691	694	376	0	0	0	0	0	10355
2017	28724	27355	27315	24442	18432	4176	2421	2683	1349	1165	26	5	0	0	0	0	7649
2018	15656	17302	41274	26023	17040	6786	1437	1013	803	36	163	38	0	0	0	0	3490
2019	4515	29380	24143	39669	17364	7151	3087	1063	555	273	76	0	0	0	0	0	5054
2020	26813	41940	32019	29992	19892	6596	2286	640	166	35	1	1	0	0	0	0	3129
2021	39017	36443	57381	21416	12120	8437	2943	944	71	21	0	1	0	0	0	0	3980
2022	1212	7317	28772	44524	14021	5758	3092	971	366	3	0	0	0	0	0	0	4432
2023	6561	18429	10377	30184	32967	7191	2913	1607	451	71	2	1	0	0	0	0	5045
2024	103385	6800	15919	17182	46137	26309	2293	614	458	43	5	0	0	0	0	0	3413

Table 22.5. Whiting in Subarea 4 and Division 7.d. Landings numbers-at-age (thousands), as estimated by ICES. Age 6 is a plus-group. Data used to calculate the landing fraction in the model estimates of catches.

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1978	0	14793	99836	155424	76829	6693	7202	1837	253	11	9	4	0	0	0	0	9316
1979	8	8488	108548	144343	89093	26584	3011	1617	250	35	1	4	0	0	0	0	4918
1980	0	3656	62405	152570	68422	41430	9911	1135	689	58	14	5	1	0	0	0	11813
1981	6	4240	69211	104348	78253	23698	12036	3530	244	45	37	1	0	0	0	0	15893

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1982	0	10890	46703	124656	59393	21376	5664	2058	400	53	26	4	1	0	0	0	8206
1983	1	10568	68640	67312	101342	31266	8330	1730	784	101	35	0	0	0	0	0	10980
1984	0	14388	62693	99204	41277	51745	12735	2813	410	182	21	0	0	0	0	0	16161
1985	1	2288	51194	57049	32340	12974	16361	3238	805	89	9	1	0	0	0	0	20503
1986	29	12879	44500	111527	37287	11285	3379	3912	485	71	0	1	0	0	0	0	7848
1987	22	11074	72372	70504	73742	10808	4506	928	899	98	7	0	0	0	0	0	6438
1988	0	7462	61360	94163	29147	16556	2158	544	56	72	37	0	0	0	0	0	2867
1989	52	8636	28406	77009	44307	9249	3888	420	208	35	6	1	0	0	0	0	4558
1990	23	6910	52533	43850	48537	16845	1341	605	91	16	0	0	0	0	0	0	2053
1991	410	11565	42525	88974	25738	21261	4581	396	249	17	1	0	0	0	0	0	5244
1992	298	9565	44697	47843	59208	9784	6099	1453	99	7	1	0	0	0	0	0	7659
1993	720	5957	28935	63383	32819	33741	2932	1339	682	56	15	0	0	0	0	0	5024
1994	77	17124	31351	45492	36289	13920	14407	914	366	73	0	0	0	0	0	0	15760
1995	277	8829	28027	58046	27775	13652	4911	4359	308	101	54	0	0	0	0	0	9733
1996	1015	12517	26611	47125	35828	11861	4396	1103	897	166	24	6	2	0	0	0	6594
1997	608	6511	23436	47717	31503	15615	2931	1010	289	135	15	0	0	0	0	0	4380
1998	1202	17071	19828	24860	24473	14579	5395	1204	219	64	16	0	0	0	0	0	6898
1999	68	16661	26669	25504	23465	14483	6554	1854	514	61	12	0	0	0	0	0	8995

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2000	0	15384	31808	28283	14241	11775	6618	3758	862	244	47	3	0	0	0	0	11532
2001	150	12260	28476	27293	17491	8633	4503	2091	877	246	124	2	0	0	0	0	7843
2002	0	2610	10346	30890	22353	6712	1710	1330	511	99	10	9	9	0	0	0	3678
2003	20	403	11613	13990	18974	9513	1861	443	329	50	16	0	0	0	0	0	2699
2004	0	3973	2812	9629	13302	11846	4409	747	174	84	12	1	4	0	0	0	5431
2005	74	11009	10414	5669	10926	10283	5933	2343	321	78	29	1	0	0	0	0	8705
2006	11	11055	11023	8494	5362	12259	10161	4118	1080	105	6	0	0	0	0	0	15470
2007	140	10378	14740	16491	7666	3310	6681	4227	2179	383	77	0	0	0	0	0	13547
2008	0	13234	12334	14120	9106	3564	1519	2505	1481	568	143	43	0	0	0	0	6259
2009	79	3056	17397	11259	10762	6411	3072	1994	2408	1679	1846	19	1	0	0	0	11019
2010	2	1368	8848	15426	6939	6296	3922	1922	1331	1378	979	24	1	0	0	0	9557
2011	32	4524	17621	14180	10021	2811	2303	1741	820	441	1215	30	1	0	0	0	6551
2012	0	2540	10148	11200	11692	6127	2020	1331	902	557	401	14	35	0	0	0	5260
2013	0	1724	7008	15154	11656	9344	2774	937	556	405	232	0	0	0	0	0	4904
2014	1	3211	7422	9439	12082	8031	3221	1673	806	566	329	0	0	0	0	0	6595
2015	136	3022	15736	7802	6584	9232	3800	1617	887	523	358	4	0	0	0	0	7189
2016	0	1405	9098	16279	5922	4187	4104	1747	550	573	312	0	0	0	0	0	7286
2017	0	731	6509	10287	12841	2666	1711	1640	1092	962	23	5	0	0	0	0	5433

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2018	0	1264	12061	13819	11797	5389	1159	798	729	33	150	35	0	0	0	0	2904
2019	0	2387	6217	21428	13320	6133	2529	963	500	227	69	0	0	0	0	0	4288
2020	1188	4844	9771	13246	13016	4932	1868	444	129	26	1	1	0	0	0	0	2469
2021	36	3237	10514	12467	9090	6549	1636	745	27	11	0	1	0	0	0	0	2420
2022	5	1142	7796	16097	7807	4666	2514	472	320	3	0	0	0	0	0	0	3309
2023	11	1226	3696	9339	18313	4902	2485	1073	358	24	2	1	0	0	0	0	3943
2024	49	811	4541	5341	16526	10525	1232	418	362	26	3	0	0	0	0	0	2041

Table 22.6. Whiting in Subarea 4 and Division 7.d. Discards numbers-at-age (thousands), as estimated by ICES. Age 6 is a plus-group. Data used to calculate the discard fraction from the model estimate of catches.

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1978	28587	52684	114965	37682	7154	255	110	0	0	0	0	0	0	0	0	0	110
1979	4577	473830	126724	31601	7322	1263	27	7	0	0	0	0	0	0	0	0	34
1980	3144	103203	250735	88399	14135	10795	786	0	0	0	0	0	0	0	0	0	786
1981	867	50407	96509	57403	7313	1285	149	10	0	0	0	0	0	0	0	0	159
1982	18639	53753	26922	52349	18230	2972	343	22	0	0	0	0	0	0	0	0	365
1983	71016	152488	85318	33325	23442	4309	295	25	9	0	0	0	0	0	0	0	329
1984	16724	200589	82563	16814	4437	4495	1034	151	0	0	0	0	0	0	0	0	1185
1985	8497	154232	48791	15117	2985	761	801	65	0	0	0	0	0	0	0	0	866
1986	7966	404604	120492	43479	5242	627	108	63	12	0	0	0	0	0	0	0	183

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1987	9978	158531	202154	34824	9776	582	49	0	31	0	0	0	0	0	0	0	80
1988	21321	65021	87197	51135	5877	846	16	0	3	0	0	0	0	0	0	0	19
1989	6898	150598	36712	61442	21267	3276	103	8	12	0	0	0	0	0	0	0	123
1990	147764	83152	241924	33084	23009	11665	246	85	0	0	0	0	0	0	0	0	331
1991	7208	81678	82053	75035	5176	1885	91	60	0	0	0	0	0	0	0	0	151
1992	7587	105838	63830	27659	23115	1231	355	1064	2	0	0	0	0	0	0	0	1421
1993	48873	128248	104844	51054	9205	10727	521	131	0	0	0	0	0	0	0	0	652
1994	8352	96890	102020	37751	9867	2885	2338	7	0	0	0	0	0	0	0	0	2345
1995	33363	53830	81783	50019	7136	1336	206	113	6	0	0	0	0	0	0	0	325
1996	4575	43126	86878	49817	11506	2205	240	179	0	0	0	0	0	0	0	0	419
1997	11525	26188	34948	32473	9398	2412	400	2	16	0	1	0	0	0	0	0	419
1998	6098	50703	24200	17053	11076	2987	936	213	87	2	18	0	0	0	0	0	1256
1999	14762	96413	56365	15228	9016	3104	862	167	51	34	0	0	0	0	0	0	1114
2000	1682	48162	81086	24082	3075	2311	1560	478	107	0	0	0	0	0	0	0	2145
2001	17352	39826	52156	23055	2795	471	283	142	19	0	0	0	0	0	0	0	444
2002	1158	10597	33371	45125	10136	1182	218	131	110	3	3	0	0	0	0	0	465
2003	3584	65829	94497	39301	21654	4314	449	30	0	0	0	1	0	0	0	0	480
2004	10478	31169	15698	21879	16951	9909	3922	605	24	9	0	0	0	0	0	0	4560

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2005	5499	25753	23486	6041	7192	4616	2992	688	211	5	0	0	0	0	0	0	3896
2006	15662	51961	25906	10935	2474	2595	1598	493	219	37	8	0	0	0	0	0	2355
2007	2350	22508	16283	7153	1784	572	940	1037	380	93	5	0	0	0	0	0	2455
2008	5631	48929	15967	8621	4465	741	328	1449	653	63	0	0	0	0	0	0	2493
2009	11540	51883	12179	3192	1382	653	139	52	64	32	24	0	0	0	0	0	311
2010	3701	30464	22610	8713	2444	1038	1988	99	2775	34	18	4	0	0	3	0	4921
2011	3430	25925	23211	13753	2053	862	760	272	24	13	29	0	0	0	0	0	1098
2012	3471	49677	21362	6943	2497	493	633	154	259	37	59	0	0	0	0	0	1142
2013	4149	17715	7711	8710	2899	693	343	40	44	217	43	0	0	0	0	0	687
2014	5943	25159	6425	7025	6438	2597	1193	239	155	38	79	0	0	0	0	0	1704
2015	7249	43271	34943	6950	2940	3947	888	313	238	39	13	0	0	0	0	0	1491
2016	14941	22682	22342	15500	1889	2536	1075	432	42	23	11	0	0	0	0	0	1583
2017	26493	24515	18650	11973	3735	1111	476	804	129	100	0	0	0	0	0	0	1509
2018	14985	15331	27274	10665	4071	914	172	145	13	1	0	0	0	0	0	0	331
2019	4130	25433	16810	15830	2913	453	342	18	21	34	0	0	0	0	0	0	415
2020	24349	34907	19827	13998	4592	838	115	119	19	5	0	0	0	0	0	0	258
2021	37729	31992	44507	7367	1973	1132	1090	113	40	9	0	0	0	0	0	0	1252
2022	1184	6036	20315	27233	5755	863	456	469	31	0	0	0	0	0	0	0	956

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2023	6166	16121	5992	18540	11458	1525	80	369	42	42	0	0	0	0	0	0	533
2024	100993	5777	10854	11124	27175	14287	904	146	55	14	2	0	0	0	0	0	1121

Table 22.7. Whiting in Subarea 4 and Division 7.d. Industrial bycatch numbers-at-age (thousands), as estimated by ICES. Data used to calculate the IBC fraction in the model estimates of catches.

[illegible]

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1992	251505	114899	58952	12318	8758	639	180	29	3	0	0	0	0	0	0	0	212
1993	578708	89219	38270	11162	4157	832	445	31	0	0	0	0	0	0	0	0	476
1994	209858	77530	24998	14316	4885	1878	1160	337	75	0	0	0	0	0	0	0	1572
1995	1564260	85510	34213	4351	738	73	0	0	0	0	0	0	0	0	0	0	0
1996	90925	30675	5421	2702	970	21	2	0	0	0	0	0	0	0	0	0	2
1997	6868	28247	22087	4146	1074	276	2	0	0	0	0	0	0	0	0	0	2
1998	64989	24782	6334	1511	746	62	12	0	0	0	0	0	0	0	0	0	12
1999	62145	76088	12381	5188	1440	684	27	0	0	0	0	0	0	0	0	0	27
2000	288	19000	16688	11341	6597	2113	580	73	0	0	0	0	0	0	0	0	653
2001	510	481	2453	1728	514	152	40	0	0	0	0	0	0	0	0	0	40
2002	134690	38131	18745	8585	2170	205	120	0	0	0	0	0	0	0	0	0	120
2003	57140	17448	5034	2575	1213	390	49	0	0	0	0	0	0	0	0	0	49
2004	23732	12824	4499	1049	147	0	11	0	0	0	0	0	0	0	0	0	11
2005	12049	11043	726	494	28	32	54	10	8	0	0	0	0	0	0	0	72
2006	0	10892	5270	2222	806	223	63	7	1	0	0	0	0	0	0	0	71
2007	0	6155	2978	1256	456	126	36	4	1	0	0	0	0	0	0	0	41
2008	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2009	520	2473	1428	730	638	368	169	107	129	90	97	1	0	0	0	0	593

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2010	227	1924	1862	1377	549	442	353	115	241	79	56	2	0	0	0	0	846
2011	101	928	1369	970	463	140	115	77	33	18	49	1	0	0	0	0	293
2012	77	1228	999	739	673	332	120	73	52	30	22	1	2	0	0	0	300
2013	192	939	829	1498	1038	775	226	71	43	38	17	0	0	0	0	0	395
2014	281	1415	776	986	1163	723	296	126	57	37	23	0	0	0	0	0	539
2015	320	2056	2666	962	696	984	380	156	85	45	30	0	0	0	0	0	696
2016	2267	3552	4725	5009	1318	1090	867	369	99	98	53	0	0	0	0	0	1486
2017	2231	2109	2156	2182	1856	399	234	239	128	103	3	0	0	0	0	0	707
2018	671	707	1939	1539	1172	483	106	70	61	2	13	3	0	0	0	0	255
2019	385	1560	1116	2411	1131	565	216	82	34	12	7	0	0	0	0	0	351
2020	1276	2189	2421	2748	2284	826	303	77	18	4	0	0	0	0	0	0	402
2021	1252	1214	2360	1582	1057	756	217	86	4	1	0	0	0	0	0	0	308
2022	23	139	661	1194	459	229	122	30	15	0	0	0	0	0	0	0	167
2023	384	1082	689	2305	3196	764	348	165	51	5	0	0	0	0	0	0	569
2024	2343	212	524	717	2436	1497	157	50	41	3	0	0	0	0	0	0	251

Table 22.8. Whiting in Subarea 4 and Division 7.d. Total catch mean weights-at-age (kg), as estimated by ICES. Age 6 is a plus-group. Ages 0–6+ are included in the final assessment. Model input.

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1978	0.01	0.074	0.182	0.234	0.321	0.428	0.428	0.466	0.615	0.702	1.539	0.589	0	0	0	0	0.442

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1979	0.009	0.098	0.167	0.259	0.301	0.411	0.455	0.492	0.578	0.617	0.737	0.515	0	0	0	0	0.475
1980	0.013	0.075	0.176	0.252	0.328	0.337	0.457	0.459	0.568	0.539	0.79	0.688	1.711	0	0	0	0.464
1981	0.011	0.083	0.168	0.242	0.322	0.379	0.411	0.444	0.651	0.833	1.041	0.695	0	0	0	0	0.424
1982	0.029	0.061	0.184	0.253	0.314	0.376	0.478	0.504	0.702	0.772	1.141	0.853	1.081	0	0	0	0.497
1983	0.015	0.107	0.191	0.273	0.325	0.384	0.426	0.452	0.52	0.677	0.516	0	0	0	0	0	0.439
1984	0.02	0.089	0.189	0.271	0.337	0.381	0.39	0.462	0.575	0.514	0.871	0	0	0	0	0	0.409
1985	0.014	0.094	0.192	0.284	0.332	0.401	0.435	0.494	0.426	0.507	0.852	0.976	0	0	0	0	0.444
1986	0.015	0.105	0.183	0.255	0.318	0.378	0.475	0.468	0.54	1.226	0.99	0.535	0	0	0	0	0.482
1987	0.013	0.077	0.148	0.247	0.297	0.375	0.38	0.542	0.555	0.857	0.603	1.193	0	0	0	0	0.435
1988	0.013	0.054	0.146	0.223	0.301	0.346	0.424	0.506	0.856	0.585	0.648	0	0	0	0	0	0.455
1989	0.023	0.07	0.157	0.225	0.267	0.318	0.391	0.431	0.37	0.515	0.857	0.609	0	0	0	0	0.395
1990	0.016	0.084	0.137	0.21	0.252	0.279	0.411	0.498	0.636	0.351	0.918	0	0	0	0	0	0.441
1991	0.018	0.104	0.168	0.217	0.289	0.306	0.339	0.365	0.385	0.589	0.996	2.756	0	0	0	0	0.344
1992	0.013	0.085	0.185	0.257	0.277	0.331	0.346	0.313	0.481	0.763	1.728	0	0	0	0	0	0.339
1993	0.012	0.073	0.174	0.25	0.316	0.328	0.346	0.4	0.376	0.417	0.359	0	0	0	0	0	0.363
1994	0.013	0.084	0.167	0.255	0.328	0.382	0.376	0.419	0.438	0.392	0.499	0	0	0	0	0	0.381
1995	0.010	0.089	0.18	0.257	0.340	0.384	0.429	0.434	0.445	0.346	0.406	0	0	0	0	0	0.431
1996	0.018	0.094	0.167	0.235	0.302	0.388	0.407	0.431	0.439	0.404	0.376	0.398	0.287	0	0	0	0.415

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1997	0.028	0.096	0.178	0.242	0.295	0.334	0.384	0.386	0.394	0.479	0.458	0	0	0	0	0	0.388
1998	0.018	0.090	0.179	0.236	0.281	0.314	0.340	0.333	0.335	0.494	0.434	0.600	0	0	0	0	0.340
1999	0.023	0.078	0.174	0.232	0.256	0.289	0.305	0.311	0.286	0.315	0.344	0	0	0	0	0	0.305
2000	0.034	0.117	0.182	0.238	0.287	0.286	0.276	0.275	0.268	0.264	0.280	0.321	0	0	0	0	0.275
2001	0.024	0.101	0.192	0.244	0.282	0.267	0.298	0.284	0.286	0.301	0.315	0.505	0	0	0	0	0.293
2002	0.01	0.069	0.155	0.218	0.273	0.303	0.350	0.343	0.327	0.411	0.289	0.231	0.304	0.643	0	0	0.345
2003	0.012	0.057	0.118	0.193	0.259	0.299	0.354	0.385	0.342	0.462	0.620	0	0	0	0	0	0.360
2004	0.031	0.111	0.150	0.213	0.253	0.286	0.285	0.286	0.346	0.351	0.352	1.463	0.337	0	0	0	0.288
2005	0.032	0.124	0.199	0.239	0.250	0.282	0.305	0.298	0.271	0.376	0.316	0.337	0.670	0	0	0	0.302
2006	0.093	0.131	0.180	0.231	0.274	0.288	0.360	0.345	0.318	0.299	0.289	0	0	0	0	0	0.352
2007	0.059	0.098	0.206	0.257	0.325	0.345	0.309	0.309	0.325	0.288	0.328	0	0	0	0	0	0.311
2008	0.027	0.104	0.218	0.282	0.315	0.402	0.407	0.317	0.359	0.337	0.334	0.433	0	0	0	0	0.348
2009	0.042	0.091	0.213	0.286	0.370	0.374	0.373	0.344	0.351	0.335	0.330	0.350	0.419	0	0	0	0.350
2010	0.049	0.111	0.234	0.373	0.406	0.456	0.355	0.459	0.272	0.475	0.471	0.399	0.259	0	0.368	0	0.366
2011	0.048	0.114	0.214	0.298	0.374	0.415	0.424	0.364	0.341	0.372	0.320	0.550	0.894	0	0	0	0.379
2012	0.038	0.105	0.195	0.311	0.445	0.411	0.430	0.428	0.366	0.418	0.406	0.552	0.733	0	0	0	0.417
2013	0.028	0.11	0.222	0.273	0.390	0.468	0.496	0.465	0.424	0.340	0.406	0	0	0	0	0	0.461
2014	0.055	0.137	0.227	0.294	0.331	0.442	0.465	0.469	0.403	0.403	0.359	1.754	0	0	0	0	0.449

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2015	0.044	0.125	0.218	0.307	0.368	0.386	0.469	0.464	0.374	0.372	0.400	0.778	0	0	0	0	0.447
2016	0.03	0.120	0.210	0.291	0.399	0.389	0.415	0.488	0.452	0.460	0.472	1.293	0	0	0	0	0.440
2017	0.026	0.078	0.212	0.320	0.409	0.436	0.487	0.444	0.457	0.419	0.528	0.489	0	0	0	0	0.457
2018	0.029	0.108	0.197	0.275	0.373	0.407	0.514	0.458	0.485	0.598	0.448	0.583	0	0	0	0	0.490
2019	0.021	0.106	0.204	0.279	0.354	0.420	0.436	0.440	0.368	0.355	0.577	0.736	0	0	0	0	0.427
2020	0.094	0.107	0.238	0.287	0.374	0.424	0.479	0.542	0.376	0.492	0.656	0.564	0	0	0	0	0.487
2021	0.039	0.118	0.191	0.322	0.401	0.479	0.431	0.529	0.459	0.509	1.065	0.774	0	0	0	0	0.455
2022	0.041	0.132	0.211	0.261	0.366	0.470	0.515	0.428	0.526	0.407	0.736	0.739	0	0	0	0	0.497
2023	0.024	0.107	0.216	0.270	0.335	0.434	0.537	0.524	0.554	0.358	0.687	0.650	0	0	0	0	0.532
2024	0.044	0.133	0.195	0.254	0.294	0.322	0.445	0.567	0.492	0.581	0.576	0	0	0	0	0	0.475

Table 22.9. Whiting in Subarea 4 and Division 7.d. Landings mean weights-at-age (kg), as estimated by ICES. Age 6 is a plus-group.

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1978	0	0.185	0.233	0.250	0.334	0.426	0.434	0.466	0.615	0.702	1.539	0.589	0	0	0	0	0.447
1979	0.113	0.206	0.231	0.277	0.304	0.416	0.456	0.491	0.583	0.617	0.737	0.515	0	0	0	0	0.475
1980	0	0.204	0.239	0.273	0.335	0.358	0.473	0.457	0.568	0.539	0.790	0.688	1.711	0	0	0	0.478
1981	0.144	0.194	0.242	0.292	0.331	0.378	0.411	0.445	0.651	0.833	1.041	0.695	0	0	0	0	0.425
1982	0	0.186	0.230	0.282	0.340	0.396	0.461	0.507	0.702	0.772	1.141	0.853	1.081	0	0	0	0.489
1983	0.132	0.199	0.240	0.282	0.332	0.383	0.429	0.452	0.522	0.677	0.516	0	0	0	0	0	0.442

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1984	0	0.194	0.231	0.279	0.346	0.391	0.403	0.472	0.575	0.514	0.871	0	0	0	0	0	0.421
1985	0.137	0.187	0.248	0.307	0.337	0.408	0.443	0.498	0.426	0.507	0.852	0.976	0	0	0	0	0.452
1986	0.131	0.189	0.230	0.279	0.327	0.376	0.484	0.472	0.546	1.226	0.990	0.535	0	0	0	0	0.489
1987	0.135	0.188	0.226	0.286	0.310	0.381	0.381	0.542	0.564	0.857	0.603	1.193	0	0	0	0	0.437
1988	0.117	0.194	0.226	0.256	0.328	0.351	0.425	0.506	0.887	0.585	0.648	0	0	0	0	0	0.456
1989	0.171	0.178	0.226	0.253	0.288	0.345	0.370	0.440	0.373	0.522	0.857	0.609	0	0	0	0	0.378
1990	0.167	0.206	0.222	0.263	0.296	0.337	0.455	0.533	0.640	0.351	0.918	0	0	0	0	0	0.485
1991	0.139	0.202	0.249	0.252	0.308	0.317	0.349	0.387	0.385	0.589	0.996	2.756	0	0	0	0	0.354
1992	0.145	0.194	0.246	0.289	0.306	0.340	0.356	0.383	0.473	0.763	1.728	0	0	0	0	0	0.363
1993	0.153	0.194	0.248	0.284	0.345	0.358	0.385	0.418	0.376	0.417	0.359	0	0	0	0	0	0.393
1994	0.132	0.182	0.248	0.297	0.346	0.392	0.382	0.412	0.414	0.392	0.499	0	0	0	0	0	0.385
1995	0.140	0.171	0.256	0.299	0.367	0.397	0.437	0.437	0.448	0.346	0.406	0	0	0	0	0	0.436
1996	0.143	0.169	0.222	0.274	0.329	0.408	0.415	0.452	0.439	0.404	0.376	0.398	0.287	0	0	0	0.424
1997	0.149	0.171	0.206	0.260	0.315	0.349	0.401	0.386	0.398	0.479	0.437	0	0	0	0	0	0.400
1998	0.138	0.164	0.208	0.259	0.304	0.331	0.361	0.348	0.392	0.504	0.603	0.600	0	0	0	0	0.362
1999	0.135	0.184	0.237	0.271	0.281	0.303	0.316	0.320	0.292	0.368	0.344	0	0	0	0	0	0.316
2000	0	0.166	0.227	0.272	0.299	0.292	0.313	0.276	0.269	0.264	0.280	0.321	0	0	0	0	0.296
2001	0.138	0.160	0.216	0.268	0.285	0.267	0.301	0.288	0.287	0.301	0.315	0.505	0	0	0	0	0.296

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2002	0	0.183	0.214	0.260	0.293	0.313	0.364	0.350	0.325	0.390	0.311	0.231	0.304	0.643	0	0	0.354
2003	0.128	0.208	0.228	0.258	0.308	0.311	0.374	0.391	0.342	0.462	0.620	0	0	0	0	0	0.376
2004	0	0.210	0.216	0.242	0.290	0.326	0.330	0.334	0.366	0.351	0.352	1.463	0.337	0	0	0	0.332
2005	0.164	0.205	0.253	0.277	0.270	0.308	0.339	0.313	0.296	0.381	0.316	0.337	0.670	0	0	0	0.331
2006	0.133	0.217	0.254	0.285	0.295	0.298	0.377	0.353	0.334	0.306	0.290	0	0	0	0	0	0.367
2007	0.202	0.199	0.264	0.28	0.351	0.361	0.319	0.332	0.342	0.318	0.334	0	0	0	0	0	0.327
2008	0	0.223	0.265	0.324	0.356	0.431	0.424	0.359	0.389	0.339	0.334	0.433	0	0	0	0	0.380
2009	0.114	0.184	0.239	0.299	0.375	0.376	0.373	0.346	0.349	0.336	0.327	0.350	0.419	0	0	0	0.349
2010	0.069	0.312	0.303	0.424	0.433	0.468	0.413	0.468	0.459	0.478	0.470	0.409	0.259	0	0.368	0	0.446
2011	0.046	0.194	0.263	0.363	0.397	0.455	0.459	0.367	0.342	0.374	0.322	0.550	0.894	0	0	0	0.389
2012	0.046	0.203	0.236	0.362	0.478	0.420	0.483	0.431	0.376	0.387	0.356	0.552	0.733	0	0	0	0.433
2013	0.038	0.203	0.247	0.295	0.417	0.477	0.515	0.460	0.419	0.413	0.391	0	0	0	0	0	0.479
2014	0.064	0.194	0.259	0.330	0.363	0.49	0.508	0.457	0.375	0.393	0.358	1.754	0	0	0	0	0.461
2015	0.103	0.197	0.253	0.355	0.401	0.428	0.495	0.466	0.406	0.380	0.400	0.778	0	0	0	0	0.465
2016	0.050	0.169	0.265	0.339	0.434	0.463	0.448	0.537	0.463	0.466	0.477	1.293	0	0	0	0	0.473
2017	0.035	0.146	0.249	0.394	0.434	0.493	0.552	0.498	0.465	0.432	0.528	0.489	0	0	0	0	0.497
2018	0.035	0.171	0.239	0.318	0.416	0.427	0.529	0.480	0.488	0.607	0.448	0.583	0	0	0	0	0.503
2019	0.033	0.194	0.269	0.324	0.375	0.429	0.458	0.438	0.373	0.351	0.577	0.736	0	0	0	0	0.440

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2020	0.132	0.214	0.330	0.358	0.416	0.444	0.491	0.603	0.400	0.535	0.656	0.564	0	0	0	0	0.507
2021	0.068	0.175	0.235	0.368	0.443	0.517	0.538	0.577	0.809	0.721	1.065	0.774	0	0	0	0	0.554
2022	0.070	0.175	0.262	0.323	0.445	0.505	0.556	0.601	0.552	0.407	0.736	0.739	0	0	0	0	0.562
2023	0.037	0.183	0.255	0.329	0.382	0.482	0.543	0.593	0.575	0.553	0.687	0.650	0	0	0	0	0.560
2024	0.057	0.179	0.221	0.288	0.354	0.401	0.520	0.626	0.511	0.711	0.714	0	0	0	0	0	0.543

Table 22.10. Whiting in Subarea 4 and Division 7.d. Discards mean weights-at-age (kg), as estimated by ICES. Age 6 is a plus-group.

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1978	0.036	0.145	0.158	0.185	0.209	0.222	0.239	0	0	0	0	0	0	0	0	0	0.239
1979	0.080	0.104	0.158	0.191	0.189	0.234	0.265	0.295	0	0	0	0	0	0	0	0	0.271
1980	0.030	0.107	0.166	0.202	0.244	0.253	0.264	0	0	0	0	0	0	0	0	0	0.264
1981	0.071	0.131	0.164	0.197	0.230	0.289	0.252	0.268	0	0	0	0	0	0	0	0	0.253
1982	0.047	0.091	0.182	0.211	0.225	0.241	0.244	0.261	0	0	0	0	0	0	0	0	0.245
1983	0.036	0.114	0.167	0.235	0.264	0.29	0.317	0.277	0.365	0	0	0	0	0	0	0	0.315
1984	0.038	0.101	0.162	0.216	0.246	0.265	0.248	0.278	0	0	0	0	0	0	0	0	0.252
1985	0.022	0.105	0.169	0.213	0.238	0.242	0.253	0.255	0	0	0	0	0	0	0	0	0.253
1986	0.028	0.123	0.166	0.190	0.208	0.227	0.194	0.217	0.311	0	0	0	0	0	0	0	0.210
1987	0.016	0.09	0.149	0.206	0.205	0.263	0.257	0	0.292	0	0	0	0	0	0	0	0.271
1988	0.030	0.063	0.146	0.181	0.210	0.219	0.235	0	0.284	0	0	0	0	0	0	0	0.243

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1989	0.033	0.083	0.164	0.191	0.213	0.227	0.241	0.351	0.221	0	0	0	0	0	0	0	0.246
1990	0.024	0.095	0.130	0.183	0.186	0.196	0.249	0.302	0	0	0	0	0	0	0	0	0.263
1991	0.041	0.089	0.154	0.177	0.213	0.230	0.253	0.268	0	0	0	0	0	0	0	0	0.259
1992	0.037	0.093	0.173	0.210	0.215	0.241	0.245	0.220	1.183	0	0	0	0	0	0	0	0.228
1993	0.023	0.087	0.160	0.205	0.237	0.235	0.225	0.213	0	0	0	0	0	0	0	0	0.223
1994	0.040	0.090	0.151	0.203	0.230	0.244	0.254	0.332	0	0	0	0	0	0	0	0	0.254
1995	0.032	0.102	0.163	0.204	0.233	0.247	0.247	0.332	0.290	0	0	0	0	0	0	0	0.277
1996	0.031	0.094	0.151	0.198	0.225	0.281	0.265	0.304	0	0	0	0	0	0	0	0	0.282
1997	0.031	0.125	0.181	0.213	0.225	0.233	0.256	0.617	0.320	0.601	0.773	0	0	0	0	0	0.261
1998	0.026	0.086	0.173	0.204	0.228	0.234	0.224	0.247	0.191	0.180	0.284	0	0	0	0	0	0.226
1999	0.062	0.100	0.166	0.197	0.201	0.225	0.231	0.212	0.231	0.220	0	0	0	0	0	0	0.228
2000	0.033	0.127	0.167	0.195	0.226	0.209	0.219	0.222	0.264	0	0	0	0	0	0	0	0.222
2001	0.023	0.084	0.183	0.217	0.259	0.248	0.240	0.225	0.243	0	0	0	0	0	0	0	0.235
2002	0.039	0.130	0.167	0.196	0.224	0.224	0.225	0.272	0.334	1.120	0.217	0	0	0	0	0	0.270
2003	0.048	0.062	0.105	0.170	0.214	0.262	0.257	0.293	0.237	0	0	0	0	0	0	0	0.259
2004	0.079	0.131	0.158	0.203	0.223	0.239	0.235	0.227	0.204	0.351	0	0	0	0	0	0	0.234
2005	0.070	0.124	0.177	0.207	0.221	0.223	0.235	0.245	0.222	0.293	0	0	0	0	0	0	0.236
2006	0.093	0.131	0.161	0.193	0.229	0.233	0.247	0.273	0.239	0.279	0.289	0	0	0	0	0	0.252

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2007	0.050	0.065	0.170	0.214	0.225	0.247	0.237	0.215	0.229	0.166	0.241	0.350	0	0	0	0	0.224
2008	0.027	0.072	0.181	0.213	0.230	0.265	0.328	0.244	0.291	0.317	0.057	0	0	0	0	0	0.269
2009	0.042	0.086	0.177	0.240	0.333	0.360	0.375	0.265	0.426	0.273	0.594	0	0	0	0	0	0.374
2010	0.049	0.102	0.207	0.283	0.331	0.381	0.242	0.277	0.182	0.362	0.521	0.337	0	0	0.368	0	0.211
2011	0.048	0.100	0.176	0.231	0.264	0.285	0.316	0.346	0.291	0.305	0.251	0	0	0	0	0	0.321
2012	0.038	0.100	0.175	0.229	0.290	0.296	0.261	0.405	0.333	0.877	0.746	0	0	0	0	0	0.342
2013	0.028	0.101	0.199	0.236	0.283	0.353	0.346	0.578	0.484	0.205	0.484	0	0	0	0	0	0.332
2014	0.055	0.130	0.189	0.245	0.270	0.294	0.348	0.556	0.547	0.550	0.361	0	0	0	0	0	0.400
2015	0.043	0.120	0.202	0.254	0.293	0.289	0.358	0.454	0.253	0.271	0.393	0	0	0	0	0	0.359
2016	0.03	0.117	0.188	0.241	0.291	0.267	0.287	0.29	0.309	0.305	0.315	0	0	0	0	0	0.289
2017	0.026	0.076	0.199	0.257	0.322	0.298	0.255	0.335	0.392	0.291	0.362	0.459	0	0	0	0	0.312
2018	0.029	0.103	0.178	0.219	0.247	0.292	0.411	0.340	0.316	0.296	0.311	0.369	0	0	0	0	0.376
2019	0.021	0.098	0.180	0.219	0.259	0.297	0.270	0.544	0.251	0.384	0	0	0	0	0	0	0.290
2020	0.092	0.092	0.192	0.219	0.253	0.307	0.285	0.313	0.216	0.266	0	0	0	0	0	0	0.292
2021	0.039	0.112	0.181	0.244	0.209	0.261	0.270	0.214	0.223	0.251	1.065	1.395	0	0	0	0	0.263
2022	0.041	0.124	0.192	0.225	0.258	0.28	0.289	0.254	0.26	0.339	0.739	0.743	0	0	0	0	0.271
2023	0.024	0.101	0.192	0.240	0.259	0.279	0.342	0.322	0.372	0.247	0	0	0	0	0	0	0.323
2024	0.044	0.127	0.184	0.238	0.257	0.263	0.343	0.398	0.365	0.339	0.369	0	0	0	0	0	0.351

[illegible]

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
1996	0.016	0.064	0.151	0.239	0.233	0.347	0.250	0	0	0	0	0	0	0	0	0	0.250
1997	0.012	0.051	0.145	0.252	0.321	0.348	0.588	0	0	0	0	0	0	0	0	0	0.588
1998	0.015	0.049	0.115	0.220	0.304	0.286	0	0	0	0	0	0	0	0	0	0	0
1999	0.013	0.027	0.077	0.144	0.194	0.286	0	0	0	0	0	0	0	0	0	0	0
2000	0.038	0.051	0.166	0.242	0.289	0.339	0	0.588	0	0	0	0	0	0	0	0	0.066
2001	0.012	0.055	0.118	0.225	0.320	0.351	0.386	0	0	0	0	0	0	0	0	0	0.386
2002	0.010	0.044	0.101	0.185	0.294	0.415	0.380	0	0	0	0	0	0	0	0	0	0.380
2003	0.010	0.035	0.102	0.189	0.302	0.418	0.462	0	0	0	0	0	0	0	0	0	0.462
2004	0.010	0.032	0.083	0.143	0.264	0	0.380	0	0	0	0	0	0	0	0	0	0.380
2005	0.014	0.043	0.133	0.196	0.205	0.366	0.438	0.541	0.530	0	0	0	0	0	0	0	0.463
2006	0	0.046	0.119	0.208	0.277	0.362	0.401	0.564	0.530	0	0	0	0	0	0	0	0.419
2007	0	0.046	0.119	0.208	0.277	0.362	0.401	0.564	0.530	0	0	0	0	0	0	0	0.420
2008	0	0.046	0.119	0.208	0.277	0.362	0.401	0.564	0.530	0	0	0	0	0	0	0	0
2009	0.042	0.092	0.213	0.286	0.370	0.374	0.373	0.343	0.351	0.335	0.331	0.350	0.419	0	0	0	0.350
2010	0.049	0.111	0.234	0.373	0.407	0.455	0.355	0.458	0.272	0.475	0.471	0.398	0.259	0	0.368	0	0.364
2011	0.048	0.114	0.214	0.298	0.374	0.415	0.424	0.364	0.340	0.372	0.320	0.550	0.894	0	0	0	0.379
2012	0.038	0.105	0.194	0.311	0.445	0.411	0.430	0.428	0.366	0.418	0.407	0.552	0.733	0	0	0	0.418
2013	0.028	0.110	0.222	0.273	0.391	0.468	0.496	0.464	0.424	0.341	0.406	0	0	0	0	0	0.464

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	6+
2014	0.055	0.137	0.227	0.294	0.331	0.442	0.465	0.469	0.403	0.402	0.359	1.754	0	0	0	0	0.451
2015	0.044	0.125	0.218	0.308	0.368	0.386	0.469	0.464	0.374	0.372	0.400	0.778	0	0	0	0	0.447
2016	0.03	0.120	0.210	0.291	0.399	0.389	0.415	0.488	0.452	0.460	0.472	1.293	0	0	0	0	0.441
2017	0.026	0.078	0.212	0.320	0.409	0.436	0.487	0.444	0.457	0.419	0.526	0.488	0	0	0	0	0.457
2018	0.029	0.108	0.196	0.275	0.373	0.407	0.514	0.458	0.485	0.594	0.448	0.583	0	0	0	0	0.490
2019	0.021	0.107	0.204	0.279	0.354	0.420	0.435	0.440	0.369	0.355	0.577	0.736	0	0	0	0	0.430
2020	0.094	0.106	0.237	0.287	0.374	0.425	0.479	0.542	0.376	0.494	0.656	0.564	0	0	0	0	0.487
2021	0.039	0.118	0.191	0.322	0.401	0.479	0.431	0.529	0.458	0.503	1.065	0.774	0	0	0	0	0.459
2022	0.041	0.132	0.211	0.262	0.366	0.470	0.515	0.428	0.527	0.404	0.736	0.739	0	0	0	0	0.500
2023	0.024	0.107	0.216	0.270	0.335	0.434	0.537	0.524	0.554	0.358	0.687	0.650	0	0	0	0	0.533
2024	0.044	0.133	0.195	0.254	0.293	0.321	0.445	0.567	0.492	0.58	0.588	0	0	0	0	0	0.479

Table 22.12. Whiting in Subarea 4 and Division 7.d. Catch components as estimated by ICES in tonnes. Discards include BMS.

Year	Catch	Landings	Discards	IBC
1978	188222	97553	35382	55287
1979	243570	107231	77391	58948
1980	223361	100775	77003	45584
1981	192119	89583	35894	66641
1982	140250	80576	26620	33055
1983	161316	88002	49562	23753
1984	145636	86275	40483	18878
1985	100330	56059	28961	15310
1986	161494	64019	79523	17953
1987	138737	68317	53901	16519
1988	133215	56100	28146	48969
1989	123533	45103	35787	42643
1990	152602	45662	55603	51337
1991	126742	51929	35058	39755
1992	108555	50946	32564	25045
1993	116911	51818	44370	20723
1994	101650	48486	35692	17473
1995	105494	45938	32176	27379
1996	76123	40503	30505	5116
1997	61435	35563	19660	6213
1998	47475	28288	15693	3494
1999	60845	30130	25677	5038
2000	63806	28583	26063	9160
2001	45242	25061	19237	944
2002	46450	20675	18501	7275
2003	45640	16161	26745	2734
2004	33557	13295	19048	1214
2005	28883	15471	12525	888

Year	Catch	Landings	Discards	IBC
2006	36769	18535	16310	1924
2007	26974	18915	6971	1088
2008	28247	17951	10296	0
2009	28430	18403	8684	1344
2010	34436	19846	12683	1907
2011	30668	18461	11173	1035
2012	30221	17407	11697	1117
2013	26660	18211	6795	1654
2014	28375	17027	9725	1623
2015	36287	17299	16891	2097
2016	33396	16118	12726	4551
2017	29344	15361	11348	2635
2018	28407	16160	10588	1658
2019	30523	18579	10080	1864
2020	34924	18014	13795	3115
2021	34394	16384	15930	2080
2022	28792	15138	12824	831
2023	29356	15809	10946	2601
2024	36577	13865	20980	1733

Table 22.13. Whiting in Subarea 4 and Division 7.d. Stock weights-at-age (kg), as estimated from scaled (using IBTS Q1) commercial catch weights at age. Age 6 is a plus-group. Model input.

Year/Age	0	1	2	3	4	5	6+
1978	0.003	0.025	0.095	0.165	0.259	0.388	0.427
1979	0.003	0.033	0.087	0.182	0.243	0.373	0.459
1980	0.004	0.026	0.092	0.177	0.265	0.306	0.449
1981	0.004	0.028	0.088	0.170	0.260	0.344	0.410
1982	0.010	0.021	0.096	0.178	0.253	0.341	0.481
1983	0.005	0.036	0.100	0.192	0.262	0.348	0.425
1984	0.007	0.030	0.099	0.191	0.272	0.346	0.396
1985	0.005	0.032	0.100	0.200	0.268	0.364	0.429
1986	0.005	0.036	0.095	0.180	0.257	0.343	0.466
1987	0.004	0.026	0.077	0.174	0.240	0.340	0.421
1988	0.004	0.018	0.076	0.157	0.243	0.314	0.440
1989	0.008	0.024	0.082	0.158	0.216	0.288	0.382
1990	0.005	0.029	0.071	0.148	0.203	0.253	0.426
1991	0.006	0.035	0.088	0.153	0.233	0.278	0.333
1992	0.004	0.029	0.096	0.181	0.224	0.300	0.328
1993	0.004	0.025	0.091	0.176	0.255	0.298	0.351
1994	0.004	0.029	0.087	0.180	0.265	0.347	0.368
1995	0.003	0.030	0.094	0.181	0.274	0.348	0.417
1996	0.006	0.032	0.087	0.165	0.244	0.352	0.401
1997	0.010	0.033	0.093	0.170	0.238	0.303	0.375
1998	0.006	0.031	0.093	0.166	0.227	0.285	0.329
1999	0.008	0.027	0.091	0.163	0.207	0.262	0.295
2000	0.012	0.040	0.095	0.168	0.232	0.259	0.266
2001	0.008	0.034	0.100	0.172	0.228	0.242	0.283
2002	0.003	0.024	0.081	0.154	0.220	0.275	0.334
2003	0.004	0.019	0.062	0.136	0.209	0.271	0.348
2004	0.011	0.038	0.078	0.150	0.204	0.259	0.279
2005	0.011	0.042	0.104	0.168	0.202	0.256	0.292

Year/Age	0	1	2	3	4	5	6+
2006	0.032	0.045	0.094	0.163	0.221	0.261	0.340
2007	0.020	0.033	0.107	0.181	0.262	0.313	0.301
2008	0.009	0.035	0.114	0.199	0.254	0.365	0.337
2009	0.014	0.031	0.111	0.201	0.299	0.339	0.338
2010	0.017	0.038	0.122	0.263	0.328	0.414	0.354
2011	0.016	0.039	0.112	0.210	0.302	0.376	0.367
2012	0.013	0.036	0.102	0.219	0.359	0.373	0.403
2013	0.010	0.037	0.116	0.192	0.315	0.425	0.446
2014	0.019	0.047	0.118	0.207	0.267	0.401	0.434
2015	0.015	0.043	0.114	0.216	0.297	0.350	0.432
2016	0.010	0.041	0.109	0.205	0.322	0.353	0.426
2017	0.009	0.027	0.111	0.225	0.330	0.396	0.442
2018	0.010	0.037	0.103	0.194	0.301	0.369	0.474
2019	0.007	0.036	0.106	0.196	0.286	0.381	0.413
2020	0.032	0.036	0.124	0.202	0.302	0.385	0.471
2021	0.013	0.040	0.100	0.227	0.324	0.435	0.440
2022	0.014	0.045	0.110	0.184	0.295	0.426	0.481
2023	0.008	0.036	0.113	0.190	0.270	0.394	0.514
2024	0.015	0.045	0.102	0.179	0.237	0.292	0.459

Table 22.14a. Whiting in Subarea 4 and Division 7.d. Smoothed estimated proportion mature-at-age as used in the assessment. Model input.

Year/Age	0	1	2	3	4	5	6+
1978	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1979	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1980	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1981	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1982	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1983	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1984	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1985	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1986	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1987	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1988	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1989	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1990	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1991	0.000	0.125	0.880	1.000	1.000	1.000	1.000
1992	0.000	0.129	0.869	0.998	1.000	1.000	1.000
1993	0.000	0.135	0.858	0.993	1.000	1.000	1.000
1994	0.000	0.141	0.846	0.988	0.998	1.000	1.000
1995	0.000	0.148	0.833	0.983	0.997	0.999	1.000
1996	0.000	0.156	0.820	0.977	0.995	0.999	1.000
1997	0.000	0.167	0.806	0.970	0.994	0.998	1.000
1998	0.000	0.179	0.793	0.964	0.992	0.998	1.000
1999	0.000	0.193	0.780	0.958	0.991	0.998	1.000
2000	0.000	0.209	0.769	0.953	0.990	0.998	1.000
2001	0.000	0.226	0.762	0.951	0.989	0.998	1.000
2002	0.000	0.242	0.761	0.951	0.990	0.998	1.000
2003	0.000	0.257	0.765	0.953	0.991	0.998	1.000
2004	0.000	0.271	0.773	0.956	0.992	0.999	1.000
2005	0.000	0.283	0.784	0.960	0.993	0.999	1.000

Year/Age	0	1	2	3	4	5	6+
2006	0.000	0.294	0.797	0.965	0.995	0.999	1.000
2007	0.000	0.303	0.810	0.970	0.996	1.000	1.000
2008	0.000	0.310	0.823	0.975	0.997	1.000	1.000
2009	0.000	0.315	0.834	0.979	0.998	1.000	1.000
2010	0.000	0.319	0.844	0.982	0.999	1.000	1.000
2011	0.000	0.322	0.850	0.985	1.000	1.000	1.000
2012	0.000	0.325	0.854	0.987	1.000	1.000	1.000
2013	0.000	0.329	0.856	0.987	1.000	1.000	1.000
2014	0.000	0.333	0.856	0.988	1.000	1.000	1.000
2015	0.000	0.340	0.856	0.987	1.000	1.000	1.000
2016	0.000	0.350	0.855	0.986	0.999	1.000	1.000
2017	0.000	0.361	0.853	0.984	0.998	1.000	1.000
2018	0.000	0.376	0.849	0.980	0.996	0.999	1.000
2019	0.000	0.394	0.845	0.975	0.994	0.998	0.999
2020	0.000	0.416	0.840	0.969	0.991	0.996	0.999
2021	0.000	0.443	0.834	0.961	0.987	0.994	0.998
2022	0.000	0.474	0.828	0.952	0.982	0.991	0.998
2023	0.000	0.507	0.821	0.942	0.977	0.988	0.997
2024	0.000	0.541	0.813	0.930	0.971	0.985	0.996

Table 22.14b. Whiting in Subarea 4 and Division 7.d. Number of samples with biological information in IBTS Q1 as used for maturity ogive estimation and Q1 index calculation.

Year/Age	1	2	3	4	5	6+
1991	690	541	435	238	122	38
1992	1216	763	478	424	107	83
1993	815	650	508	269	278	59
1994	1011	925	537	450	144	124
1995	930	427	339	253	98	57
1996	634	419	311	209	95	52
1997	548	357	317	239	139	84
1998	654	338	307	253	147	83

Year/Age	1	2	3	4	5	6+
1999	588	314	170	170	147	71
2000	792	815	491	305	164	186
2001	1120	1102	848	439	199	298
2002	893	920	701	394	142	94
2003	558	859	589	589	329	123
2004	591	382	381	366	310	163
2005	580	680	307	340	355	368
2006	952	669	355	194	250	359
2007	1016	1035	593	227	126	543
2008	1323	881	704	344	142	348
2009	1174	1172	475	324	207	244
2010	1169	948	711	325	229	190
2011	1191	1350	951	732	303	392
2012	1123	1256	850	603	461	258
2013	1080	949	979	612	322	344
2014	1219	1055	788	705	348	281
2015	1475	1429	589	498	535	331
2016	664	678	590	273	184	250
2017	1410	997	929	634	232	258
2018	708	1020	797	594	323	167
2019	1207	814	915	566	263	149
2020	1182	734	554	670	308	149
2021	1231	1183	677	634	439	237
2022	655	659	706	293	207	141
2023	684	591	964	755	221	181
2024	381	699	489	1028	829	395
2025	475	274	309	228	519	409

Table 22.15. Whiting in Subarea 4 and Division 7.d. Smoothed natural mortality-at-age estimates based on ICES WGSAM (2024) as used in the assessment. Model input.

Year/Age	0	1	2	3	4	5	6+
1978	1.126	1.281	0.729	0.577	0.521	0.474	0.268
1979	1.143	1.275	0.729	0.570	0.516	0.475	0.270
1980	1.158	1.267	0.727	0.563	0.511	0.476	0.271
1981	1.168	1.257	0.722	0.555	0.506	0.475	0.273
1982	1.172	1.242	0.715	0.547	0.501	0.474	0.276
1983	1.170	1.224	0.704	0.538	0.496	0.472	0.279
1984	1.165	1.206	0.691	0.529	0.490	0.469	0.283
1985	1.159	1.191	0.678	0.521	0.485	0.465	0.290
1986	1.155	1.179	0.664	0.513	0.48	0.462	0.298
1987	1.154	1.172	0.652	0.507	0.475	0.459	0.308
1988	1.159	1.170	0.640	0.501	0.472	0.456	0.321
1989	1.171	1.171	0.631	0.496	0.468	0.454	0.335
1990	1.190	1.175	0.623	0.492	0.465	0.452	0.350
1991	1.218	1.183	0.618	0.489	0.462	0.450	0.364
1992	1.254	1.194	0.615	0.487	0.460	0.448	0.378
1993	1.298	1.208	0.615	0.485	0.459	0.447	0.391
1994	1.351	1.223	0.615	0.485	0.459	0.446	0.401
1995	1.412	1.239	0.617	0.486	0.459	0.447	0.411
1996	1.478	1.255	0.620	0.488	0.460	0.448	0.420
1997	1.549	1.269	0.622	0.49	0.463	0.450	0.428
1998	1.621	1.282	0.626	0.494	0.466	0.452	0.436
1999	1.692	1.292	0.630	0.498	0.47	0.456	0.444
2000	1.760	1.299	0.636	0.503	0.474	0.460	0.451
2001	1.822	1.303	0.643	0.509	0.48	0.466	0.459
2002	1.876	1.304	0.651	0.516	0.487	0.472	0.466
2003	1.923	1.297	0.66	0.524	0.494	0.479	0.473
2004	1.959	1.284	0.668	0.531	0.500	0.485	0.480
2005	1.987	1.267	0.676	0.537	0.507	0.491	0.484

Year/Age	0	1	2	3	4	5	6+
2006	2.006	1.247	0.681	0.543	0.512	0.496	0.487
2007	2.019	1.227	0.685	0.547	0.516	0.499	0.487
2008	2.027	1.208	0.687	0.549	0.519	0.501	0.484
2009	2.032	1.190	0.687	0.550	0.521	0.502	0.478
2010	2.037	1.174	0.687	0.549	0.522	0.502	0.469
2011	2.041	1.160	0.686	0.549	0.522	0.502	0.458
2012	2.039	1.148	0.685	0.547	0.522	0.501	0.444
2013	2.030	1.137	0.683	0.545	0.521	0.500	0.430
2014	2.012	1.127	0.681	0.542	0.519	0.498	0.415
2015	1.985	1.119	0.678	0.539	0.517	0.496	0.400
2016	1.948	1.112	0.676	0.535	0.513	0.493	0.387
2017	1.903	1.103	0.673	0.531	0.509	0.489	0.375
2018	1.851	1.091	0.671	0.526	0.504	0.483	0.365
2019	1.795	1.075	0.668	0.521	0.499	0.477	0.357
2020	1.737	1.057	0.666	0.515	0.493	0.469	0.349
2021	1.678	1.038	0.664	0.510	0.487	0.461	0.342
2022	1.618	1.018	0.662	0.504	0.481	0.452	0.335
2023	1.618	1.018	0.662	0.504	0.481	0.452	0.335
2024	1.618	1.018	0.662	0.504	0.481	0.452	0.335

Table 22.16a. Whiting in Subarea 4 and Division 7.d. NS IBTS Quarter 1 tuning series used in the assessment and forecast; model input.

IBTS–Q1					
Year/Age	1	2	3	4	5
1983	1.265	1.211	1.078	0.765	0.337
1984	4.265	1.645	0.805	0.276	0.267
1985	3.243	3.449	0.617	0.171	0.079
1986	4.511	2.826	2.127	0.349	0.093
1987	6.680	5.395	0.864	0.428	0.060
1988	4.329	8.312	2.998	0.308	0.173
1989	14.246	5.205	3.946	1.033	0.172

IBTS–Q1					
Year/Age	1	2	3	4	5
1990	5.140	8.397	1.992	0.988	0.201
1991	9.341	7.593	3.660	0.735	0.336
1992	9.984	4.501	2.423	0.748	0.573
1993	10.613	5.507	1.928	0.880	0.392
1994	7.317	5.711	1.922	0.677	0.135
1995	6.563	4.709	2.040	0.643	0.135
1996	4.796	4.686	2.174	0.676	0.351
1997	3.165	2.610	1.598	0.820	0.235
1998	5.107	1.621	1.175	0.484	0.220
1999	6.108	2.638	1.461	0.672	0.274
2000	8.133	4.628	1.857	0.317	0.181
2001	6.462	5.632	2.507	0.723	0.289
2002	5.347	3.505	2.588	0.484	0.124
2003	1.370	2.729	2.468	1.264	0.444
2004	1.874	0.932	1.599	0.778	0.435
2005	1.284	0.753	0.511	0.425	0.287
2006	1.931	1.052	0.476	0.223	0.160
2007	0.638	1.485	0.640	0.217	0.112
2008	2.571	1.993	0.556	0.183	0.095
2009	2.115	2.873	0.681	0.173	0.162
2010	3.379	1.961	1.721	0.515	0.735
2011	1.751	3.521	1.350	0.708	0.188
2012	2.204	5.620	1.001	0.396	0.293
2013	0.525	1.629	2.447	0.670	0.346
2014	2.585	1.873	0.978	0.607	0.337
2015	3.241	2.032	0.510	0.244	0.225
2016	3.510	2.933	0.849	0.241	0.140
2017	5.651	2.333	1.012	0.305	0.111

IBTS-Q1					
Year/Age	1	2	3	4	5
2018	1.215	2.304	0.736	0.328	0.121
2019	2.175	1.749	1.169	0.442	0.129
2020	5.190	2.023	0.785	0.526	0.164
2021	5.999	7.034	1.129	0.384	0.154
2022	3.800	6.376	2.235	0.376	0.134
2023	3.161	5.506	2.728	1.575	0.309
2024	0.709	4.351	2.833	1.676	0.426
2025	8.748	5.373	2.887	1.434	0.774

Table 22.16b. Whiting in Subarea 4 and Division 7.d. NS IBTS Quarter 3 tuning series used in the assessment and forecast, model input.

IBTS-Q3						
Year/Age	0	1	2	3	4	5
1991	5.065	6.776	1.478	0.858	0.297	0.169
1992	13.232	5.468	2.504	0.709	0.539	0.316
1993	8.781	6.247	1.803	0.426	0.246	0.169
1994	5.687	6.932	2.358	0.494	0.186	0.106
1995	7.035	6.252	2.730	0.712	0.209	0.090
1996	2.832	4.446	3.279	1.267	0.347	0.099
1997	19.735	2.902	1.655	1.192	0.265	0.202
1998	25.563	3.176	1.386	0.539	0.315	0.124
1999	23.860	11.486	1.775	0.521	0.226	0.102
2000	18.681	8.953	3.048	0.582	0.172	0.084
2001	34.265	6.447	2.677	0.845	0.220	0.081
2002	2.566	7.703	2.390	1.275	0.344	0.075
2003	3.481	2.502	2.735	1.193	0.676	0.189
2004	6.800	1.377	0.597	0.629	0.428	0.246
2005	1.639	1.451	0.810	0.314	0.429	0.315
2006	1.894	1.653	0.775	0.287	0.228	0.183
2007	7.773	0.853	0.611	0.336	0.155	0.082

IBTS-Q3						
Year/Age	0	1	2	3	4	5
2008	7.281	3.425	0.615	0.294	0.131	0.066
2009	5.553	5.414	3.361	0.504	0.131	0.089
2010	4.725	2.160	1.336	0.433	0.125	0.123
2011	2.311	4.031	1.360	0.593	0.191	0.082
2012	2.828	2.494	2.097	0.630	0.215	0.146
2013	3.083	0.627	0.575	0.624	0.198	0.072
2014	19.385	2.073	0.908	0.580	0.329	0.097
2015	19.307	2.926	2.093	0.539	0.265	0.176
2016	9.005	2.752	2.226	0.663	0.200	0.089
2017	1.710	8.764	1.926	0.825	0.260	0.114
2018	1.687	2.363	2.842	0.807	0.317	0.210
2019	13.649	4.285	1.461	0.831	0.220	0.150
2020	12.224	14.487	2.086	0.594	0.424	0.346
2021	6.020	11.442	4.037	0.633	0.204	0.102
2022	19.898	4.954	3.753	2.513	0.708	0.196
2023	7.863	10.117	3.065	3.220	2.207	0.290
2024	14.899	5.522	3.212	1.230	1.618	0.909

Table 22.17. Whiting in Subarea 4 and Division 7.d. Final fishing mortality estimates from SAM, model output.

Year/Age	0	1	2	3	4	5	6+
1978	0.025	0.096	0.283	0.522	0.662	0.848	0.848
1979	0.027	0.103	0.304	0.563	0.680	0.861	0.861
1980	0.024	0.094	0.286	0.586	0.752	0.959	0.959
1981	0.025	0.097	0.272	0.557	0.728	0.921	0.921
1982	0.025	0.101	0.253	0.487	0.618	0.779	0.779
1983	0.029	0.127	0.317	0.583	0.706	0.857	0.857
1984	0.031	0.141	0.351	0.669	0.845	1.026	1.026
1985	0.027	0.121	0.285	0.559	0.778	0.978	0.978
1986	0.029	0.141	0.341	0.635	0.898	1.094	1.094

Year/Age	0	1	2	3	4	5	6+
1987	0.027	0.131	0.345	0.645	0.902	1.124	1.124
1988	0.028	0.138	0.342	0.584	0.779	0.986	0.986
1989	0.024	0.120	0.319	0.532	0.704	0.916	0.916
1990	0.026	0.132	0.371	0.561	0.666	0.815	0.815
1991	0.022	0.110	0.312	0.468	0.515	0.599	0.599
1992	0.022	0.113	0.309	0.463	0.505	0.528	0.528
1993	0.021	0.117	0.329	0.534	0.598	0.627	0.627
1994	0.018	0.105	0.300	0.520	0.620	0.656	0.656
1995	0.016	0.092	0.262	0.459	0.549	0.592	0.592
1996	0.013	0.079	0.232	0.410	0.500	0.536	0.536
1997	0.011	0.071	0.209	0.360	0.452	0.480	0.480
1998	0.009	0.066	0.189	0.315	0.408	0.451	0.451
1999	0.009	0.071	0.215	0.354	0.450	0.496	0.496
2000	0.007	0.055	0.182	0.324	0.426	0.489	0.489
2001	0.005	0.043	0.130	0.226	0.310	0.362	0.362
2002	0.006	0.049	0.131	0.204	0.260	0.286	0.286
2003	0.008	0.078	0.177	0.221	0.252	0.263	0.263
2004	0.006	0.070	0.148	0.182	0.214	0.239	0.239
2005	0.006	0.069	0.144	0.171	0.195	0.222	0.222
2006	0.006	0.080	0.166	0.208	0.225	0.252	0.252
2007	0.005	0.070	0.148	0.197	0.214	0.230	0.230
2008	0.005	0.065	0.139	0.196	0.219	0.234	0.234
2009	0.004	0.056	0.122	0.187	0.226	0.259	0.259
2010	0.003	0.048	0.113	0.184	0.233	0.278	0.278
2011	0.003	0.046	0.109	0.173	0.213	0.252	0.252
2012	0.004	0.051	0.107	0.166	0.213	0.260	0.260
2013	0.003	0.044	0.097	0.162	0.220	0.286	0.286
2014	0.003	0.042	0.105	0.183	0.245	0.320	0.320
2015	0.003	0.046	0.132	0.226	0.283	0.354	0.354

Year/Age	0	1	2	3	4	5	6+
2016	0.003	0.038	0.120	0.233	0.297	0.348	0.348
2017	0.003	0.029	0.099	0.205	0.277	0.294	0.294
2018	0.002	0.027	0.095	0.201	0.269	0.261	0.261
2019	0.002	0.025	0.091	0.205	0.271	0.248	0.248
2020	0.002	0.020	0.075	0.177	0.242	0.215	0.215
2021	0.002	0.015	0.053	0.127	0.194	0.185	0.185
2022	0.001	0.009	0.032	0.083	0.145	0.148	0.148
2023	0.001	0.008	0.026	0.068	0.128	0.133	0.133
2024	0.001	0.008	0.024	0.066	0.133	0.137	0.137

Table 22.18. Whiting in Subarea 4 and Division 7.d. Final abundance estimates from SAM, model output. *Survivors at the beginning of the year; no recruitment estimate available in final year.

Year/Age	0	1	2	3	4	5	6+
1978	28806230	8542641	1752712	837151	219575	16009	18593
1979	22939653	9161630	2168902	636835	281623	66864	10351
1980	11232364	7223786	2276480	765826	204596	86920	20994
1981	10332505	3369272	1947803	813389	239013	57453	26891
1982	9079163	3127842	860403	755338	264088	68303	22238
1983	13251306	2702391	805984	330217	279610	86182	27080
1984	10327947	4061233	695986	288752	107149	87122	31570
1985	17870726	3047816	1069471	243812	86004	28209	28385
1986	16603720	5573228	806487	427194	83246	24330	14649
1987	12930872	5112556	1500440	291447	140097	20731	8820
1988	17942372	3906960	1450087	550414	91245	35544	6322
1989	11197133	5609755	1017005	560326	185896	26348	10093
1990	9947328	3357250	1575838	392044	204362	57608	9409
1991	10921500	2923883	898089	573448	137884	66300	19000
1992	12679436	3182906	803195	348985	219311	53089	30085
1993	12408704	3539629	837938	318954	135900	86134	31395
1994	11489725	3319395	927004	317421	117079	46741	40839
1995	8953575	2961456	870793	368003	116139	39519	29441

Year/Age	0	1	2	3	4	5	6+
1996	7472558	2126000	784550	357752	143591	42648	24453
1997	11581108	1671176	551685	336694	141761	56206	24877
1998	19200006	2392073	436281	240277	140651	56900	32012
1999	21228150	3751878	601322	197292	109513	58016	36137
2000	17482387	3871189	919243	252168	84396	44103	36685
2001	16500278	2970264	1023710	385466	104676	34866	31216
2002	8042256	2647560	784355	493875	179703	45456	28451
2003	8143018	1207780	690678	373485	241921	83932	33964
2004	9586420	1180887	286702	292952	183354	115334	56462
2005	9382231	1340169	302730	125858	141697	91201	83955
2006	7325400	1317375	358513	132808	63012	70150	87617
2007	11706084	971237	349912	159140	63011	29799	76748
2008	11748029	1567427	274858	152846	77509	30405	52881
2009	11414558	1543517	443098	120701	73237	38560	41631
2010	11668511	1494352	438571	194724	56911	37266	39024
2011	7857955	1548407	447829	198300	92083	26737	35875
2012	5742322	1030966	496094	196761	94642	44660	30325
2013	9386073	729007	297917	233099	97395	46008	35605
2014	11517380	1252159	227817	136377	114590	46911	38334
2015	10430632	1515320	402400	106161	64712	53663	38932
2016	12449642	1382173	457289	177935	50403	29200	41034
2017	7828541	1794203	425724	197855	81673	22701	31968
2018	10191710	1160773	571090	194624	91549	37417	26164
2019	22107964	1612606	385905	256456	94196	41540	31124
2020	20265321	3782815	532535	183212	118764	44313	35891
2021	10987241	3626019	1330656	249319	91787	55307	41738
2022	13561044	1997905	1264814	657665	132029	46108	52452
2023	8629733	2710535	726505	637053	372500	69163	56504
2024	16065660	1643087	962256	372671	371719	202943	71785

Year/Age	0	1	2	3	4	5	6+
2025*		3204235	598057	477106	215079	198374	157394

Table 22.19. Whiting in Subarea 4 and Division 7.d. Final SAM summary table. Model output. Recruitment is in thousands, all weights are in tonnes. Low and High refer to 95% confidence intervals.

Year	Rec.age.0	Rec_Low	Rec_High	SSB	SSB_Low	SSB_High	Fbar.2.5	Fbar.2.5_Low	Fbar.2.5_High	TSB	TSB_Low	TSB_High
1978	28806230	22437348	36982932	382041	333715	437365	0.579	0.462	0.725	688020	613197	771973
1979	22939653	17497539	30074383	418385	368734	474722	0.602	0.495	0.732	778377	691368	876336
1980	11232364	8853552	14250325	432707	380651	491882	0.646	0.533	0.782	668605	595789	750320
1981	10332505	8168875	13069201	393367	345744	447549	0.619	0.508	0.755	535671	478801	599296
1982	9079163	7225054	11409077	316122	278373	358990	0.534	0.440	0.649	472309	425014	524867
1983	13251306	10455467	16794766	261168	231711	294369	0.616	0.519	0.731	424411	383008	470288
1984	10327947	8184217	13033194	202474	180306	227368	0.723	0.613	0.851	388478	349640	431630
1985	17870726	14221506	22456331	200545	177175	226997	0.650	0.548	0.771	383690	344124	427805
1986	16603720	13063478	21103379	205727	182903	231398	0.742	0.626	0.879	473758	421970	531902
1987	12930872	10282013	16262132	213545	189027	241242	0.754	0.627	0.907	401715	359442	448961
1988	17942372	14280110	22543854	228537	201299	259462	0.673	0.555	0.816	383850	344292	427953
1989	11197133	8965242	13984651	230140	203250	260588	0.618	0.500	0.763	444527	398458	495922
1990	9947328	7986767	12389161	229066	201999	259761	0.603	0.491	0.741	380585	340901	424888
1991	10921500	8787610	13573562	226568	199489	257324	0.473	0.386	0.581	393320	352320	439090
1992	12679436	10218721	15732703	217037	191201	246364	0.452	0.370	0.551	363411	325412	405848

Year	Rec.age.0	Rec_Low	Rec_High	SSB	SSB_Low	SSB_High	Fbar.2.5	Fbar.2.5_Low	Fbar.2.5_High	TSB	TSB_Low	TSB_High
1993	12408704	9984048	15422196	204021	179671	231672	0.522	0.432	0.630	341880	305661	382391
1994	11489725	9233168	14297777	200027	175784	227614	0.524	0.428	0.640	345457	307922	387567
1995	8953575	7171570	11178375	204495	178783	233904	0.466	0.376	0.576	326202	289135	368021
1996	7472558	5954699	9377321	184040	160092	211569	0.419	0.337	0.523	300943	265874	340638
1997	11581108	8987624	14922971	165887	143460	191818	0.375	0.300	0.469	333432	290273	383008
1998	19200006	14914131	24717512	142193	122389	165203	0.341	0.271	0.428	329934	283825	383534
1999	21228150	16605545	27137583	140874	120634	164510	0.379	0.300	0.478	400729	342799	468449
2000	17482387	13678334	22344377	180044	153378	211345	0.355	0.274	0.460	526251	447183	619299
2001	16500278	13001948	20939877	204851	172303	243547	0.257	0.194	0.340	446368	381017	522928
2002	8042256	6294615	10275111	196419	164366	234721	0.220	0.168	0.288	290167	247567	340096
2003	8143018	6448667	10282549	171477	142605	206194	0.228	0.178	0.293	234988	199891	276247
2004	9586420	7516144	12226939	154104	127940	185619	0.195	0.152	0.251	295020	251753	345723
2005	9382231	7333458	12003378	137147	114182	164730	0.183	0.143	0.234	287575	244559	338158
2006	7325400	5770467	9299331	126878	106078	151755	0.213	0.168	0.269	407605	342733	484756
2007	11706084	9220157	14862262	117036	98550	138989	0.197	0.156	0.249	382465	320238	456784
2008	11748029	9242292	14933113	121011	102643	142666	0.197	0.157	0.247	273418	233768	319792
2009	11414558	8981751	14506320	128889	109424	151818	0.199	0.157	0.251	333269	282521	393133

Year	Rec.age.0	Rec_Low	Rec_High	SSB	SSB_Low	SSB_High	Fbar.2.5	Fbar.2.5_Low	Fbar.2.5_High	TSB	TSB_Low	TSB_High
2010	11668511	8929371	15247898	161183	136588	190207	0.202	0.158	0.259	403266	338747	480073
2011	7857955	6178029	9994685	153745	130092	181699	0.187	0.145	0.240	330794	282019	388006
2012	5742322	4506722	7316684	160370	135203	190221	0.186	0.145	0.239	267341	228652	312575
2013	9386073	7347735	11989867	148728	124784	177266	0.191	0.149	0.245	261932	222837	307887
2014	11517380	8760979	15141006	136441	114581	162471	0.213	0.166	0.274	394930	325490	479183
2015	10430632	7919084	13738719	138512	115659	165881	0.249	0.193	0.321	343925	282922	418082
2016	12449642	9572012	16192372	142405	117622	172409	0.249	0.19	0.327	313854	259793	379165
2017	7828541	6021567	10177759	151145	123863	184436	0.219	0.164	0.292	258443	214731	311054
2018	10191710	7789647	13334488	156387	127450	191894	0.206	0.154	0.276	293175	241647	355691
2019	22107964	16500071	29621816	162059	131428	199830	0.204	0.151	0.275	362942	294354	447512
2020	20265321	15139492	27126619	218019	175749	270456	0.177	0.131	0.240	958138	750978	1222444
2021	10987241	8054995	14986908	300822	240971	375540	0.140	0.102	0.190	552195	443594	687383
2022	13561044	9655647	19046050	355582	282352	447805	0.102	0.074	0.140	622452	493725	784740
2023	8629733	5837221	12758174	385309	303413	489311	0.089	0.065	0.121	528874	417534	669903
2024	16065660	9508580	27144477	358464	278529	461339	0.090	0.065	0.125	659479	492051	883877
2025				409187	310901	538545				693446	467153	1029358

Table 22.20. Whiting in Subarea 4 and Division 7.d. Final summary catch table estimated by SAM, model output. Units: tonnes.

Year	Catch	Low	High
1978	190683	158548	229331
1979	226891	193681	265796
1980	218109	185568	256357
1981	188129	159634	221711
1982	145642	123760	171393
1983	144882	124673	168366
1984	132739	114072	154461
1985	109774	93762	128520
1986	144993	123562	170140
1987	134674	115000	157712
1988	130101	110564	153092
1989	131830	112732	154165
1990	132435	112458	155960
1991	112497	96015	131808
1992	105100	90223	122429
1993	107384	92220	125042
1994	102627	88164	119464
1995	93567	80196	109166
1996	75959	65191	88504
1997	63001	54093	73375
1998	50948	43960	59047
1999	56672	48700	65950
2000	63412	54258	74109
2001	51658	43639	61150
2002	45496	38842	53290
2003	42293	36122	49518
2004	33754	29156	39077

Year	Catch	Low	High
2005	29795	25753	34472
2006	33059	28326	38584
2007	26884	23107	31278
2008	27137	23411	31455
2009	27429	23656	31804
2010	32794	28212	38122
2011	29373	25301	34099
2012	30010	25871	34810
2013	27765	23902	32252
2014	29027	25097	33573
2015	32872	28324	38151
2016	31781	27339	36944
2017	29674	25406	34657
2018	29117	24986	33930
2019	30435	26053	35554
2020	33658	28927	39162
2021	32802	28212	38138
2022	29343	24979	34470
2023	31502	26714	37149
2024	30717	25664	36765

Table 22.21. Whiting in Subarea 4 and Division 7.d. SAM model parameters, model output.

	par	sd(par)	exp(par)	Low	High
logFpar_0	-13.216	0.077	0	0	0
logFpar_1	-12.132	0.077	0	0	0
logFpar_2	-12.069	0.078	0	0	0
logFpar_3	-12.265	0.083	0	0	0
logFpar_4	-13.176	0.097	0	0	0
logFpar_5	-12.31	0.097	0	0	0
logFpar_6	-12.12	0.098	0	0	0

	par	sd(par)	exp(par)	Low	High
logFpar_7	-12.287	0.102	0	0	0
logFpar_8	-12.353	0.112	0	0	0
logSdLogFsta_0	-1.575	0.13	0.207	0.16	0.268
logSdLogN_0	-0.971	0.129	0.379	0.293	0.49
logSdLogN_1	-2.836	0.496	0.059	0.022	0.158
logSdLogObs_0	0.242	0.118	1.274	1.006	1.614
logSdLogObs_1	-1.57	0.091	0.208	0.174	0.249
logSdLogObs_2	-0.88	0.075	0.415	0.357	0.482
logSdLogObs_3	-0.75	0.08	0.472	0.403	0.554
logSdLogObs_4	-1.132	0.17	0.322	0.23	0.453
transfIRARdist_0	-0.23	0.229	0.794	0.503	1.255
transfIRARdist_1	0.879	0.425	2.409	1.029	5.639
transfIRARdist_2	-1.001	0.272	0.367	0.213	0.633
itrans_rho_0	1.139	0.19	3.123	2.135	4.568

Table 22.22. Whiting in Subarea 4 and Division 7.d. Mohn’s rho.

Mohn’s rho	
R(age 0)	-0.143
SSB	-0.083
Fbar(2-5)	-0.027

Table 22.23. Whiting in Subarea 4 and Division 7.d. Reference points as determined in during WGNSSK 2024 (ICES, 2024b).

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY B _{trigger}	167 419	B _{pa} ; in tonnes.	ICES (2024b)
	F _{MSY}	0.68	Stochastic simulation (EqSim) with segmented regression with a fixed breakpoint at B _{lim} based on recruitment period 1983–2023. Capped at F _{pa} .	ICES (2024b)
Precautionary approach	B _{lim}	119 585	B _{loss} (SSB in 2007, as estimated in the 2024 assessment); in tonnes.	ICES (2024b)
	B _{pa}	167 419	B _{lim} × 1.4; in tonnes.	ICES (2024b)
	F _{lim}	1.18	The F that on average leads to B _{lim} from EqSim.	ICES (2024b)

Framework	Reference point	Value	Technical basis	Source
	F_{pa}	0.68	The F that provides a 95% probability for SSB to be above B_{lim} ($F_{P,0.5}$ with advice rule).	ICES (2024b)
EU Management Plan*	MAP MSY $B_{trigger}$	167 419	MSY $B_{trigger}$, in tonnes.	ICES (2024b)
	MAP B_{lim}	119 585	B_{lim} , in tonnes.	ICES (2024b)
	MAP F_{MSY}	0.68	F_{MSY}	ICES (2024b)
	MAP range F_{lower}	0.46–0.68	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY.	ICES (2024b)
	MAP range F_{upper}	0.68–0.68	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with MSY. Capped at F_{pa} .	ICES (2024b)

* EU multiannual plan (MAP) for the North Sea (EU, 2018).

Table 22.24. Whiting in Subarea 4 and Division 7.d. Recruitment assumption (in millions) as used in the short-term forecast.

Year	Geometric mean of recruitment Time-series 2003–2024
2025	10 718
2026	10 718
2027	10 718

Table 22.25. Whiting in Subarea 4 and Division 7.d. Short-term forecast inputs. Forecasted SSB in the intermediate year used average maturities and stock weights at age (2022–2024), age 6 is a plusgroup.

MFDP version 1a						
MFDP version 1a						
Run: run1						
Time and date: 16:06 16/04/2025						
Fbar age range (Total) : 2-5						
Fbar age range Fleet 1 : 2-5						
2025						
Age	N	M	Mat	PF	PM	SWt
0	10717930	1.6185	0	0	0	0.012353
1	3204235	1.0182	0.5074	0	0	0.04216
2	598057	0.6623	0.8204	0	0	0.108021

3	477106	0.5044	0.9413	0	0	0.184213
4	215079	0.481	0.9764	0	0	0.267655
5	198374	0.4525	0.9881	0	0	0.370661
6	157394	0.3352	0.9967	0	0	0.484789
Catch						
Age	Sel	CWt	DSel	DCWt		
0	0	0.05467	0.00093	0.03633		
1	0.00091	0.179	0.00681	0.11733		
2	0.00797	0.246	0.01716	0.18933		
3	0.02271	0.31333	0.04333	0.23433		
4	0.06384	0.39367	0.05848	0.258		
5	0.08461	0.46267	0.04047	0.274		
6	0.09508	0.555	0.02906	0.315		
IBC						
Age	Sel	CWt				
0	0.00003	0.03633				
1	0.00029	0.124				
2	0.00107	0.20733				
3	0.00335	0.262				
4	0.00792	0.33133				
5	0.00907	0.40833				
6	0.01002	0.504				
2026						
Age	N	M	Mat	PF	PM	SWt
0	10717930	1.6185	0	0	0	0.012353
1	.	1.0182	0.5074	0	0	0.04216
2	.	0.6623	0.8204	0	0	0.108021
3	.	0.5044	0.9413	0	0	0.184213
4	.	0.481	0.9764	0	0	0.267655
5	.	0.4525	0.9881	0	0	0.370661

6	.	0.3352	0.9967	0	0	0.484789
Catch						
Age	Sel	CWt	DSel	DCWt		
0	0	0.05467	0.00093	0.03633		
1	0.00091	0.179	0.00681	0.11733		
2	0.00797	0.246	0.01716	0.18933		
3	0.02271	0.31333	0.04333	0.23433		
4	0.06384	0.39367	0.05848	0.258		
5	0.08461	0.46267	0.04047	0.274		
6	0.09508	0.555	0.02906	0.315		
IBC						
Age	Sel	CWt				
0	0.00003	0.03633				
1	0.00029	0.124				
2	0.00107	0.20733				
3	0.00335	0.262				
4	0.00792	0.33133				
5	0.00907	0.40833				
6	0.01002	0.504				
2027						
Age	N	M	Mat	PF	PM	SWt
0	10717930	1.6185	0	0	0	0.012353
1	.	1.0182	0.5074	0	0	0.04216
2	.	0.6623	0.8204	0	0	0.108021
3	.	0.5044	0.9413	0	0	0.184213
4	.	0.481	0.9764	0	0	0.267655
5	.	0.4525	0.9881	0	0	0.370661
6	.	0.3352	0.9967	0	0	0.484789
Catch						
Age	Sel	CWt	DSel	DCWt		

0	0	0.05467	0.00093	0.03633
1	0.00091	0.179	0.00681	0.11733
2	0.00797	0.246	0.01716	0.18933
3	0.02271	0.31333	0.04333	0.23433
4	0.06384	0.39367	0.05848	0.258
5	0.08461	0.46267	0.04047	0.274
6	0.09508	0.555	0.02906	0.315
IBC				
Age	Sel	CWt		
0	0.00003	0.03633		
1	0.00029	0.124		
2	0.00107	0.20733		
Input units are thousands and kg - output in tonnes				

Table 22.26. Whiting in Subarea 4 and Division 7.d. MFDP output table for short-term forecasts.

Basis: F(2025) = average exploitation (2022-2024), scaled to 2024 Fsq = 0.090; SSB (2024) = 358 464 t; Landings 4 (2024) = 13 865 t; Discards (2024) = 20 980 t; IBC (2024) = 1733 t. Recruitment (2025-2027)=10 718 million; TAC 27.4 +7.d (2025) = 139 425 t.

Output units in tonnes

2025																						
Biomass	SSB	Catch			Landings			27.4+27.7d HC catch	27.4 HC catch	27.7d HC catch	Discards		IBC									
		FMult	FBar	Yield	FBar	Yield	FBar				Yield	FMult	FBar	Yield								
627380	409189	1	0.090	34523	0.0448	20305	32418		26009	6409	0.0399	12113	1	0.0054	2105							
2026															2027							
Biomass	SSB	Catch			Landings			27.4+27.7d HC catch	27.4 HC catch	27.7d HC catch	Discards		IBC		Landings			Biomass	SSB	27.4 TAC change	SSB change	
		FMult	FBar	Yield	FBar	Yield	FBar				Yield	FMult	FBar	Yield	FMult	FBar	Yield					
617666	413121	0	0.0054	2242	0.000	0	0		0	0	0.000	0	1.0	0.0054	2242	630527	430525	-98.4%	4.2%	No HC fishery		
.	413121	0.1	0.0139	5624	0.005	2175	3393		2722	671	0.004	1218	1.0	0.0054	2231	627793	427843	-96.0%	3.6%			
.	413121	0.2	0.0224	8974	0.009	4327	6754		5419	1335	0.008	2427	1.0	0.0054	2220	625087	425189	-93.6%	2.9%			

2026										2027									
Catch					Landings					Discards		IBC	Landings						
Biomass	SSB	FMult	FBar	Yield	FBar	Yield	27.4+27.7d HC catch	27.4 HC catch	27.7d HC catch	FBar	Yield	FMult	FBar	Yield	Biomass	SSB	27.4 TAC change	SSB change	
.	413121	0.3	0.0308	12293	0.013	6457	10084		8090	1994	0.012	3627	1.0	0.0054	2209	622408	422563	-91.2%	2.3%
.	413121	0.4	0.0392	15583	0.018	8566	13385		10739	2646	0.016	4819	1.0	0.0054	2198	619758	419964	-88.8%	1.7%
.	413121	0.5	0.0477	18843	0.022	10653	16656		13363	3293	0.020	6003	1.0	0.0054	2187	617134	417392	-86.5%	1.0%
.	413121	0.6	0.0562	22073	0.027	12718	19896		15963	3933	0.024	7178	1.0	0.0054	2177	614538	414846	-84.2%	0.4%
.	413121	0.7	0.0646	25274	0.031	14763	23108		18540	4568	0.028	8345	1.0	0.0054	2166	611968	412327	-81.9%	-0.2%
.	413121	0.8	0.0731	28445	0.036	16787	26290		21092	5198	0.032	9503	1.0	0.0054	2155	609425	409835	-79.6%	-0.8%
.	413121	0.9	0.0816	31589	0.040	18790	29444		23623	5821	0.036	10654	1.0	0.0054	2145	606908	407368	-77.3%	-1.4%
.	413121	1	0.0901	34703	0.045	20773	32569		26130	6439	0.040	11796	1.0	0.0054	2134	604416	404926	-75.1%	-2.0%
.	413121	1.1	0.0985	37790	0.049	22736	35666		28615	7051	0.044	12930	1.0	0.0054	2124	601950	402510	-72.9%	-2.6%
.	413121	1.2	0.1069	40850	0.054	24679	38736		31078	7658	0.048	14057	1.0	0.0054	2114	599509	400119	-70.7%	-3.1%

2026										2027									
		Catch			Landings						Discards		IBC	Landings					
Biomass	SSB	FMult	FBar	Yield	FBar	Yield	27.4+27.7d HC catch	27.4 HC catch	27.7d HC catch	FBar	Yield	FMult	FBar	Yield	Biomass	SSB	27.4 TAC change	SSB change	
.	413121	1.3	0.1154	43883	0.058	26603	41779	33519	8260	0.052	15176	1.0	0.0054	2104	597093	397753	-68.5%	-3.7%	
.	413121	1.4	0.1239	46888	0.063	28507	44794	35938	8856	0.056	16287	1.0	0.0054	2094	594702	395410	-66.4%	-4.3%	
.	413121	1.5	0.1324	49866	0.067	30392	47782	38335	9447	0.060	17390	1.0	0.0054	2084	592335	393092	-64.2%	-4.8%	
.	413121	1.6	0.1409	52818	0.072	32258	50744	40712	10032	0.064	18486	1.0	0.0054	2074	589992	390798	-62.1%	-5.4%	
.	413121	1.7	0.1493	55743	0.076	34105	53679	43067	10612	0.068	19574	1.0	0.0054	2064	587673	388527	-60.0%	-6.0%	
.	413121	1.8	0.1577	58643	0.081	35934	56589	45401	11188	0.072	20655	1.0	0.0054	2054	585377	386280	-57.9%	-6.5%	
.	413121	1.9	0.1662	61516	0.085	37744	59472	47714	11758	0.076	21728	1.0	0.0054	2044	583104	384055	-55.9%	-7.0%	
.	413121	2	0.1747	64364	0.090	39536	62330	50007	12323	0.080	22794	1.0	0.0054	2034	580855	381853	-53.8%	-7.6%	
617666	413121	0.75	0.069	25875	0.034	15364	23721	19032	4690	0.030	8358	1.0	0.0054	2153	607900	408306	-81%	-1.2%	0.75 * Fsq

2026										2027										
Catch				Landings				Discards				IBC	Landings							
Biomass	SSB	FMult	FBar	Yield	FBar	Yield	27.4+27.7d HC catch	27.4 HC catch	27.7d HC catch	FBar	Yield	FMult	FBar	Yield	Biomass	SSB	27.4 TAC change	SSB change		
.	413121	1.25	0.111	40962	0.056	25082	38856		31174	7682	0.050	13774	1.0	0.0054	2106	597307	397949	-71%	-3.7%	1.25 * Fsq
.	413121	5.06	0.434	139706	0.226	86762	137923		110656	27267	0.202	51161	1.0	0.0054	1782	526338	328666	0%	-20.4%	Rollover TAC
.	413121	1.00	0.090	33503	0.045	20286	31374		25171	6203	0.040	11088	1.0	0.0054	2130	602552	403077	-76%	-2.4%	Fsq
.	413121	7.97	0.680	198609	0.357	121538	197033		158080	38953	0.318	75495	1.0	0.0054	1576	482707	286198	42%	-30.7%	Fpa
.	413121	7.97	0.680	198609	0.357	121538	197033		158080	38953	0.318	75495	1.0	0.0054	1576	482707	286198	42%	-30.7%	Fmsy
.	413121	5.37	0.460	146669	0.241	90964	144910		116262	28649	0.214	53946	1.0	0.0054	1759	521226	323685	5%	-21.6%	Fmsylower
.	413121	7.97	0.680	198609	0.357	121538	197033		158080	38953	0.318	75495	1.0	0.0054	1576	482707	286198	42%	-30.7%	Fmsyupper
.	413121	26.59	2.256	385494	1.191	215820	384628		308587	76041	1.060	168808	1.0	0.0054	866	355541	167419	176%	-59.5%	Bpa, MSY Btrigger
.	413121	49.50	4.195	485115	2.217	244498	484622		388813	95810	1.973	240124	1.0	0.0054	492	282941	119585	248%	-71.1%	Blim

Table 22.27. Whiting in Subarea 4 and Division 7.d. NS IBTS tuning series for northern component used in the area-specific SURBAR analysis.

Year/Age	Q1 North					Q3 North					
	1	2	3	4	5	0	1	2	3	4	5
1983	143.401	154.856	150.829	113.598	50.897						
1984	323.567	212.552	106.415	41.278	40.292						
1985	412.895	341.159	81.823	23.344	11.227						
1986	587.697	385.153	239.606	39.83	12.625						
1987	707.64	788.303	122.369	57.297	8.179						
1988	301.643	1115.424	435.943	44.031	23.551						
1989	2049.504	668.536	580.893	160.983	20.942						
1990	490.822	1251.354	261.582	138.013	29.097						
1991	754.334	999.549	477.884	76.369	31.452	190.132	285.241	124.822	88.607	26.92	13.102
1992	1384.302	545.011	317.356	90.528	78.729	1357.232	615.218	191.926	84.976	65.436	33.848
1993	1529.746	810.122	269.711	122.998	52.18	339.611	578.148	248.966	55.832	30.695	21.417
1994	1058.43	853.101	299.173	105.475	20.999	237.937	712.663	324.467	57.501	16.051	11.43
1995	894.427	651.711	308.658	95.983	19.891	330.847	810.471	360.665	101.783	28.238	12.829
1996	603.663	651.987	314.636	96.581	45.633	83.743	444.379	388.123	165.359	48.308	13.145
1997	445.667	378.412	240.241	117.637	32.536	2750.385	330.418	225.354	161.952	35.658	29.341
1998	744.221	222.632	173.569	73.104	32.244	2484.246	405.455	197.391	75.867	44.141	17.651
1999	858.032	335.233	193.737	96.323	41.596	1723.648	810.794	242.511	74.55	33.258	15.492
2000	1127.728	652.372	272.851	45.871	27.249	1456.711	767.782	342.896	73.195	20.076	11.358
2001	413.843	588.073	343.71	77.607	29.033	291.479	642.804	296.602	111.774	25.051	9.898
2002	513.057	428.163	386.74	72.702	17.767	105.617	603.626	300.637	173.636	46.367	10.344
2003	156.456	311.894	344.993	184.118	64.629	413.41	245.277	326.312	166.634	88.931	24.592
2004	270.146	130.282	237.838	116.137	65.129	211.061	190.845	76.868	90.696	63.2	36.431
2005	160.63	70.445	71.669	61.544	43.237	154.069	195.852	97.403	45.119	64.845	47.659
2006	261.558	86.555	64.824	30.563	22.823	44.878	190.902	104.718	40.801	34.285	27.364
2007	62.938	202.914	93.486	31.871	16.757	346.981	74.776	78.557	48.2	22.754	12.043

Q1 North						Q3 North					
Year/Age	1	2	3	4	5	0	1	2	3	4	5
2008	198.753	195.499	78.913	27.568	14.458	848.142	334.74	72.776	39.989	18.66	9.79
2009	156.742	239.482	72.965	20.13	20.976	560.618	257.218	134.847	32.409	13.392	10.651
2010	302.33	269.377	239.438	76.001	110.69	70.104	248.174	175.906	57.992	16.82	16.516
2011	185.922	504.592	198.931	105.466	28.249	94.343	411.617	163.839	65.764	23.956	11.099
2012	266.626	796.159	145.62	58.537	44.488	316.803	238.565	268.773	84.896	30.912	21.17
2013	59.098	212.457	350.904	98.115	52.337	141.998	58.759	57.269	79.205	26.334	9.801
2014	367.829	274.711	147.237	91.846	51.213	2017.069	202.053	73.682	48.725	42.318	13.446
2015	423.217	250.756	67.447	34.917	33.132	2113.574	244.567	195.931	55.372	37.056	25.098
2016	263.992	199.177	97.841	31.325	18.422	729.877	318.709	194.394	72.089	26.372	11.006
2017	455.449	241.933	136.348	43.761	15.935	148.347	633.78	210.029	107.555	34.8	16.409
2018	84.998	236.167	92.087	52.645	20.466	204.112	147.061	258.238	97.385	39.992	27.824
2019	268.933	201.402	156.042	63.584	19.824	749.566	375.037	145.446	99.861	28.428	20.008
2020	473.535	186.89	100.121	70.15	21.446	654.4	1011.846	188.023	69.895	43.277	30.275
2021	483.497	830.364	143.204	53.927	21.881	48.177	538.947	416.742	73.307	28.946	17.553
2022	199.455	815.503	322.694	55.209	18.713	45.357	194.319	377.394	273.392	75.482	23.002
2023	85.576	129.587	270.776	156.587	38.762	59.164	674.879	318.557	468.036	271.126	35.897
2024	16.484	124.755	256.813	186.926	46.502	24.318	208.789	265.905	151.671	195.08	95.032
2025	122.093	49.555	108.058	77.862	72.016						

Table 22.28. Whiting in Subarea 4 and Division 7.d: NS IBTS tuning series for southern component used in the area-specific SURBAR analysis.

Q1 South						Q3 South					
Year/Age	1	2	3	4	5	0	1	2	3	4	5
1983	85.450	99.851	52.686	19.987	5.019						
1984	593.881	84.243	43.152	4.049	2.825						
1985	114.689	330.400	30.889	11.822	3.018						
1986	155.459	93.190	215.536	54.700	7.664						
1987	542.592	86.810	27.029	26.761	3.098						

Q1 South						Q3 South					
Year/Age	1	2	3	4	5	0	1	2	3	4	5
1988	487.545	262.104	50.705	6.855	6.541						
1989	291.589	229.438	71.118	4.646	11.552						
1990	470.323	118.887	87.744	32.480	4.558						
1991	1106.472	287.446	151.874	66.871	37.686	958.688	1334.419	170.203	64.644	31.132	22.847
1992	265.104	258.351	117.670	56.676	27.940	1200.775	406.283	311.477	40.846	30.723	26.147
1993	140.264	59.430	62.389	31.774	23.154	1626.475	671.101	63.728	21.692	15.256	9.817
1994	191.711	156.048	25.782	8.463	4.159	951.750	640.529	84.975	43.115	25.091	10.09
1995	222.579	239.969	49.752	19.783	6.470	1219.269	222.51	80.845	7.972	6.656	1.232
1996	231.472	233.724	70.389	33.571	37.795	499.52	417.706	205.879	47.99	11.737	6.928
1997	67.325	43.278	13.870	22.699	10.577	480.258	227.918	35.787	32.328	8.812	2.345
1998	95.505	56.861	23.986	6.323	8.272	2229.932	238.089	36.015	15.326	9.628	3.981
1999	153.527	147.624	127.128	30.833	6.278	2794.07	1724.311	49.323	13.413	4.241	0.809
2000	219.275	151.941	55.605	10.679	3.761	2456.096	1090.356	226.153	30.001	12.365	2.95
2001	942.456	448.546	84.966	70.175	31.13	8867.757	697.026	218.85	36.408	18.91	5.883
2002	457.447	120.386	34.448	13.216	7.754	385.891	989.146	113.49	32.153	12.349	3.461
2003	96.052	216.304	81.629	29.913	8.828	227.231	288.794	171.351	28.265	26.959	8.576
2004	38.818	53.641	34.870	14.430	10.014	1641.775	81.054	65.172	14.855	5.381	3.609
2005	89.895	67.155	22.92	11.112	9.571	208.437	54.154	4.017	2.917	2.161	1.504
2006	48.506	67.392	25.404	10.769	8.899	443.497	74.551	15.069	4.141	3.422	2.752
2007	77.838	58.664	12.349	5.486	3.344	2203.686	142.166	20.520	6.177	1.968	0.942
2008	427.504	247.607	26.007	4.196	2.120	546.391	596.203	54.246	16.160	4.215	0.806
2009	438.147	459.551	74.428	18.350	15.819	634.897	1044.568	664.476	76.080	11.132	6.005
2010	508.82	81.019	64.927	17.960	9.475	914.230	154.524	49.117	12.785	3.941	3.783
2011	465.753	207.833	44.203	12.609	5.268	511.566	444.079	87.814	51.98	10.342	2.203
2012	244.074	196.178	21.112	13.571	10.862	208.426	295.544	101.813	22.997	3.231	1.612
2013	137.181	93.381	52.843	10.687	10.847	772.182	100.621	55.296	26.365	5.548	1.584
2014	1129.913	147.201	35.603	17.160	13.996	1884.952	283.798	169.738	124.258	70.136	15.764

Q1 South						Q3 South					
Year/Age	1	2	3	4	5	0	1	2	3	4	5
2015	340.564	393.710	134.634	21.941	19.974	1622.776	462.836	309.691	79.912	13.378	5.747
2016	633.544	643.699	111.985	27.244	15.101	1245.384	208.678	157.555	55.207	9.166	6.349
2017	989.077	266.910	52.213	10.761	6.419	229.522	1442.214	199.056	49.837	12.495	3.198
2018	185.133	192.633	47.576	21.585	11.409	111.591	391.478	376.988	65.935	19.927	9.468
2019	152.457	74.143	38.974	21.925	3.684	2247.084	335.335	87.211	68.268	12.984	5.108
2020	531.832	171.634	32.164	24.288	10.189	2381.931	1825.626	248.69	51.606	47.583	50.993
2021	817.832	443.208	66.728	13.645	5.105	1643.416	2308.425	517.862	80.788	17.508	7.127
2022	1139.12	547.176	104.463	25.711	12.994	5821.460	933.007	300.835	160.140	55.881	11.422
2023	587.428	1177.124	255.787	150.679	17.017	2100.358	1405.675	259.491	57.295	128.207	18.437
2024	244.043	1969.023	545.237	211.41	66.725	3820.796	1042.855	343.835	54.64	83.145	72.798
2025	2595.934	616.08	362.181	81.093	110.451						

Table 22.29. Whiting in Subarea 4 and Division 7.d. Maturity estimates for northern component used in the area-specific SURBAR analysis. Prior to 1991 values of 1991 were used.

Year/Age	0	1	2	3	4	5	6+
1991	0	0.121	0.861	0.996	1	1	1
1992	0	0.126	0.856	0.993	1	1	1
1993	0	0.132	0.85	0.991	1	1	1
1994	0	0.138	0.842	0.988	0.999	1	1
1995	0	0.146	0.834	0.985	0.998	0.999	1
1996	0	0.155	0.825	0.981	0.997	0.999	1
1997	0	0.165	0.816	0.977	0.997	0.999	1
1998	0	0.177	0.806	0.973	0.996	0.999	1
1999	0	0.19	0.796	0.97	0.995	0.999	1
2000	0	0.204	0.788	0.967	0.995	0.999	1
2001	0	0.218	0.784	0.965	0.995	0.999	1
2002	0	0.23	0.784	0.966	0.995	0.999	1
2003	0	0.24	0.789	0.968	0.996	0.999	1
2004	0	0.248	0.796	0.97	0.996	0.999	1
2005	0	0.253	0.806	0.974	0.997	1	1
2006	0	0.256	0.817	0.978	0.998	1	1
2007	0	0.257	0.829	0.982	0.999	1	1
2008	0	0.256	0.84	0.986	1	1	1
2009	0	0.255	0.85	0.989	1	1	1
2010	0	0.254	0.857	0.992	1	1	1
2011	0	0.254	0.862	0.994	1	1	1
2012	0	0.255	0.865	0.995	1	1	1
2013	0	0.256	0.865	0.996	1	1	1
2014	0	0.26	0.864	0.995	1	1	1
2015	0	0.266	0.862	0.994	1	1	1
2016	0	0.276	0.858	0.991	1	1	1
2017	0	0.289	0.853	0.987	1	1	1
2018	0	0.307	0.846	0.98	0.997	0.998	0.999

Year/Age	0	1	2	3	4	5	6+
2019	0	0.328	0.837	0.971	0.992	0.995	0.998
2020	0	0.354	0.826	0.96	0.985	0.991	0.996
2021	0	0.385	0.812	0.945	0.976	0.985	0.994
2022	0	0.418	0.797	0.928	0.964	0.978	0.991
2023	0	0.454	0.779	0.908	0.951	0.97	0.988
2024	0	0.49	0.76	0.885	0.936	0.961	0.984

Table 22.30. Whiting in Subarea 4 and Division 7.d. Maturity estimates for southern component used in the area-specific SURBAR analysis. Prior to 1991, values of 1991 were used.

Year/Age	0	1	2	3	4	5	6+
1991	0	0.185	0.925	1	1	1	1
1992	0	0.185	0.884	1	1	1	1
1993	0	0.183	0.845	0.991	1	1	1
1994	0	0.180	0.809	0.977	0.998	1	1
1995	0	0.174	0.773	0.959	0.991	0.998	0.999
1996	0	0.167	0.736	0.936	0.981	0.993	0.997
1997	0	0.165	0.703	0.909	0.967	0.987	0.994
1998	0	0.173	0.677	0.880	0.950	0.979	0.99
1999	0	0.191	0.651	0.849	0.931	0.969	0.985
2000	0	0.217	0.621	0.819	0.914	0.961	0.981
2001	0	0.246	0.600	0.800	0.904	0.956	0.979
2002	0	0.279	0.594	0.798	0.904	0.957	0.979
2003	0	0.314	0.604	0.807	0.911	0.960	0.981
2004	0	0.348	0.625	0.823	0.921	0.965	0.983
2005	0	0.379	0.654	0.845	0.934	0.972	0.987
2006	0	0.411	0.690	0.870	0.948	0.979	0.990
2007	0	0.443	0.731	0.896	0.961	0.986	0.994
2008	0	0.472	0.772	0.920	0.973	0.991	0.997
2009	0	0.493	0.806	0.938	0.981	0.995	0.999
2010	0	0.504	0.830	0.951	0.986	0.997	1
2011	0	0.51	0.844	0.958	0.989	0.998	1

Year/Age	0	1	2	3	4	5	6+
2012	0	0.513	0.848	0.960	0.989	0.998	1
2013	0	0.514	0.847	0.959	0.989	0.998	1
2014	0	0.513	0.845	0.959	0.988	0.998	1
2015	0	0.510	0.848	0.961	0.989	0.998	1
2016	0	0.505	0.851	0.963	0.99	0.998	1
2017	0	0.498	0.853	0.966	0.991	0.998	1
2018	0	0.493	0.853	0.967	0.992	0.998	1
2019	0	0.491	0.852	0.968	0.993	0.998	1
2020	0	0.495	0.854	0.970	0.994	0.998	1
2021	0	0.501	0.858	0.972	0.995	0.999	1
2022	0	0.508	0.861	0.973	0.996	0.999	1
2023	0	0.516	0.863	0.973	0.997	1	1
2024	0	0.527	0.865	0.973	0.997	1	1

Table 22.31. Whiting in Subarea 4 and Division 7.d. Ratio of current (WGNSSK 2025) over previous (WGNSSK 2024) numbers-at-age from stock assessment and forecast results.

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024*	2025**	2026***
0	0.98	0.97	0.98	1.02	1.12	1.19	1.08	1.42	1.57	1.05	1.05
1	0.99	0.98	0.97	0.99	1.03	1.13	1.19	1.07	1.41	1.61	1.07
2	1.00	0.98	0.98	0.97	0.98	1.05	1.13	1.19	1.05	1.44	1.62
3	1.00	0.99	0.98	0.97	0.97	0.98	1.06	1.15	1.19	1.04	1.44
4	1.00	1.00	0.98	0.97	0.96	0.96	0.97	1.08	1.18	1.23	1.05
5	0.99	1.00	0.99	0.97	0.96	0.94	0.94	0.95	1.09	1.15	1.21
6+	0.99	0.98	0.98	0.98	0.96	0.94	0.92	0.90	0.89	1.00	1.05

* 2024 was the intermediate year at WGNSSK 2024.

** 2025 was the advice year at WGNSSK 2024 and is the intermediate year at WGNSSK 2025.

*** 2026 is the advice year at WGNSSK 2025.

Table 22.32. Whiting in Subarea 4 and Division 7.d. Ratio of current (WGNSSK 2025) over previous (WGNSSK 2024) bio-mass-at-age from stock assessment and forecast results.

Age	2016	2017	2018	2019	2020	2021	2022	2023	2024*	2025**	2026***
0	1.00	0.99	1.01	1.05	1.15	1.25	0.81	1.46	1.80	0.99	0.99
1	1.01	1.01	0.99	1.01	1.06	1.09	1.04	1.08	1.49	1.59	1.05
2	1.01	0.99	0.99	0.98	1.00	1.07	1.11	1.16	0.99	1.44	1.62
3	1.01	0.99	0.98	0.98	0.97	0.99	1.02	1.12	1.05	0.94	1.31
4	1.00	1.00	0.98	0.97	0.96	0.99	0.96	1.05	0.95	1.11	0.94
5	0.99	1.00	0.99	0.97	0.96	0.97	0.94	0.92	0.76	1.02	1.07
6+	0.98	0.97	0.97	0.97	0.95	0.98	0.89	0.86	0.84	1.00	1.05

* 2024 was the intermediate year at WGNSSK 2024.

** 2025 was the advice year at WGNSSK 2023 and is the intermediate year at WGNSSK 2025.

*** 2026 is the advice year at WGNSSK 2025.

Table 22.33. Whiting in Subarea 4 and Division 7.d. Summary of forecast assumptions from previous assessment (WGNSSK 2024) and current assessment (WGNSSK 2025). '2024' is the intermediate year in previous assessment, '2025' is the advice year in the previous assessment.

	Year	Previous assessment (2024)	Current assessment (2025)
Assumed recruitment	2024	10 241 millions	16 066 millions
	2025	10 241 millions	10 718 millions
Model-estimated Catch	2024	32 578 t	30 717 t
SSB	2024	362 538 t	358 464 t
	2025	364 877 t	409 189 t
F(2-5)	2024	0.083	0.090
Target F for TAC	2025/2026	0.68	0.68

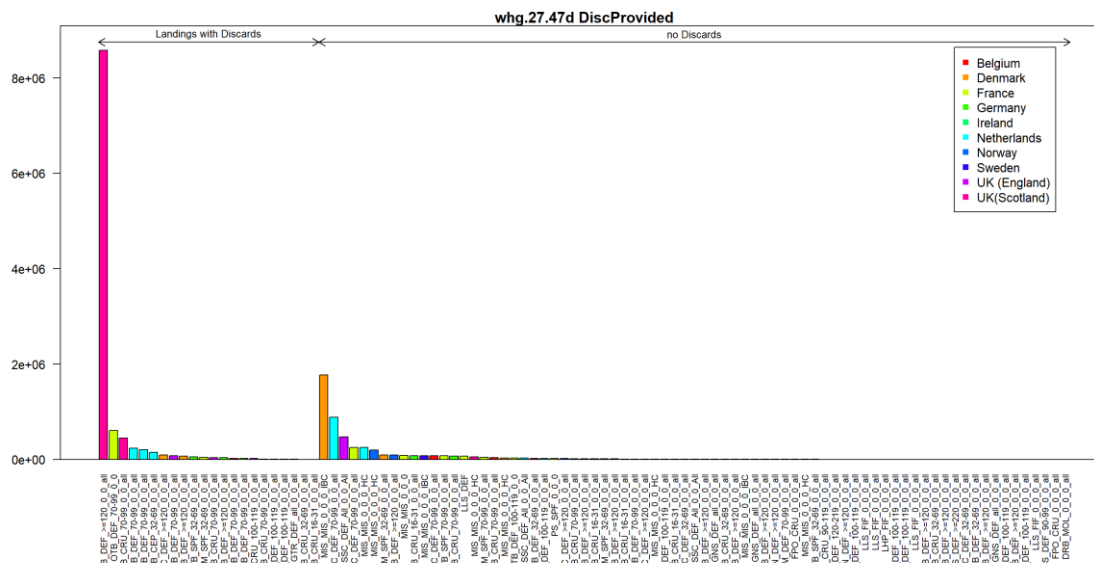


Figure 22.1. Whiting in Subarea 4 and Division 7.d. Landings with provided discards. Métier with industrial bycatch landings (MIS_MIS_0_0_0_IBC, Denmark, orange) generally does not have associated discards.

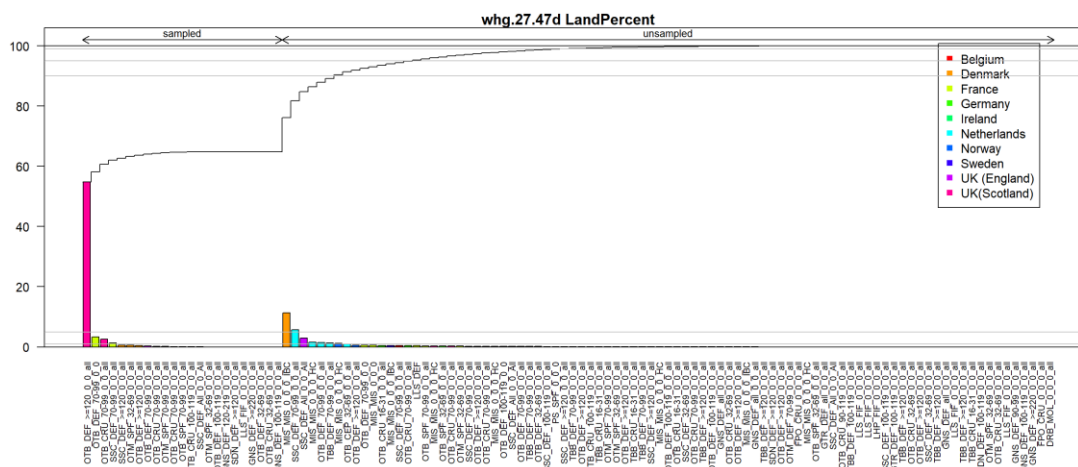


Figure 22.2a. Whiting in Subarea 4 and Division 7.d. Reported landings (in percent, coloured bars) for each sampled and unsampled fleet, along with cumulative landings (in percent, black line) for fleets in descending order of yield.

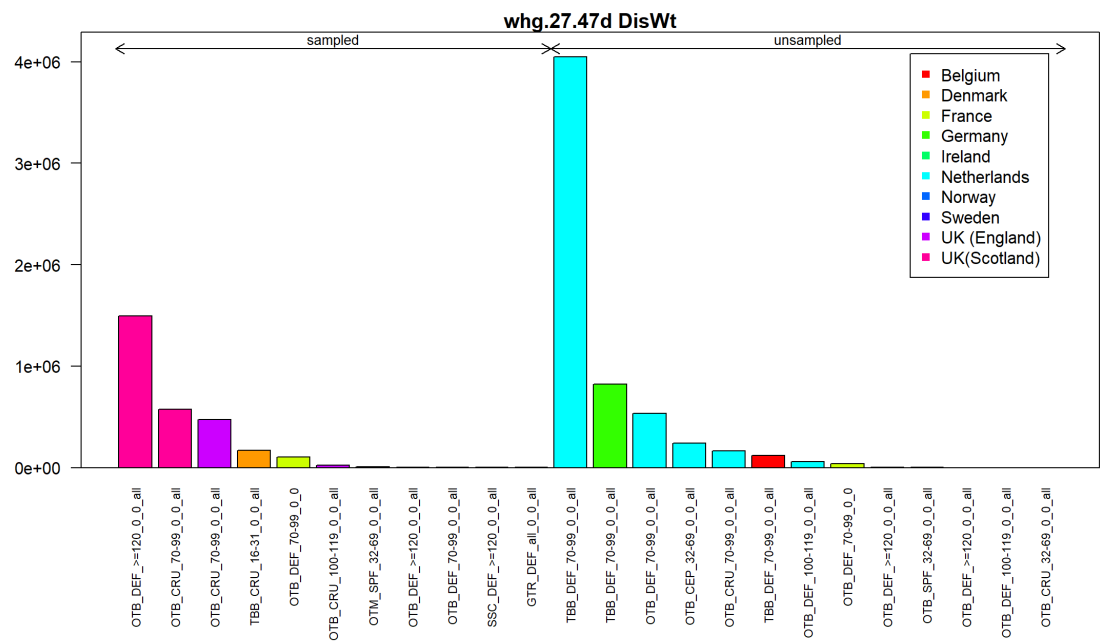


Figure 22.2b. Whiting in Subarea 4 and Division 7.d. Reported discards (in tonnes, coloured bars) for each sampled and unsampled fleet, in descending order of yield.

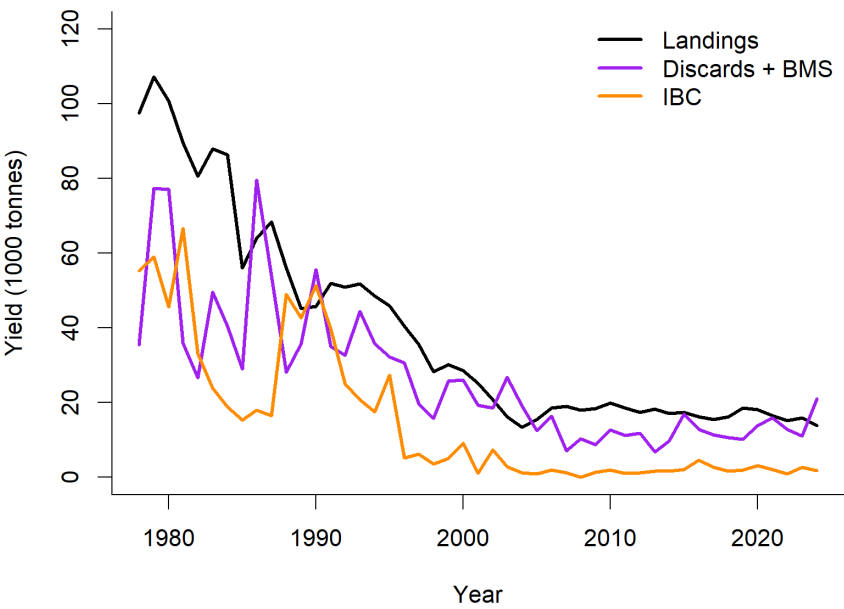


Figure 22.3. Whiting in Subarea 4 and Division 7.d. Yield by catch component as estimated by ICES.

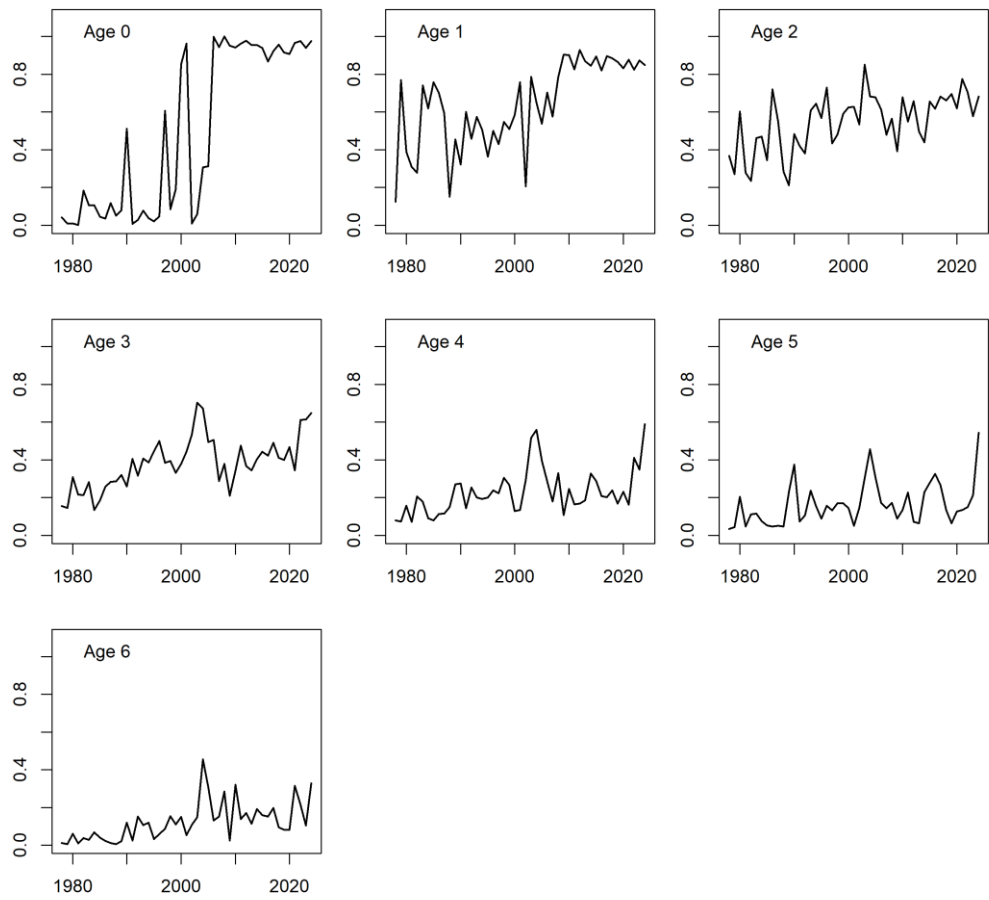


Figure 22.4. Whiting in Subarea 4 and Division 7.d. Proportion of discards in total catch, by age and year.

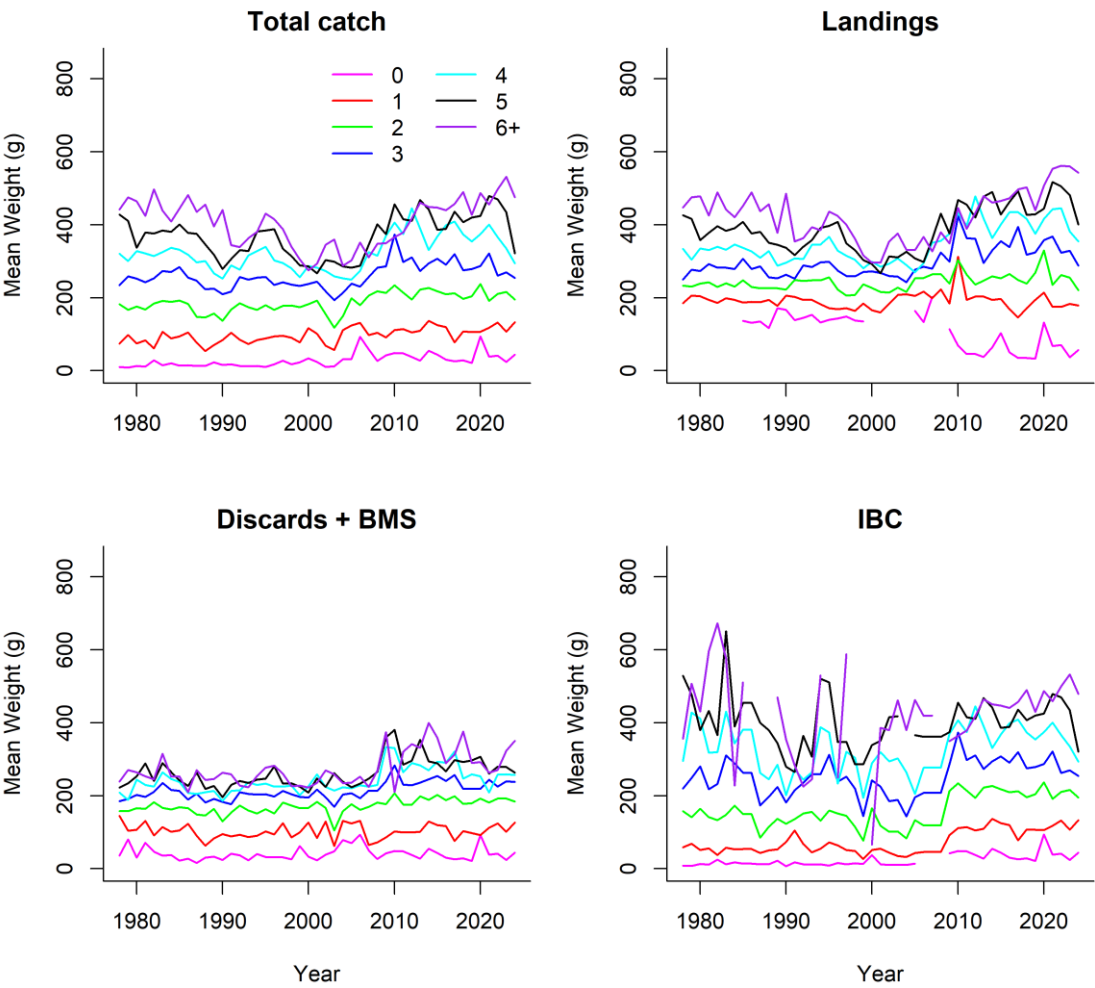


Figure 22.5. Whiting in Subarea 4 and Division 7.d. Mean weights-at-age (g) by catch component (age 0–6+).

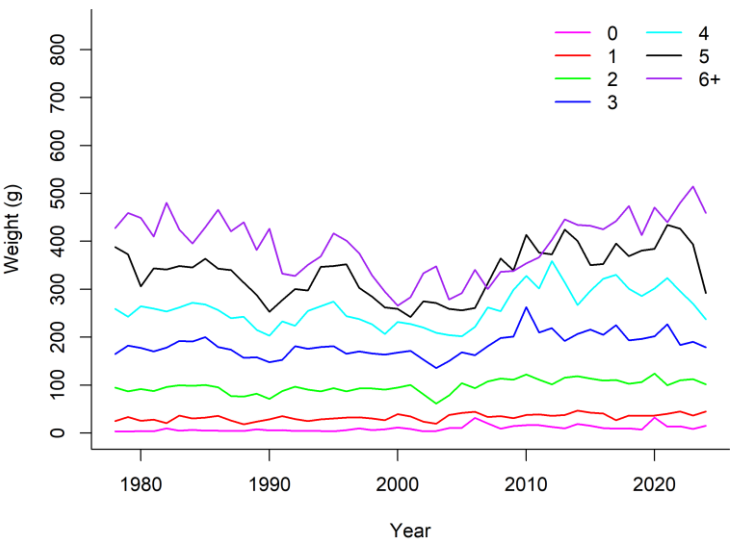


Figure 22.6. Whiting in Subarea 4 and Division 7.d. Stock mean weights-at-age (g) (age 0–6+).

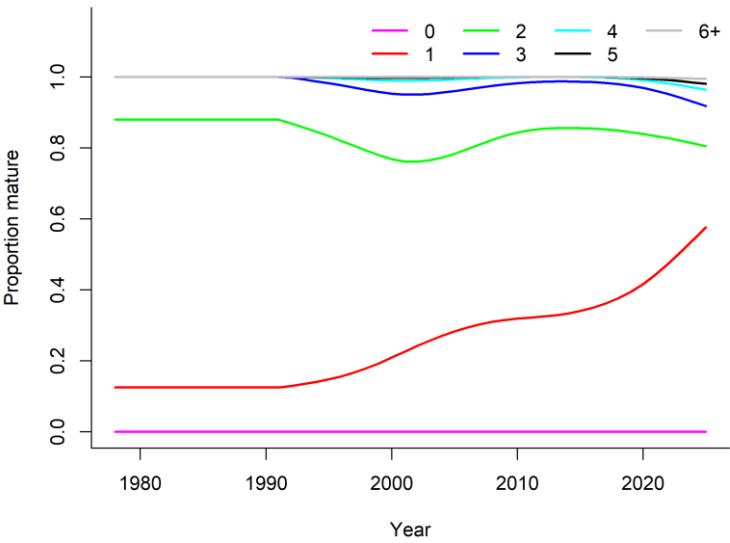


Figure 22.7. Whiting in Subarea 4 and Division 7.d. Smoothed maturity estimates from NS IBTS Q1 data as used in the assessment. Ages 6 and older have the same maturity values. Estimates prior 1991 are assumed constant using values of 1991.

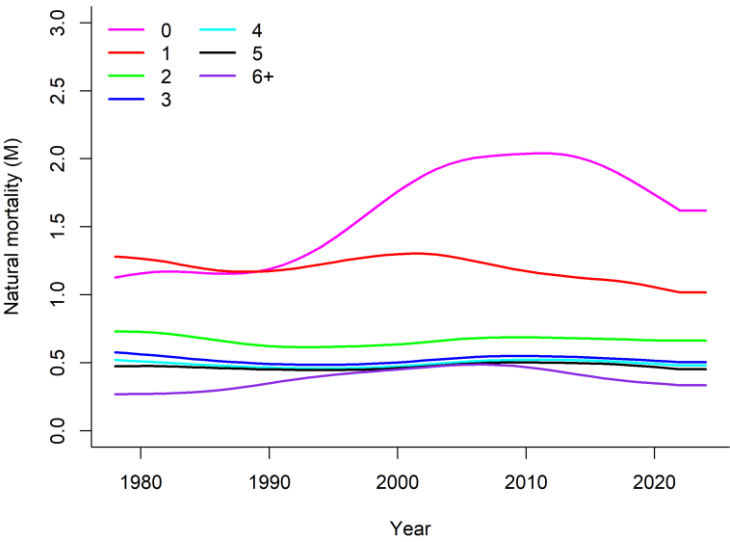


Figure 22.8. Whiting in Subarea 4 and Division 7.d. Smoothed natural mortality estimates using the 2023 update of the North Sea SMS key run (WGSAM, 2024a) used in the assessment.



Figure 22.9. Whiting in Subarea 4 and Division 7.d. Survey distribution maps for Ages 1–3+ Q1 2021–2025. Size of the bubbles indicates numbers caught per 30 minutes for each age (on a log10 scale). The maps are based on the IBTS–Q1 survey in the North Sea.



Figure 22.10. Whiting in Subarea 4 and Division 7.d. Survey distribution maps for ages 0–3+ Q3 2021–2024. Size of the bubbles indicates numbers caught per 30 minutes for each age (on a log10 scale). The maps are based on the IBTS–Q3 survey in the North Sea.

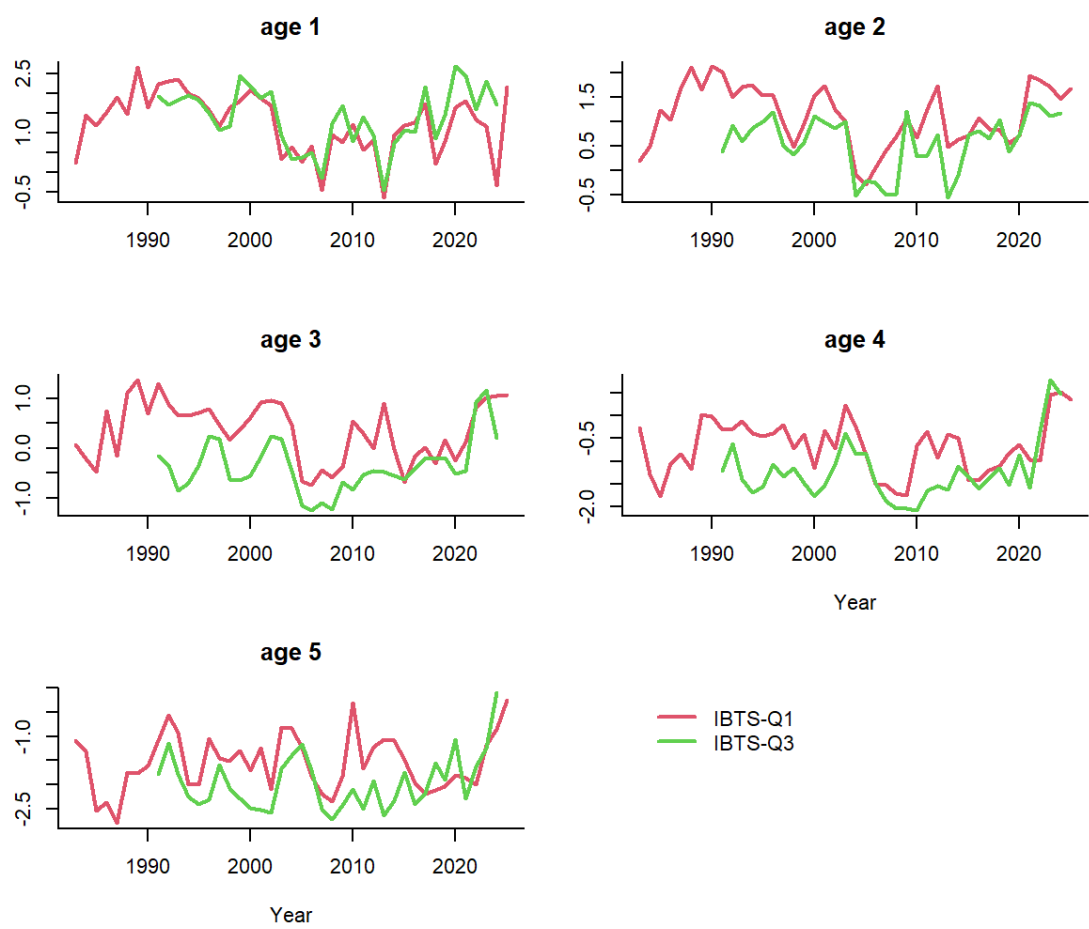


Figure 22.11. Whiting in Subarea 4 and Division 7.d. Survey log CPUE (catch per unit effort) at age.

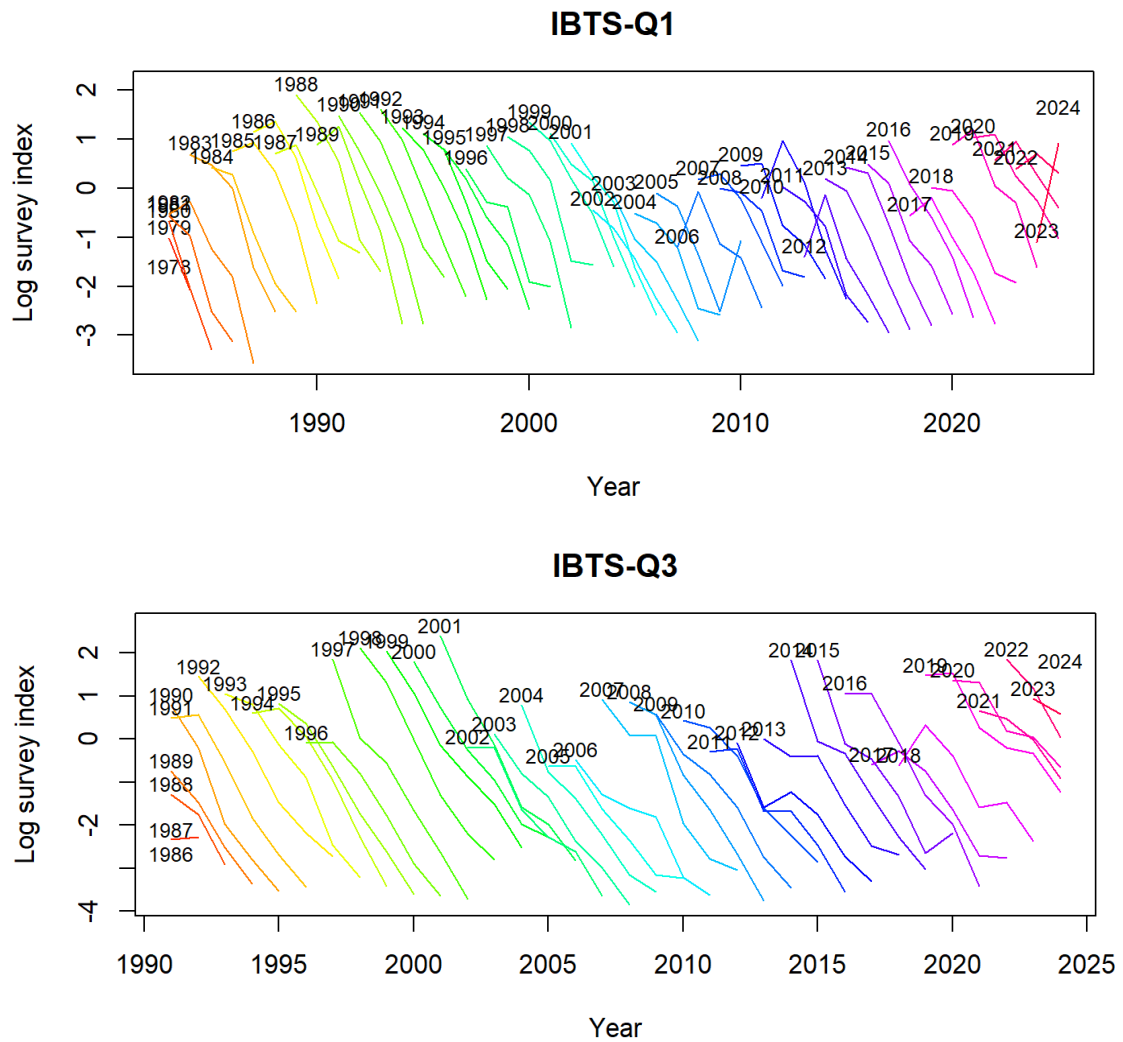


Figure 22.12. Whiting in Subarea 4 and Division 7.d. Log survey indices by cohort for each of the two surveys. The spawning year for each cohort is indicated at the start of each line.

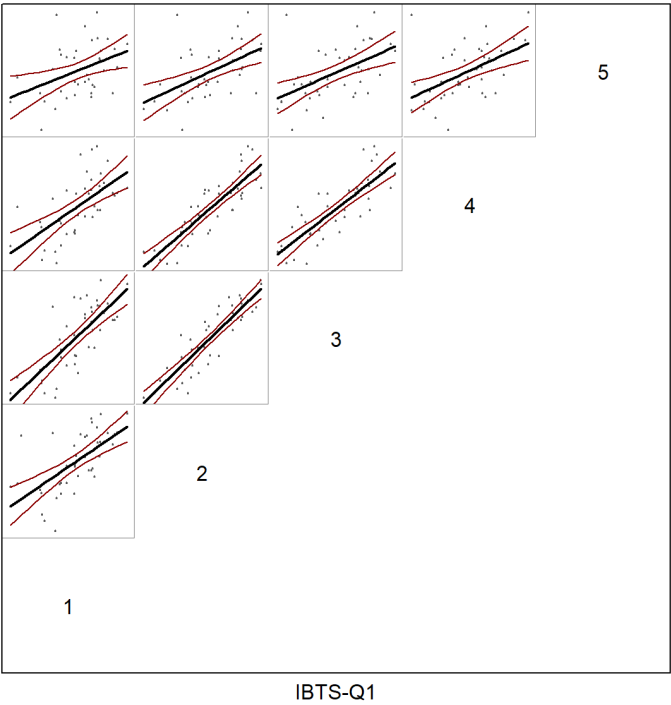


Figure 22.13. Whiting in Subarea 4 and Division 7.d. Within-survey correlations for the IBTS–Q1 survey series, comparing index values at different ages for the same year-classes (cohorts). In each plot, the straight line is a normal linear model fit: a thick line (with black points) represents a significant ($p < 0.05$) regression, while a thin line (with blue points) is not significant. Approximate 95% confidence intervals for each fit are also shown.

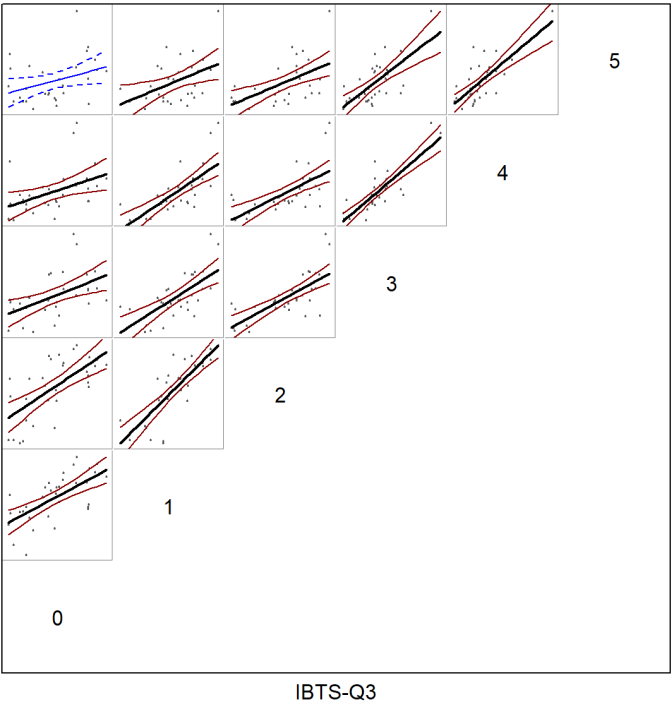


Figure 22.14. Whiting in Subarea 4 and Division 7.d. Within-survey correlations for the IBTS–Q3 survey series, comparing index values at different ages for the same year-classes (cohorts). In each plot, the straight line is a normal linear model fit: a thick line (with black points) represents a significant ($p < 0.05$) regression, while a thin line (with blue points) is not significant. Approximate 95% confidence intervals for each fit are also shown.

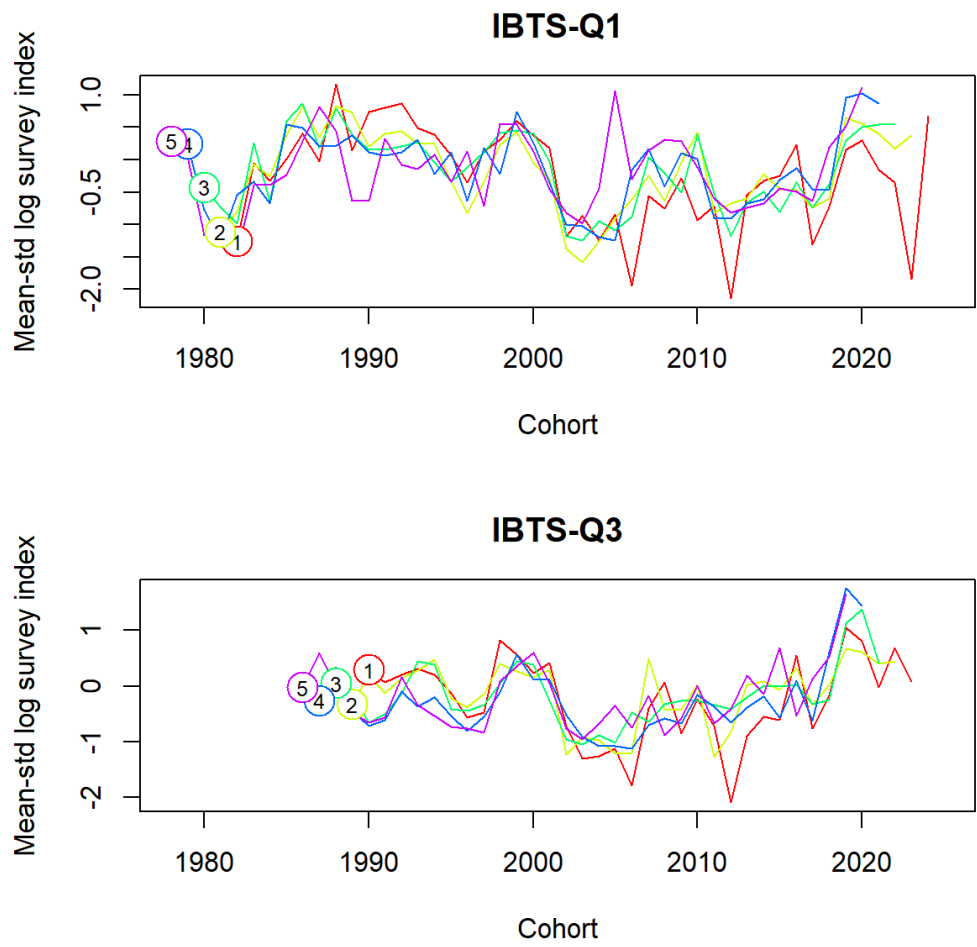


Figure 22.15. Whiting in Subarea 4 and Division 7.d. Survey log CPUE (catch per unit effort) for the IBTS–Q1 and Q3 surveys, by cohort. Each line shows the log CPUE for the age indicated at the start of the line.

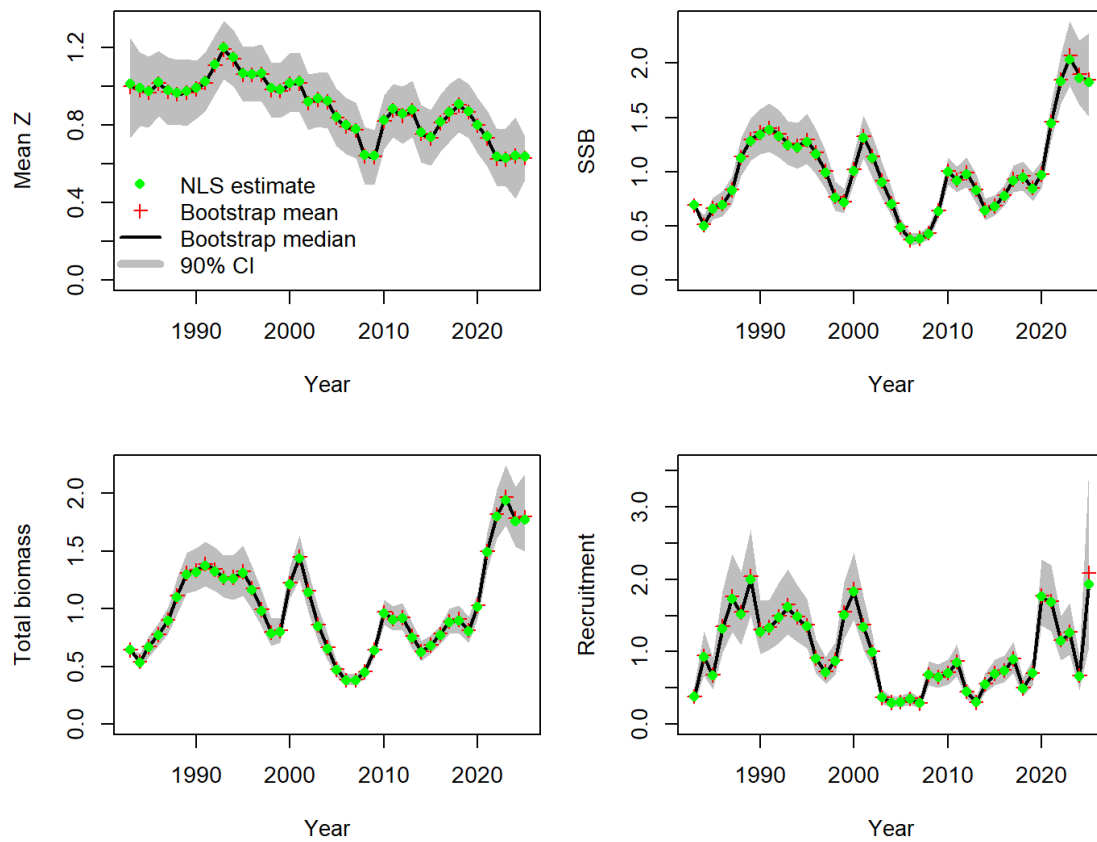


Figure 22.16. Whiting in Subarea 4 and Division 7.d. Summary plots from an exploratory SURBAR assessment, using both available surveys (IBTS–Q1 and Q3). Mean mortality Z (ages 2 to 4), relative spawning stock biomass (SSB), relative total biomass (TSB), and relative recruitment (age 1). Shaded grey areas correspond to the 90% CI. Green points give the model estimates, while red crosses and black lines give (respectively) the mean and median values from the uncertainty estimation bootstrap.

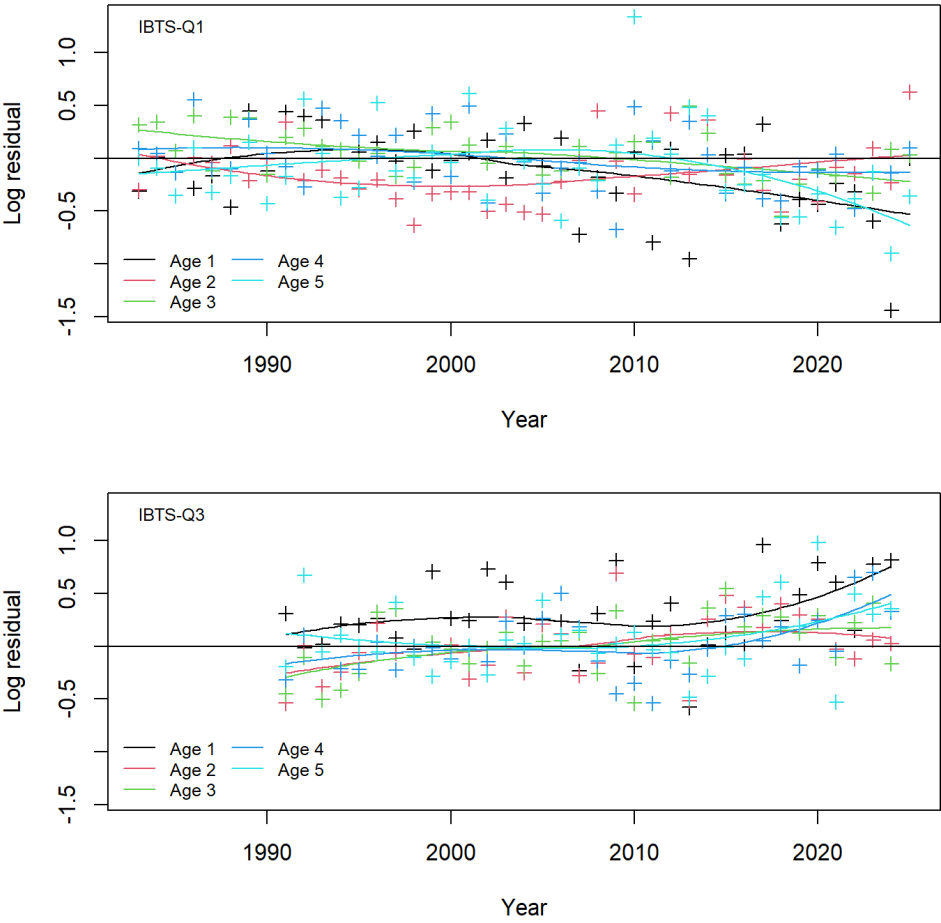


Figure 22.17. Whiting in Subarea 4 and Division 7.d. Log survey residuals from the SURBAR analysis. Ages are color-coded, and a LOESS smoother (span = 2) has been fitted through each age time-series.

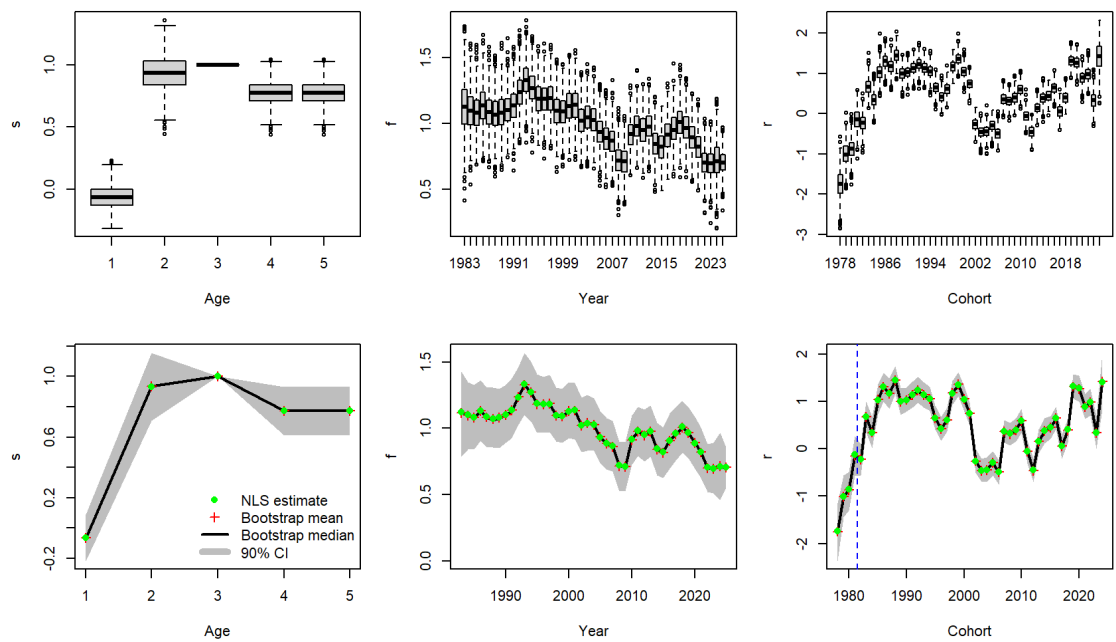


Figure 22.18. Whiting in Subarea 4 and Division 7.d. Parameter estimates from SURBAR analysis. Top row: age, year and cohort effect estimates as box-and-whisker plots. Bottom row: estimates as line plots with 90% confidence intervals.

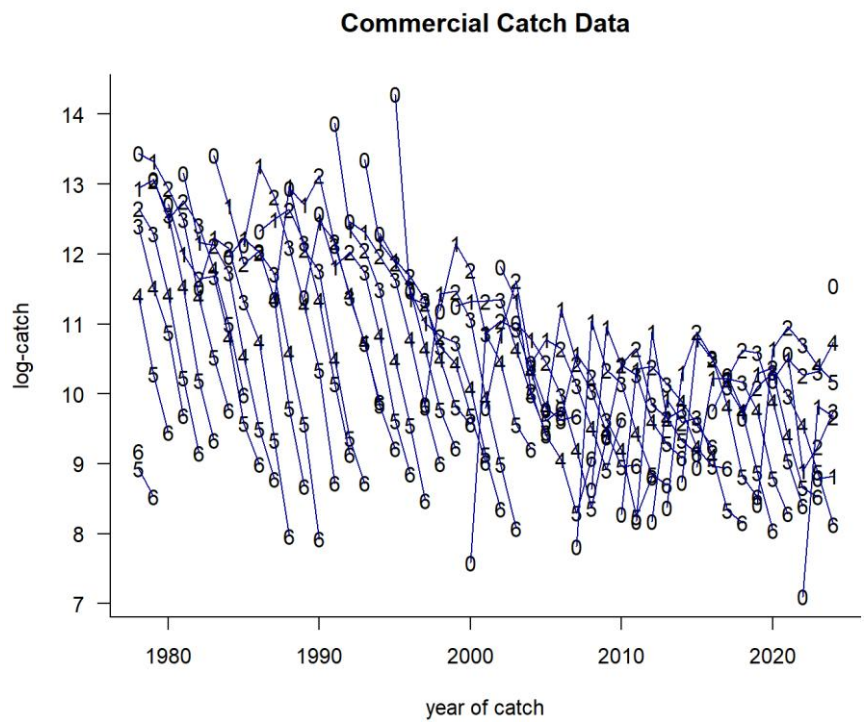


Figure 22.19. Whiting in Subarea 4 and Division 7.d. Log-catch curves by cohort for total catches (ages 0–6+).

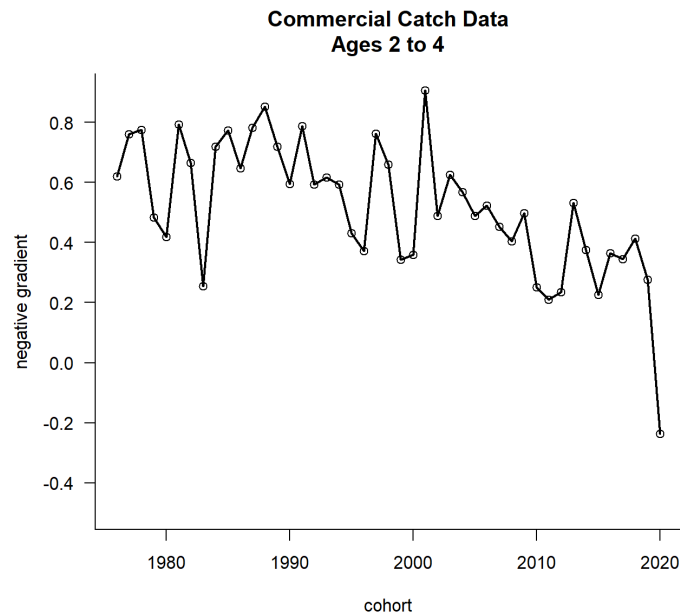


Figure 22.20. Whiting in Subarea 4 and Division 7.d. Negative gradients of log catches per cohort, averaged over ages 2–4. The x-axis represents the spawning year of each cohort.

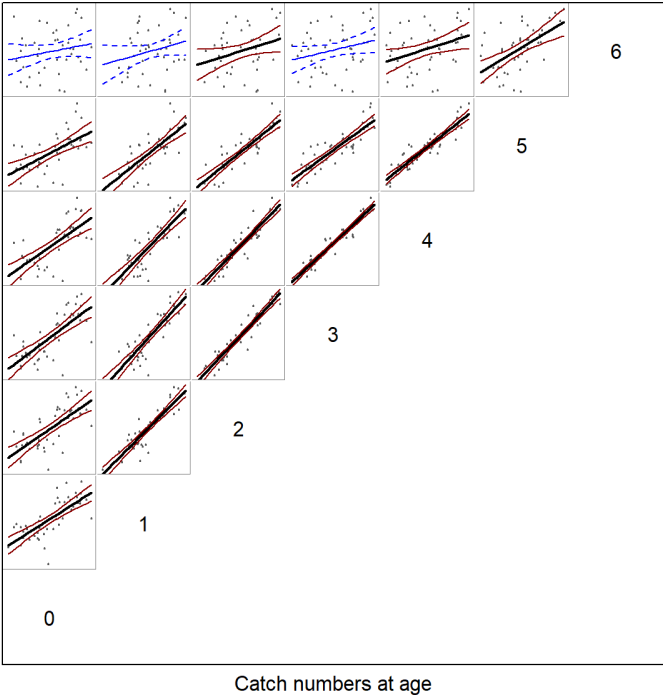


Figure 22.21. Whiting in Subarea 4 and Division 7.d. Correlations in the catch-at-age matrix (including the plus-group for ages 6 and older), comparing estimates at different ages for the same year-classes (cohorts). In each plot, the straight line is a normal linear model fit: a thick line (and black points) represents a significant ($p < 0.05$) regression, while a thin line (and blue points) is not significant. Approximate 95% confidence intervals for each fit are also shown.

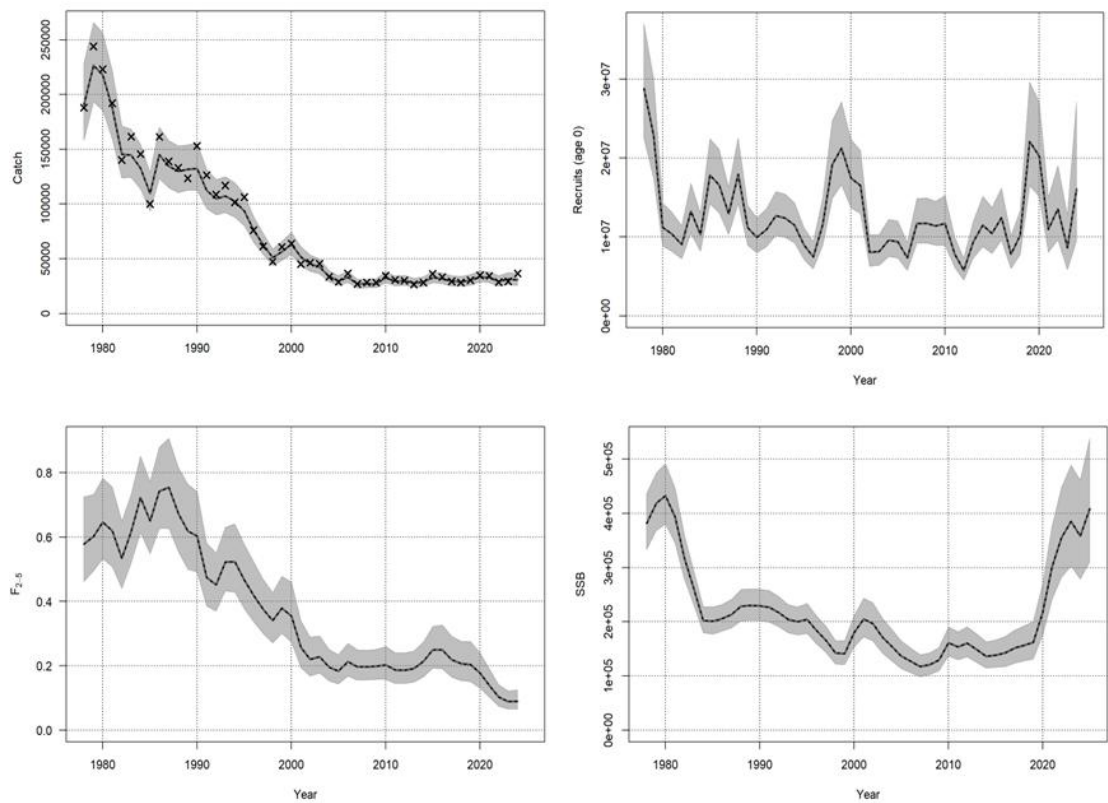


Figure 22.22. Whiting in Subarea 4 and Division 7.d. SAM assessment results using catch data series (1978–2024) with IBTS survey data starting in 1983 (Q1) and 1991 (Q3). Estimates with 95% Confidence intervals for total catch weight, SSB, mean fishing mortality and recruitment (at age 0).

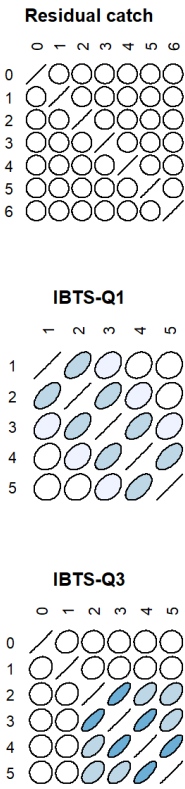


Figure 22.23. Whiting in Subarea 4 and Division 7.d. SAM estimated correlations between age groups for each fleet.

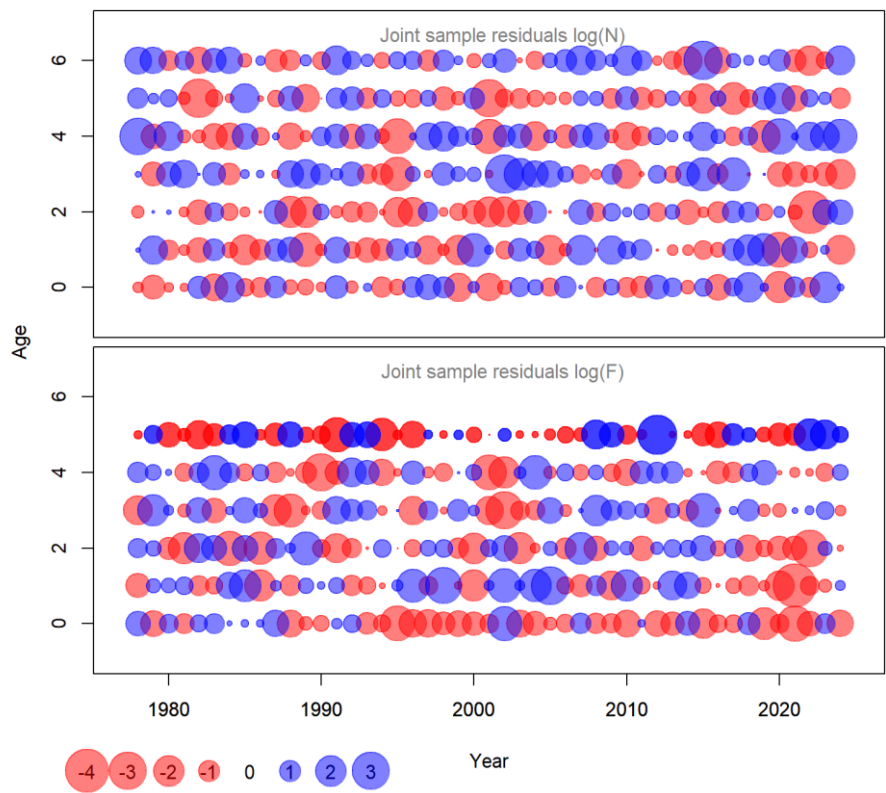


Figure 22.24. Whiting in Subarea 4 and Division 7.d. SAM standardised joint-sample residuals of process increments (for stock size N and fishing mortality F processes).

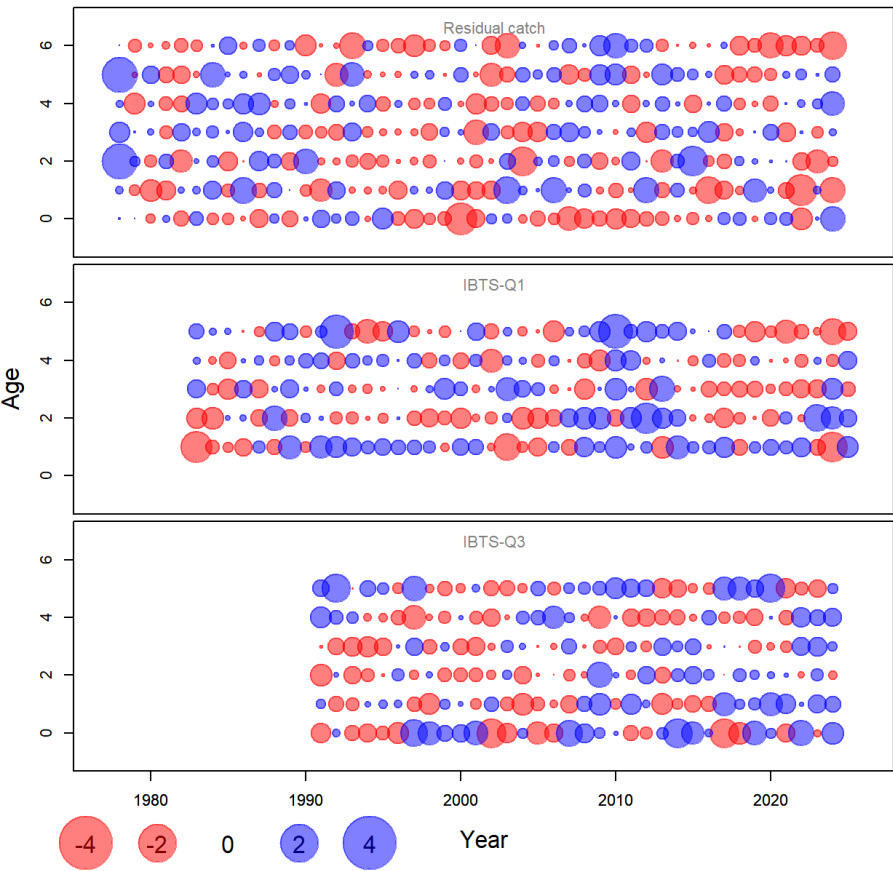


Figure 22.25. Whiting in Subarea 4 and Division 7.d. SAM standardized one-observation-ahead residuals.

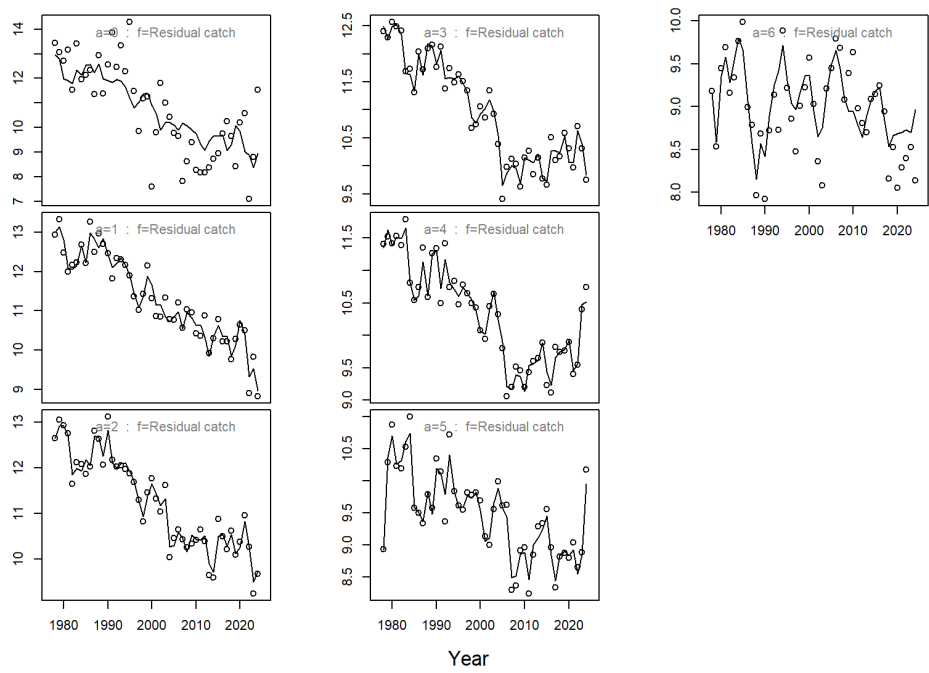


Figure 22.26. Whiting in Subarea 4 and Division 7.d. SAM predicted line and observed points (log scale) for the catch fleet.

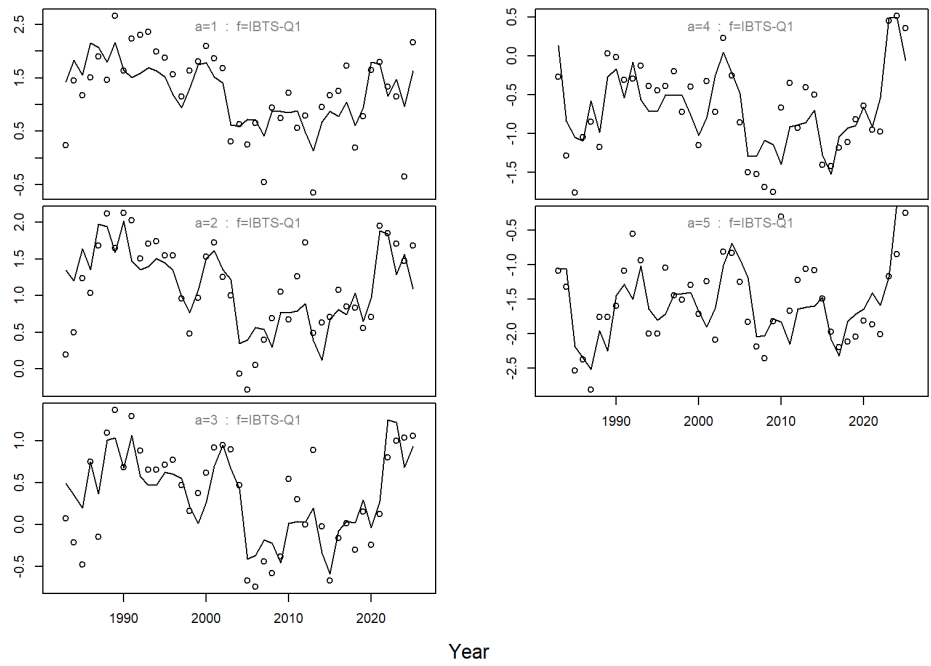


Figure 22.27. Whiting in Subarea 4 and Division 7.d. SAM predicted line and observed points (log scale), for survey fleet IBTS Q1.

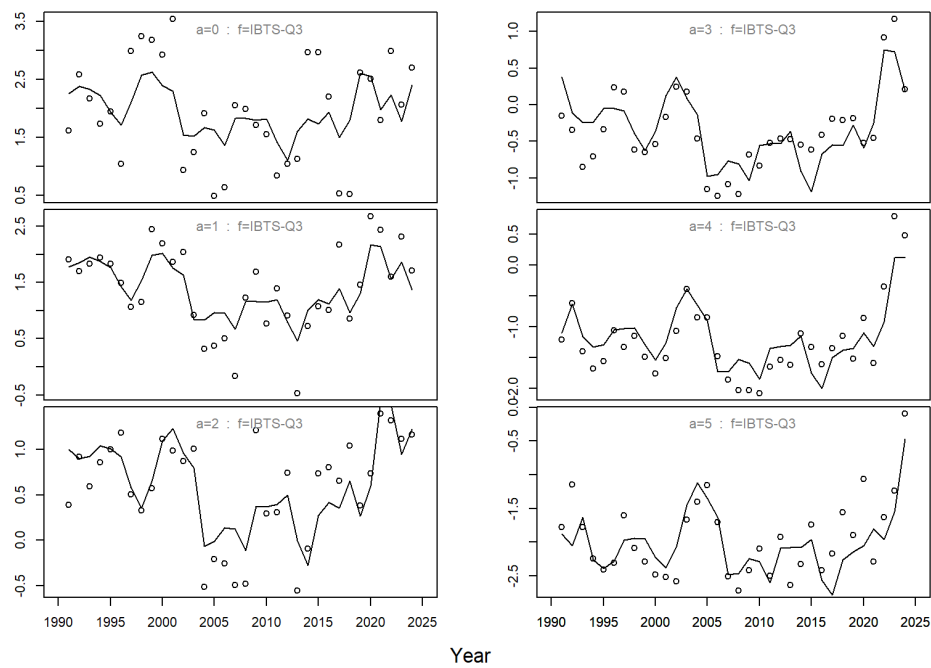


Figure 22.28. Whiting in Subarea 4 and Division 7.d. SAM predicted line and observed points (log scale), for survey fleet IBTS Q3.

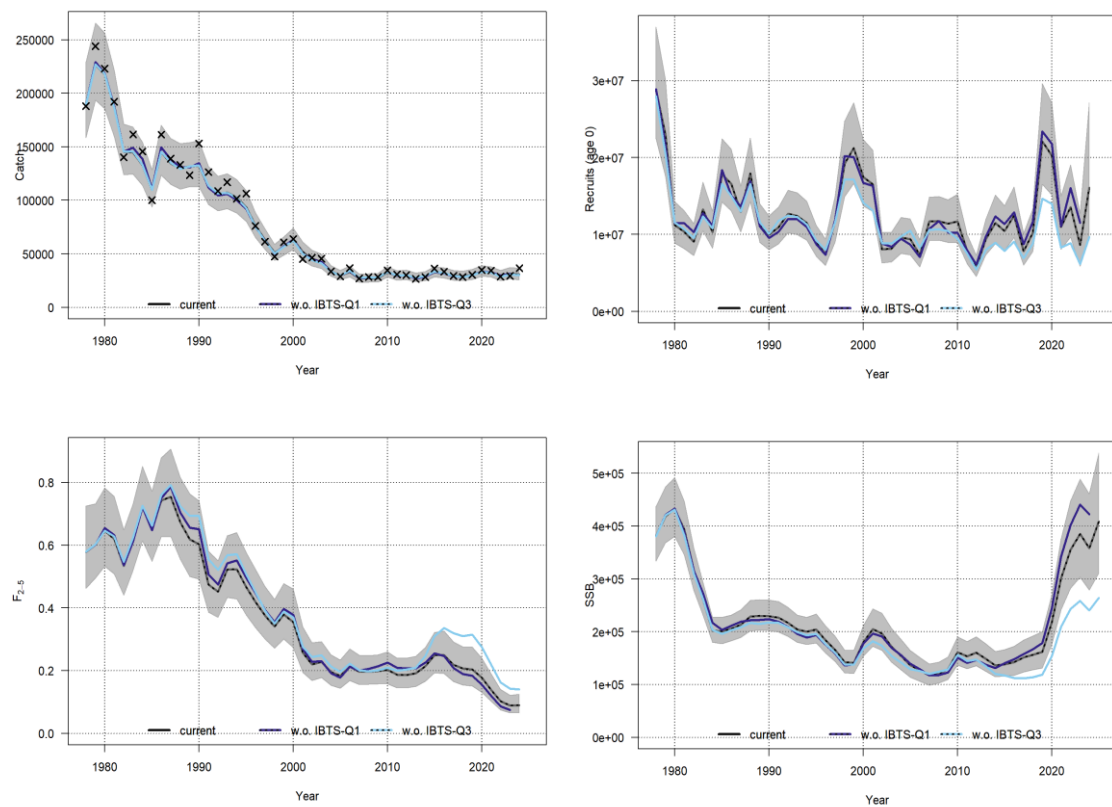


Figure 22.29. Whiting in Subarea 4 and Division 7.d. SAM leave-one-out diagnostics. Final run (black), run without IBTS Q1 (dark blue), run without IBTS Q3 (light blue).

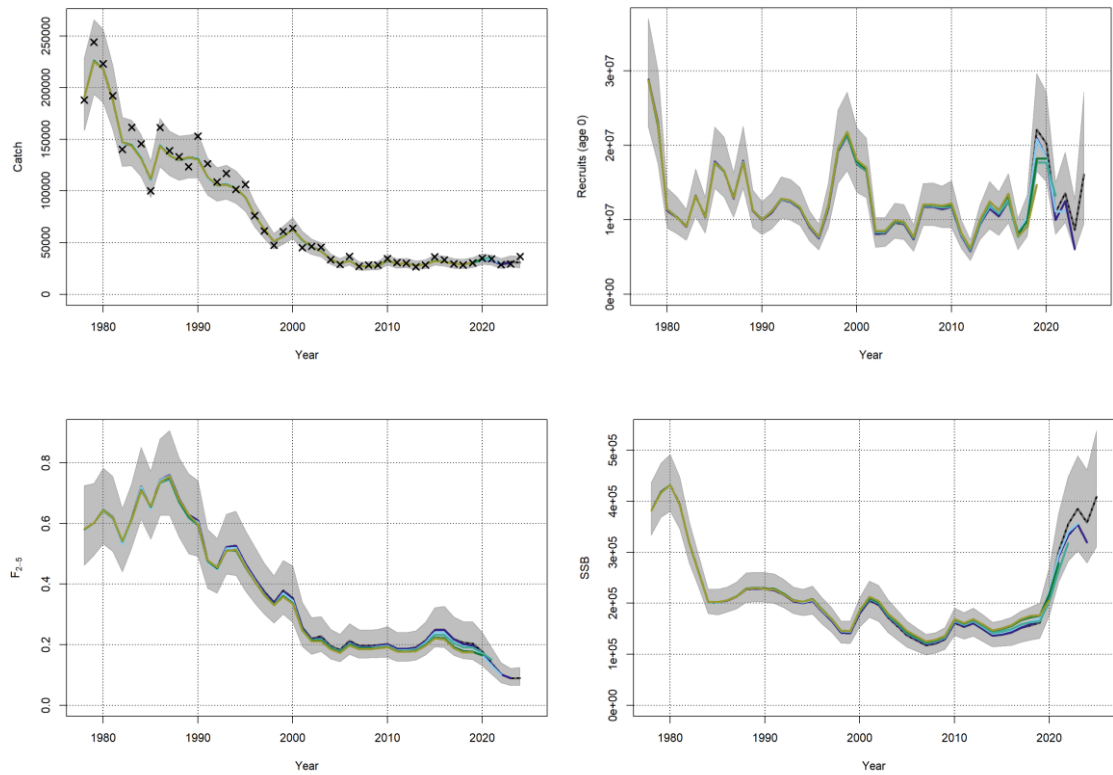


Figure 22.30. Whiting in Subarea 4 and Division 7.d. SAM Retrospective pattern in catch estimates, SSB, fishing mortality and recruitment.

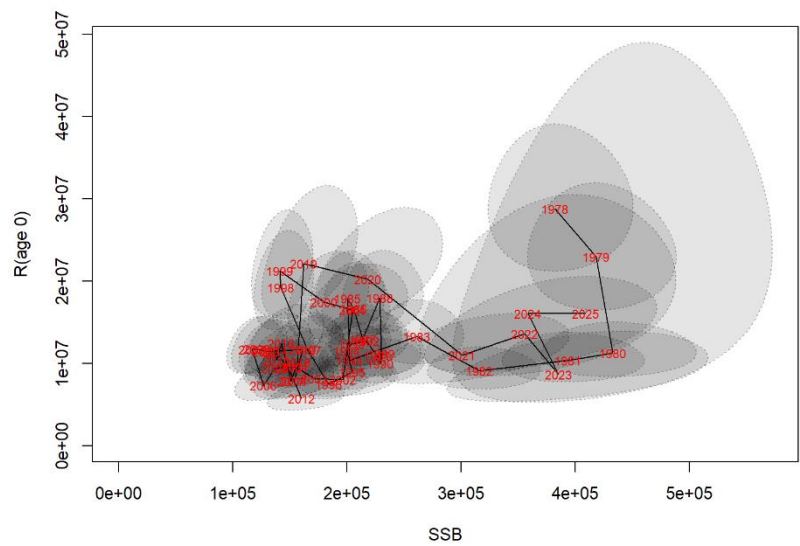


Figure 22.31. Whiting in Subarea 4 and Division 7.d. Stock-recruitment relationship.

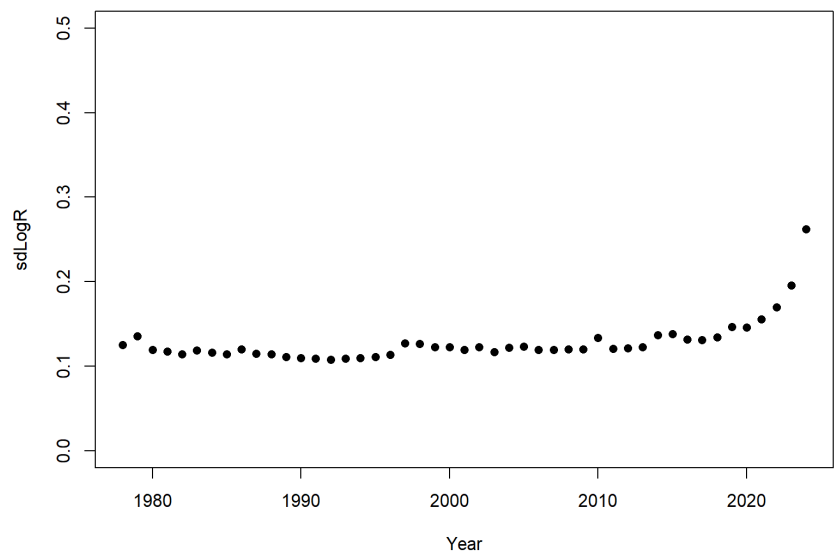


Figure 22.32. Whiting in Subarea 4 and Division 7.d. SAM estimated standard deviation in logarithmic recruitment (age 0).

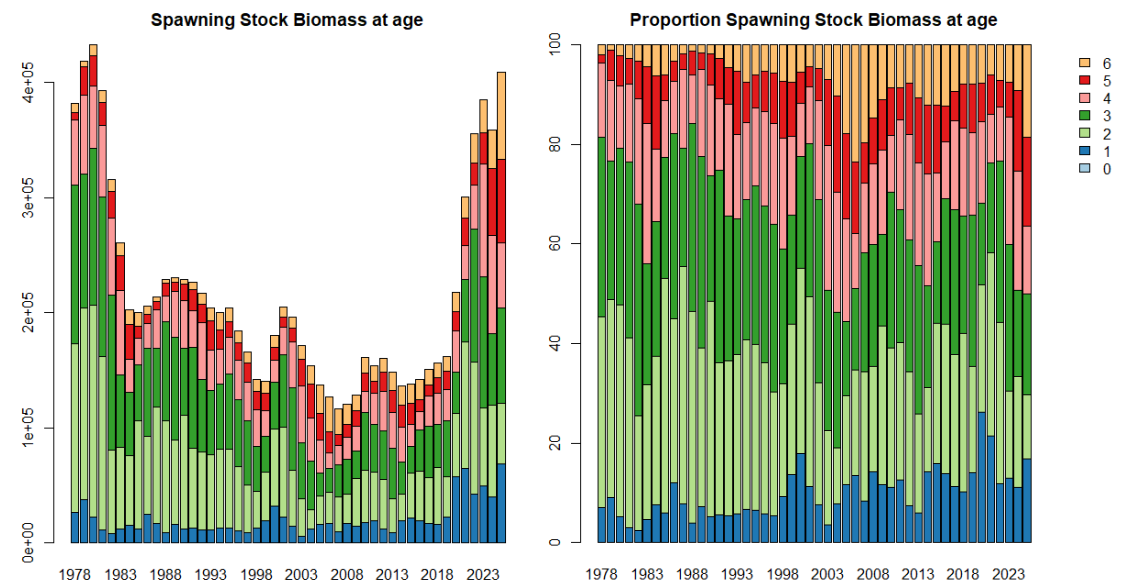


Figure 22.33. Whiting in Subarea 4 and Division 7.d. SAM estimated spawning stock biomass at age per year, in absolute numbers (left) and as a percentage of total spawning stock biomass.

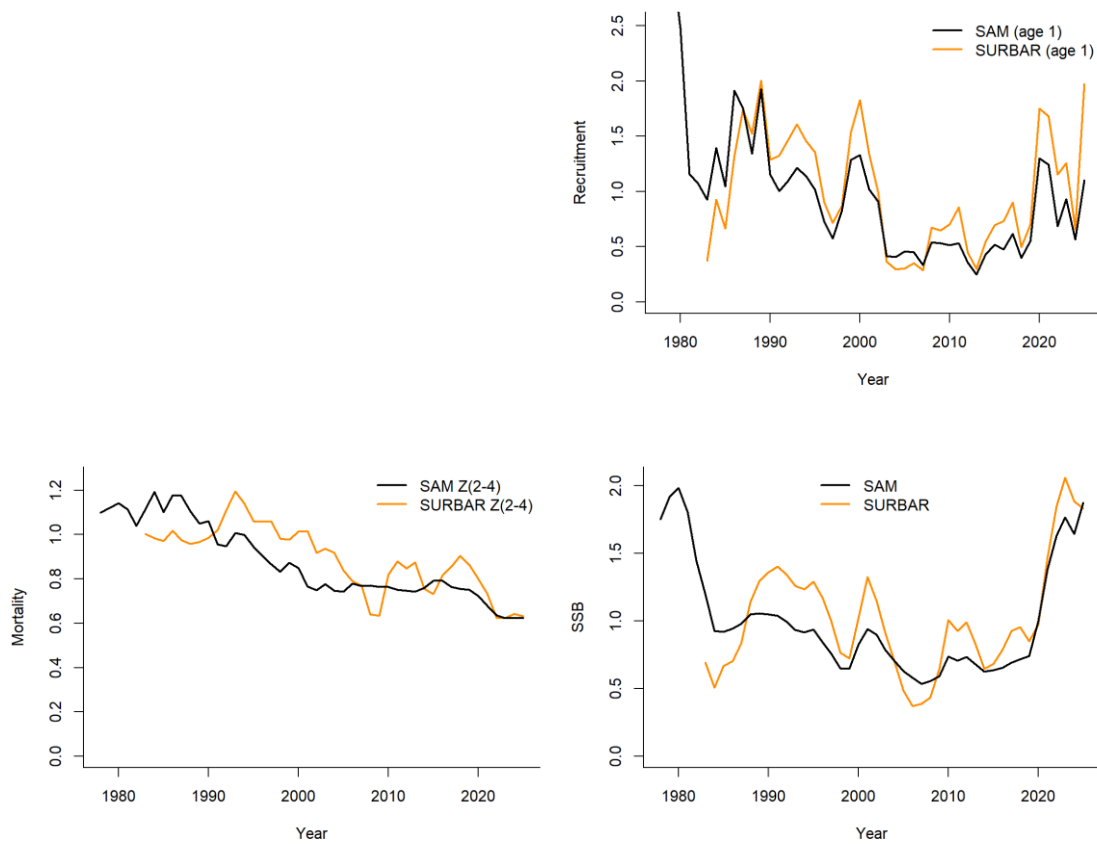


Figure 22.34. Whiting in Subarea 4 and Division 7.d. Comparisons of stock summary estimates from the final SAM (black) and SURBAR (orange) models. To facilitate comparison, recruitment and SSB values have been mean-standardised using the year range for which estimates are available from all three models. Mortality is presented as total mortality $Z(2-4)$ for SAM and for SURBAR.

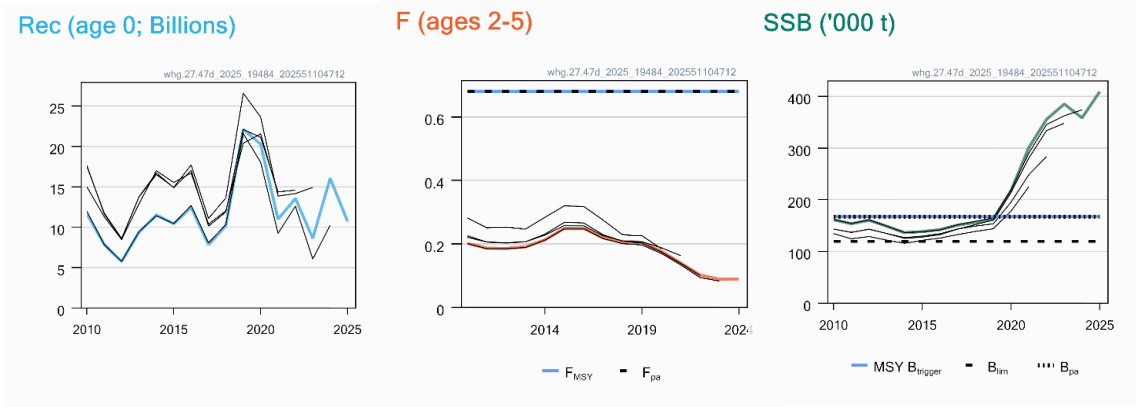


Figure 22.35. Whiting in Subarea 4 and Division 7.d. Historical assessments from Standard graphs.

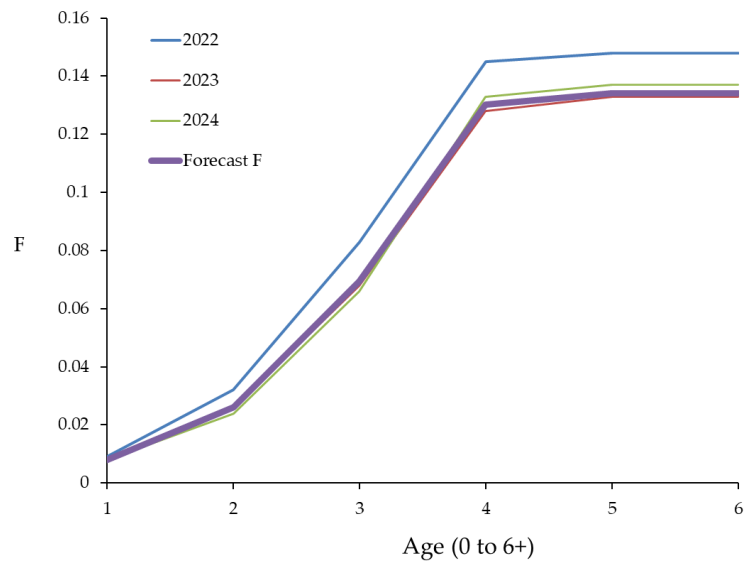


Figure 22.36. Whiting in Subarea 4 and Division 7.d. SAM F at age estimates for 2022–2024, along with scaled mean exploitation used for the forecast.

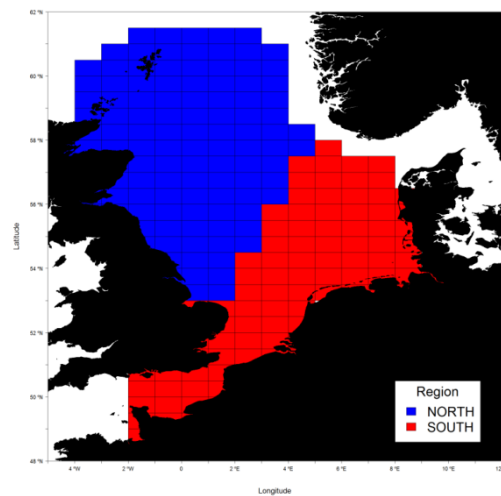


Figure 22.37. Whiting in Subarea 4 and Division 7.d. Components suggested by Holmes *et al.* (2014) to analyse spatial differences in maturation and SURBAR analysis.

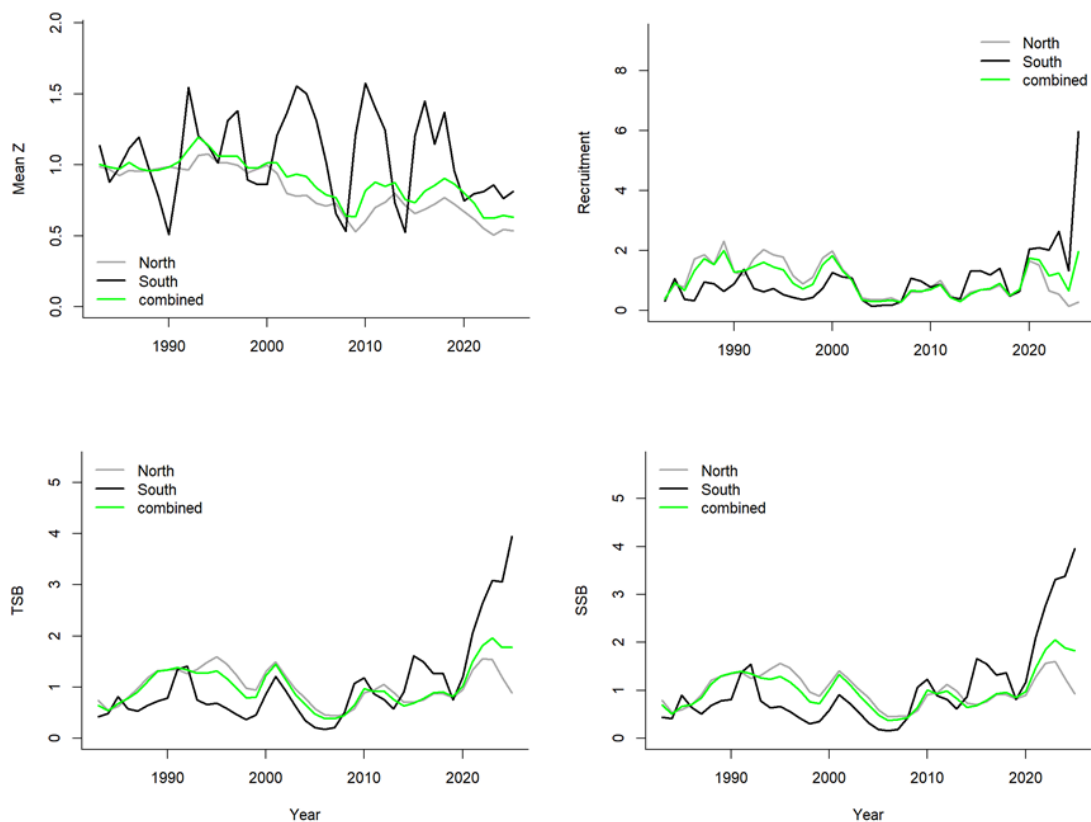


Figure 22.38. Whiting in Subarea 4 and Division 7.d. SURBAR results comparison combined (whg.27.4.47d) and northern and southern component as defined in WKNSEA 2018. Recruitment at age 1, total mortality is mean Z for ages 2–4, total stock biomass (TSB) and spawning stock biomass (SSB).

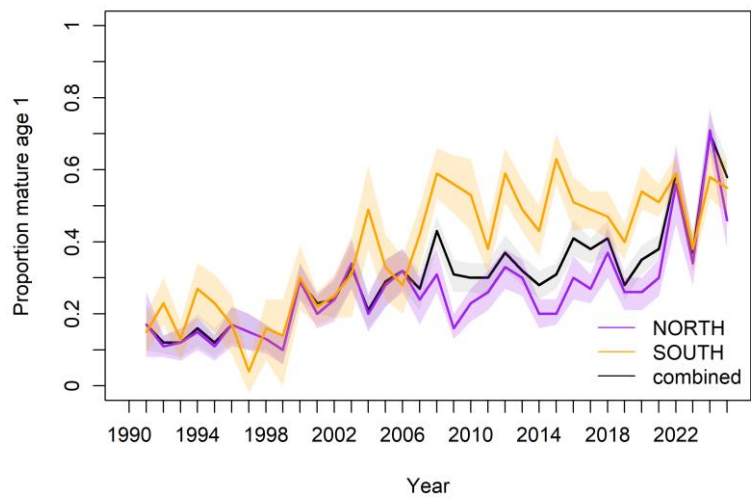


Figure 22.39a. Whiting in Subarea 4 and Division 7.d. Trends in proportion mature individuals at age 1 for combined (whg.27.4.47d) and northern and southern component as defined in WKNSEA 2018.

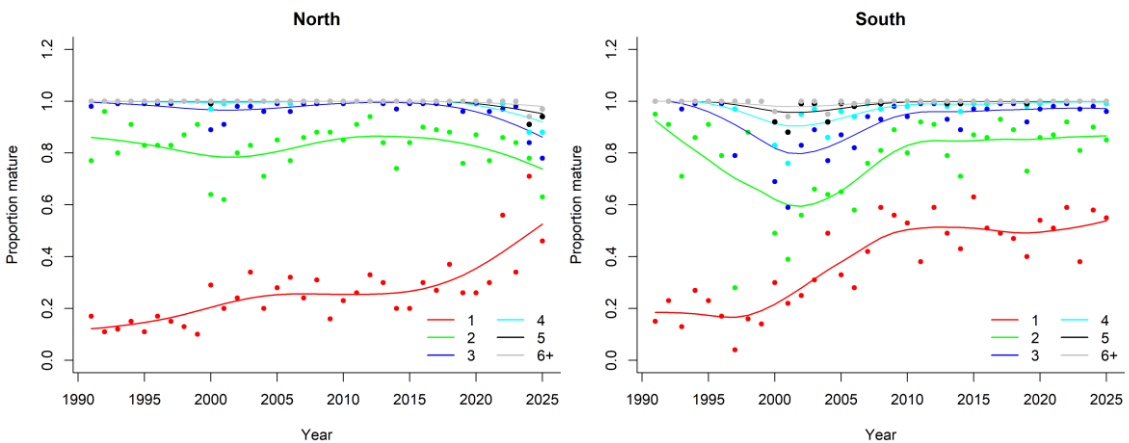


Figure 22.39b. Whiting in Subarea 4 and Division 7.d. Trends in proportion mature individuals at age for northern and southern component, as defined in WKNSEA 2018.

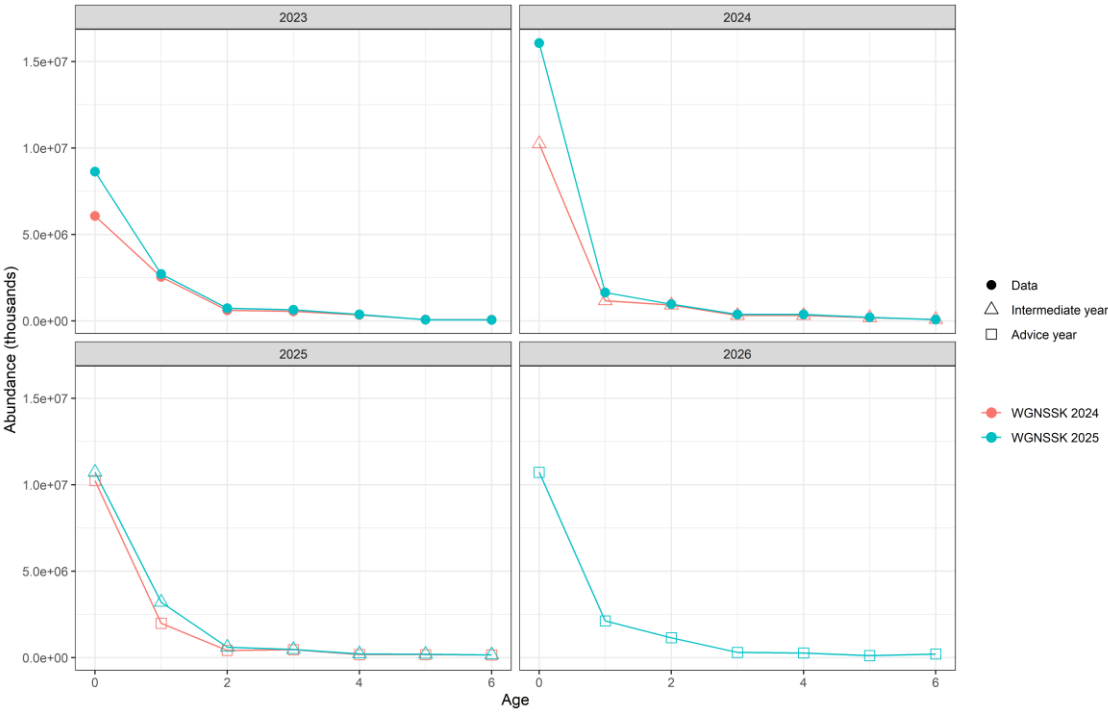


Figure 22.40. Whiting in Subarea 4 and Division 7.d. Comparison of numbers at age between previous (WGNSSK 2024) and current (WGNSSK 2025) assessments and forecasts.

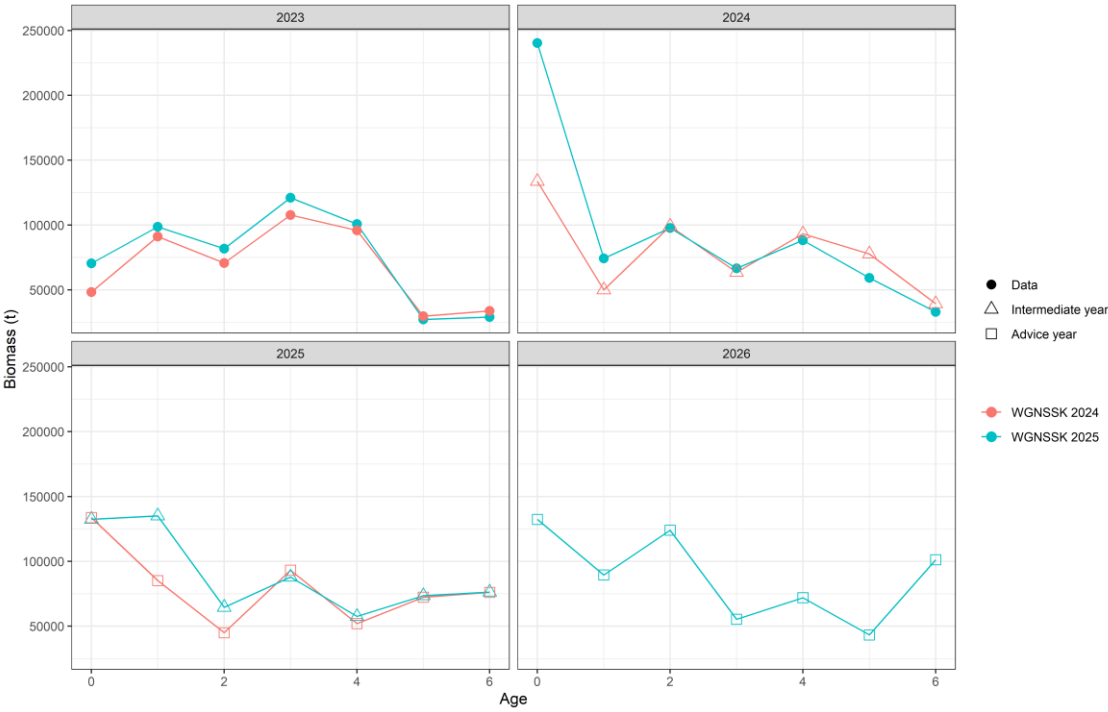


Figure 22.41. Whiting in Subarea 4 and Division 7.d. Comparison of biomass at age between previous (WGNSSK 2024) and current (WGNSSK 2025) assessments and forecasts.

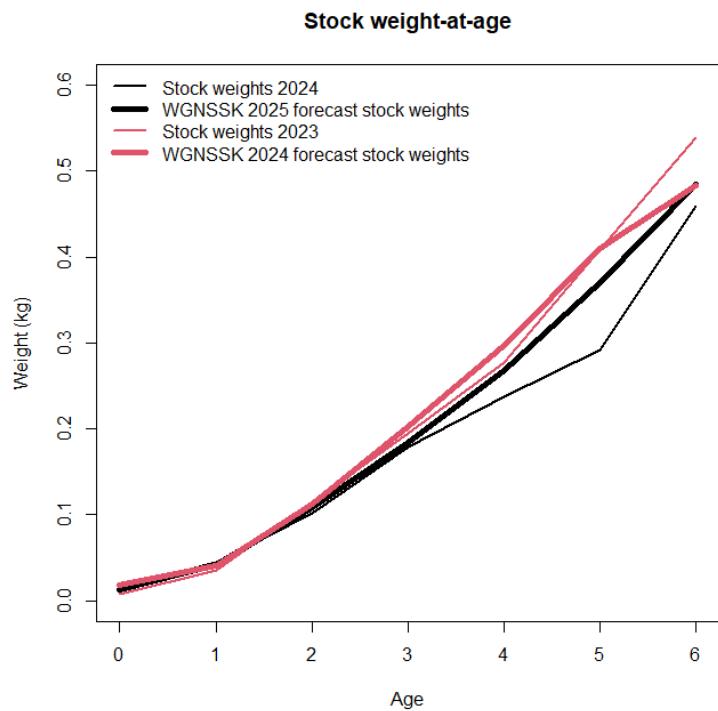


Figure 22.42. Whiting in Subarea 4 and Division 7.d. Comparison of stock weights at age used in the previous (WGNSSK 2024) and current (WGNSSK 2025) forecasts (thick line). Stock weights at age used in the last year of the previous (WGNSSK 2024) and current (WGNSSK 2025) assessments are shown as comparison (thin line).

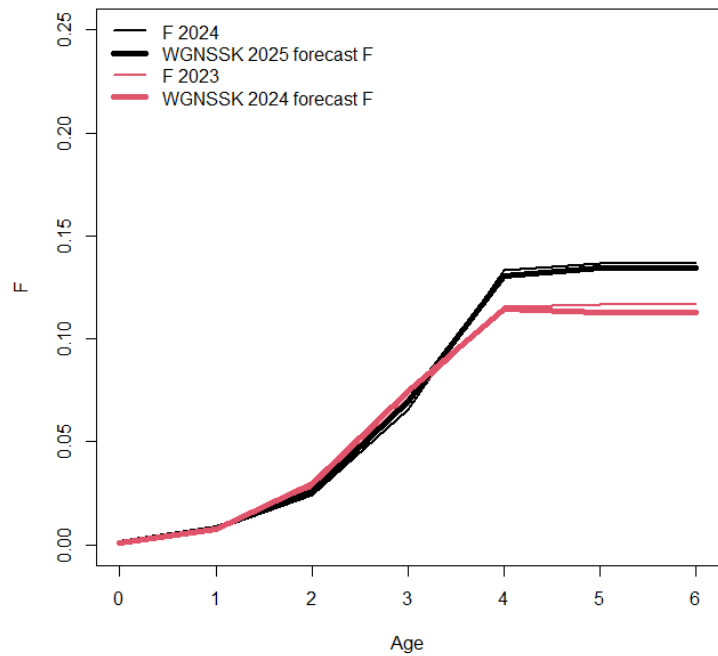


Figure 22.43. Whiting in Subarea 4 and Division 7.d. Comparison of F at age used in the previous (WGNSSK 2024) and current (WGNSSK 2025) forecasts (thick line). F at age from the last data year of the previous (WGNSSK 2024) and current (WGNSSK 2025) assessments are shown as comparison (thin line).

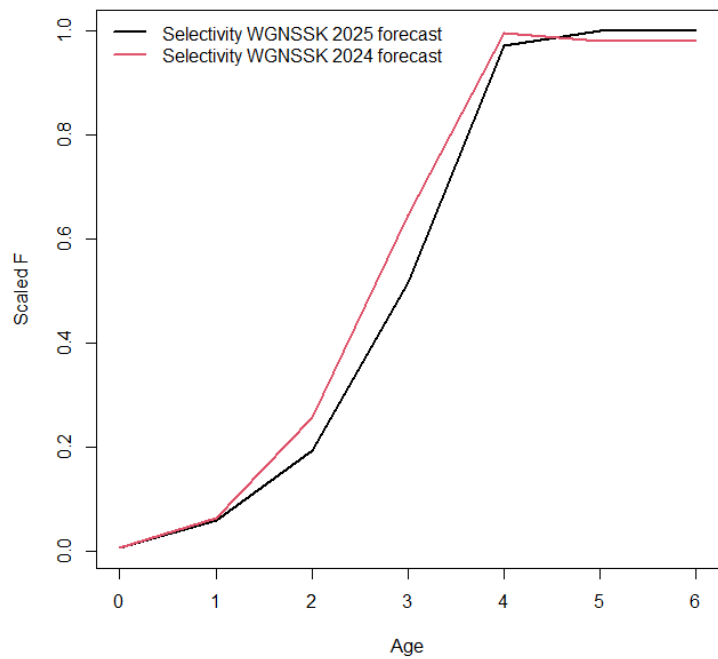


Figure 22.44. Whiting in Subarea 4 and Division 7.d. Comparison of selectivities (F scaled to 1) used in the previous (WGNSSK 2024) and current (WGNSSK 2025) forecasts.

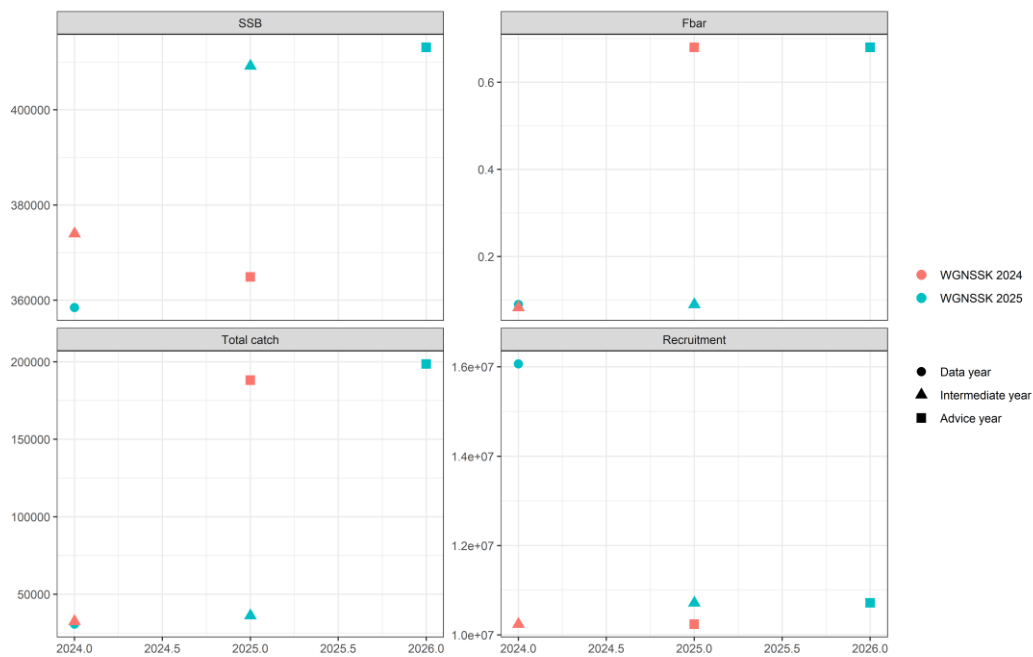


Figure 22.45. Whiting in Subarea 4 and Division 7.d. Comparison of assumptions used in the previous (WGNSSK 2024) and current (WGNSSK 2025) forecasts.

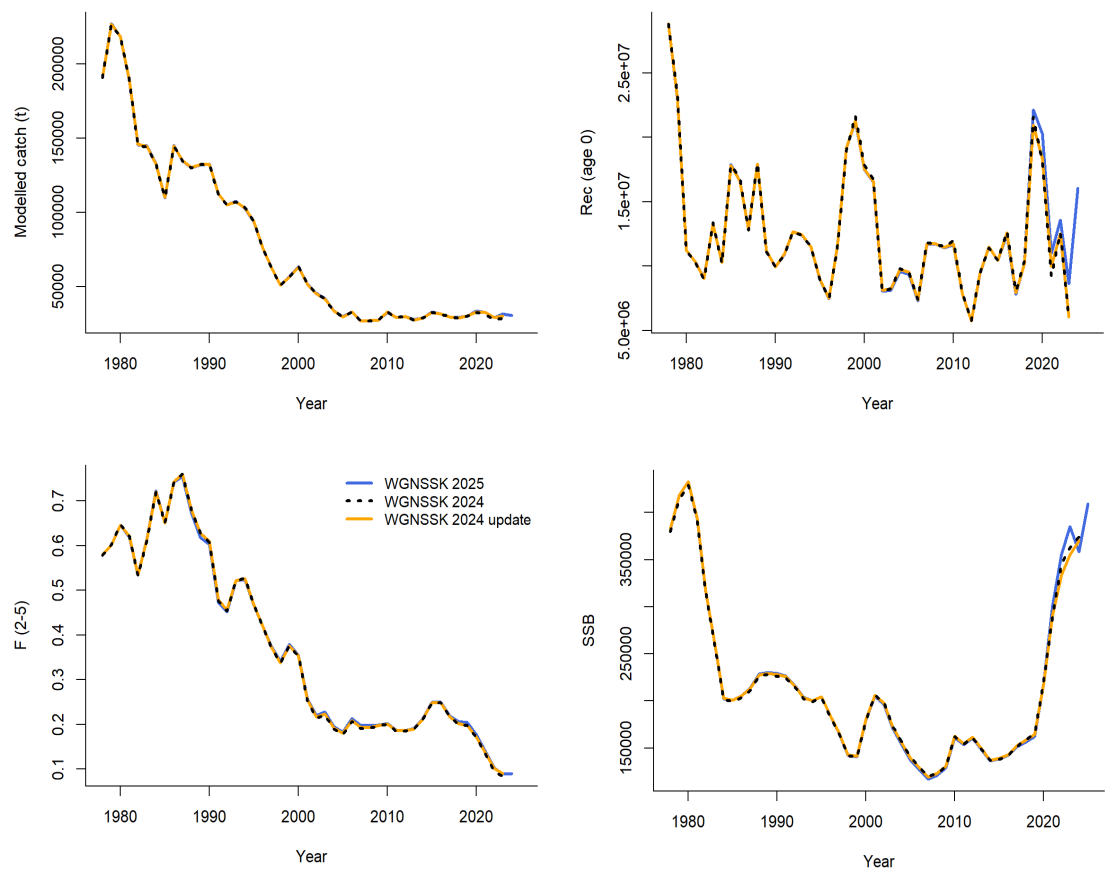


Figure 22.46. Whiting in Subarea 4 and Division 7.d. Comparison of assessments WGNSSK 2024 with updated and old catch data and WGNSSK 2025 assessment.

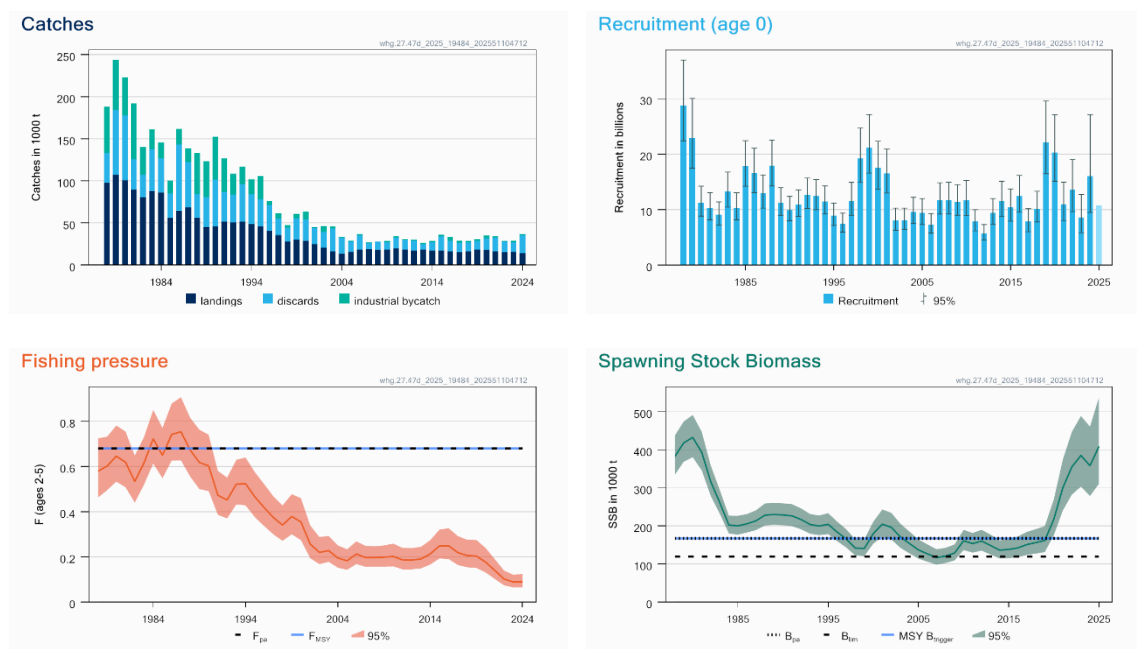


Figure 22.47. Whiting in Subarea 4 and Division 7.d. Summary of the stock assessment. The assumed recruitment value for 2025 is shaded in a lighter colour. ICES estimated catches, discards include below minimum size (BMS) landings.

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